State of EDR in the US
CDR Update Apr 2021
CDR Software Level 21.0.2



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Main Messages

More and More Data every year!

GM SDM50 – 8 sec precrash, strg, whl spds

Ford AB12 and RC8, speed @0.2 sec

Toyota 17EDR – X and Y DV sych'd

Toyota VCS data includes PICTURES ~~CDR~~

Most Subaru now covered by CDR

More Mitsubishi coverage expected Dec/Jan

GM ASCM's – more data than SDM

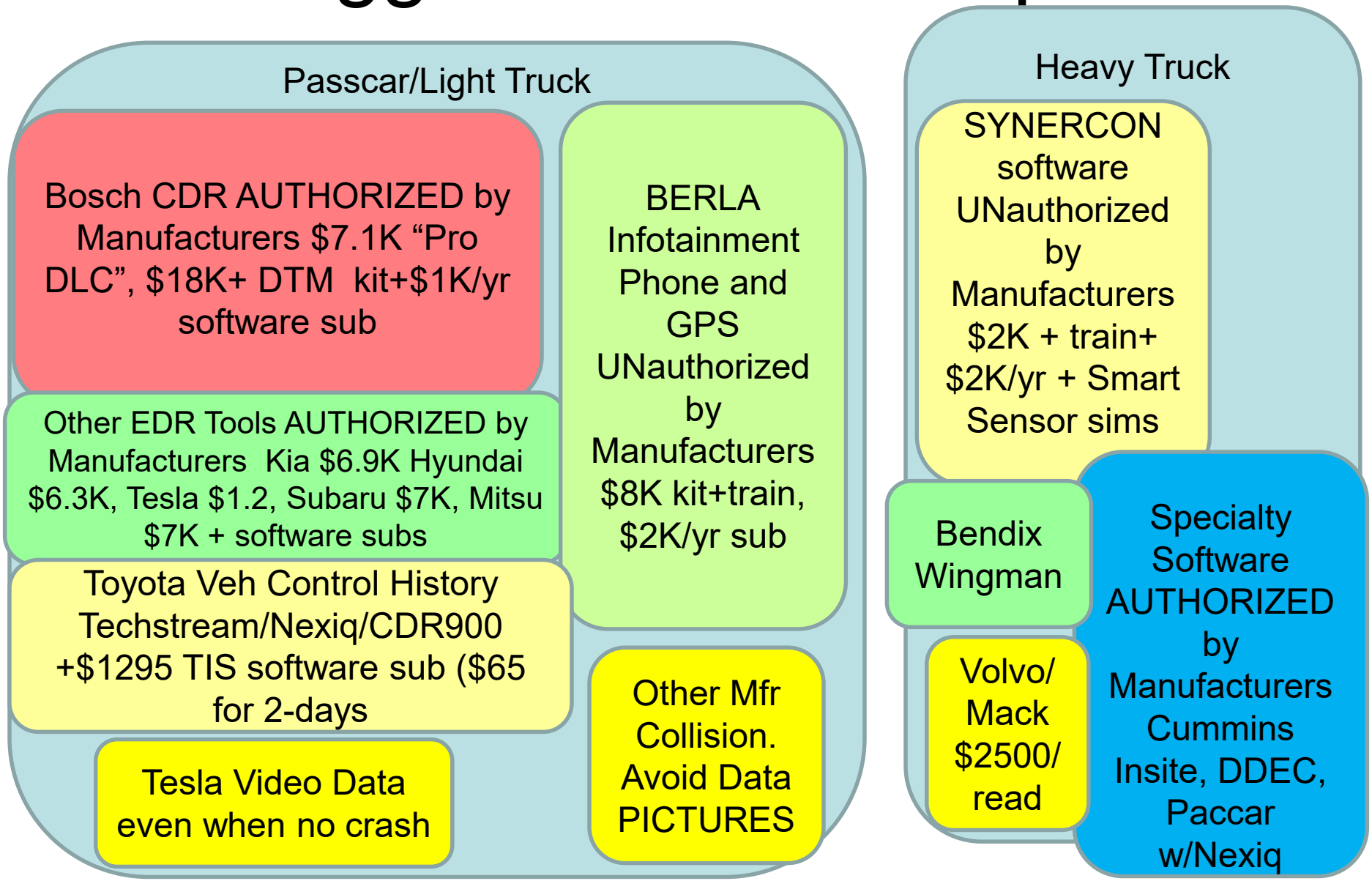
Other Mfrs storing PICTURES ~~CDR~~

Lots of anomalies to keep us on our toes

Keeping EDR's in Perspective

1. There are 305 million vehicles registered in the US. 63+% have an EDR with data accessible by the Bosch CDR system
2. 87+% of 2020's are supported by Bosch.
3. Another 12% have EDR accessible by other tools. **99+%** have EDR.
4. With *advanced* analysis techniques, we can get ***speed at impact*** for both vehicles even if only one has a 563 EDR.

The bigger vehicle data picture



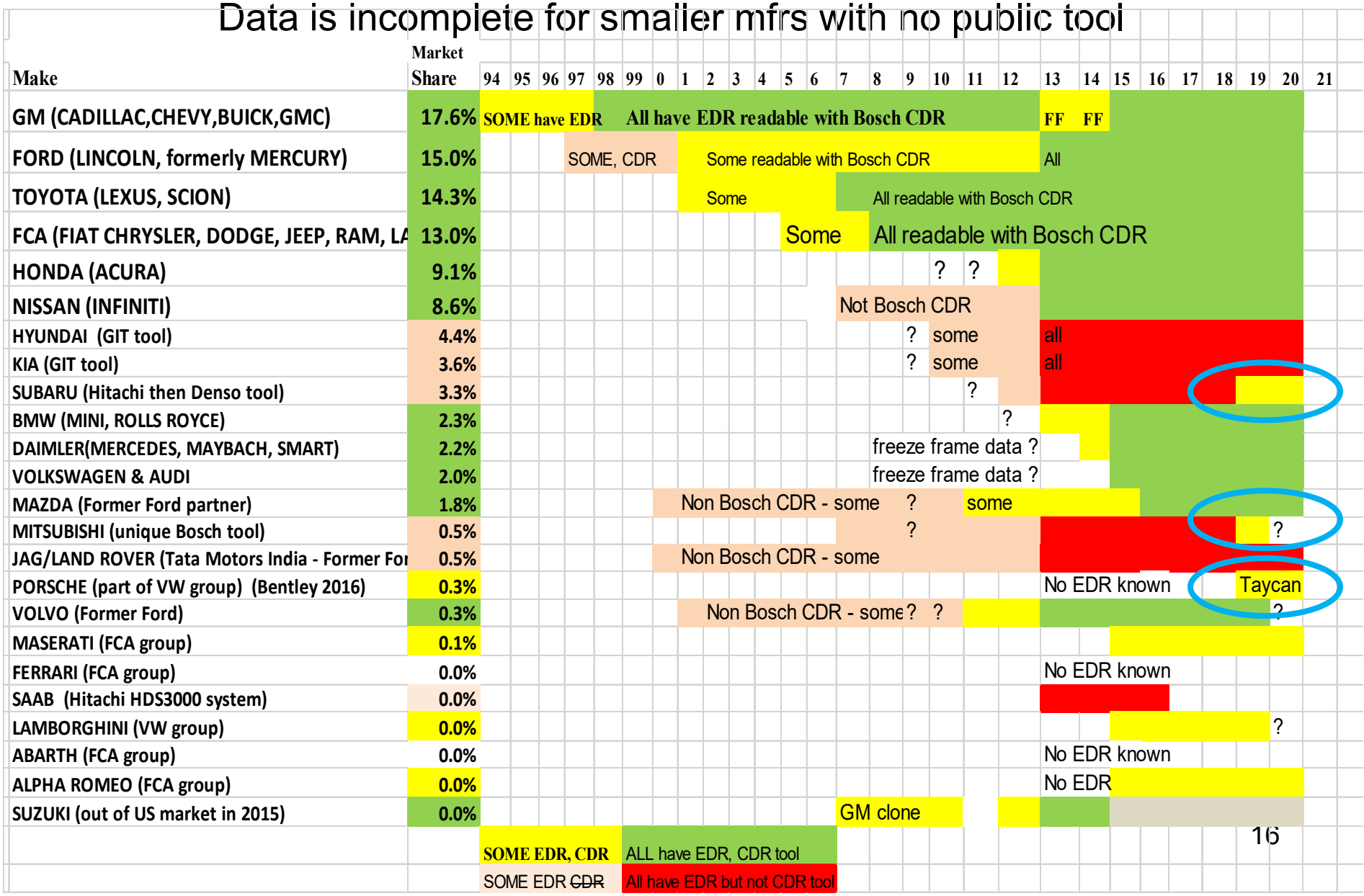
US Lt. Veh EDR & Tool Used

Manufacturer	Annual Sales 4Q 18 to 3Q 19	Percent	EDR First Known Use	Tool?	
GM (Chevy, Buick, GMC, Cadillac)	2,920,580	17.0%	1994	Bosch CDR	
Ford (Ford, Lincoln)	2,404,373	14.0%	1997	Bosch CDR 2001+	
Toyota (includes Lexus, Scion)	2,381,734	13.9%	2001	Bosch CDR	
Fiat Chrysler (includes Dodge, Ram, Jeep, Lancia	2,216,366	12.9%	2005	Bosch CDR	
Honda (includes Acura)	1,604,240	9.4%	2012	Bosch CDR 2012+	
Nissan (includes Infinity, Renault, EXCLUDES Mitsubishi)	1,413,835	8.3%	2006	Nissan Consult	CDR 2013+
Hyundai (Includes Kia, Genesis)	1,309,862	7.6%	2010	GIT tool co.	
Subaru	702,046	4.1%	2010	Denso +SSM	CDR 2019
BMW (incudes Mini, Rolls Royce)	640,003	3.7%	2013	Bosch CDR	
Volkswagen (Includes Audi, Porsche, Bentley)	355,692	2.1%	2014	Bosch CDR	
Daimler Benz (Mercedes, Smart	348,173	2.0%	2014	Bosch CDR	
Mazda	273,368	1.6%	2011	Bosch CDR 2011+	
Tesla	216,125	1.3%	2012	Tesla Tool	
Tata (Jaguar, Land Rover)	125,388	0.7%		Bosch SPX	
Mitsubishi (Part of Nissan Renault Alliance)	120,000	0.7%	2006	Bosch SPX	CDR 2019
Volvo	102,059	0.6%	early	Bosch CDR 2011+	
Industry Total	17,133,844.00	100.0%			

Porsche 2020 Taycan

EDR Phase-in by Mfr

Data is incomplete for smaller mfrs with no public tool



2021 Recording Capability

EDR Major Data Elements by Manufacturer (All Mfrs with EDR have the required 15 data elements, all are not listed)

Information is taken from Bosch CDR sample files, Ruth cases, and 2012 MY practice projected into 2019 model year

	Part 563 Table 1								Part 563 Table 2								Beyond 563									
Mfr	Event Complete	Key Cycles	Long. ΔV	Speed/Brake/ Throttle or Accel	Multi Event	Lateral ΔV	RPM	Steering Angle	ACM Acceleration	ABS on/off	Roll Angle	ESC Long Accel	ESC Lat Accel	Yaw Rate	Tire Pressure	Panic Brake Assist	Wheel Speeds	Roll Rate	Brake Pressure	OTHER
GM	X	X	X	5@0.5 TA	X	X	X	SDM50	, Z 300@2	X		50	50	50	LO		50	1 @ 10ms	50	50 Day/date/time
Ford	X	X	X	5@0.5 A	X	X	X	5@0.1	-	X		5@0.1		5@0.1	LO		-	5@0.1	**	2017 AB12 precrash 0.2 sec
Chrysler	X	X	X	5@0.1 TA	X	X	X	5@0.1	-	X		-		5@0.1	PSI	X	X			2018 ACC set speed 2018, Avoidance
Toyota	X	X	X	5@0.5 TA	X	Side	X	5@0.5	-		Peak&atTRG	5	in roll	5@0.5	-	-	-	In Roll	5@0.5	Ped 2018, Veh history rpt
Honda	X	X	X	5@0.5 TA	X	X	X	5@0.5	X, Y, Z 250@10	X	-1+5@0.1	-		-	-	-	-			AEmBrk. Lanekeep
Nissan	X	X	X	5@0.5 A	X	X	X	5@0.5	X, Y 250@10	-	-	-		-	-	-	?			
Mazda	X	X	X	5@0.5 T	X	X	X		X, Y, Z 250@10	-							-			
Volvo	X	X	X	5@0.5T	X	2016 XC90	-	2016 XC90	X,Y,Z 250@10	2016 XC90	-1+0.3@0.1									
BMW	X	X	X	5@0.5A	X		X	5@0.5	X, Y 300@10ms	X	-									
Mercedes	X	X	X	5@0.5A	X	2017 E class adds														
VW	X	X	X	5@0.5A	X	X														Time of Crash
Hyundai/Kia	X	X	X	5@0.5	X	X	X	5@0.5	X,Y,Z 250@10	X	-1+5@0.1	-		-	-	-	-		19 Forte	
Subaru	X	X	X	5@0.5A	X	X	X	5@0.5		X								-1 to+1.5		Eyesight
Mitsubishi	X	X	X	5@0.5A	X	X	X	5@0.5	X, Y 250@10		-1+5@0.1									

Bosch CDR Hardware Update

1. *CDR-900 VCI module released July 2018 needed for new vehicles*
2. *Supports w/*CAN-HD and Ethernet.
3. Derivative of proven diagnostic scan tool.
4. *New round 19 pins cables vs. flat 15 pin for new releases after 2018.*
5. *CDR 900 must be registered to your laptop (network device) before use.*



CDR 900 Upgrade Kit

$4000

CDR900

1. Needed for NEW vehicles
2. Intended to eventually read many legacy vehicles, but CAN+ still needed ..

CDR900 Upgrade Kit



Help file identifies interface needed

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
13	2020 Chevrolet	Aveo (note 6)	ACM	Data	CANplus	F00K108287	F00K108454	Center tunnel
4	2020 Chevrolet	Blazer (note 6)	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the rear cargo storage area, under the left rear trim panel, at the front of the wheel well

CAN+ only

Either one

CDR900only

CDR 900 details

1. CDR900 integrates power management (no external relays/adapters).
2. CDR900 uses the existing 2.5 amp current limited power supply, but new ACM's may drive bigger supply later.
3. Includes WIRELESS laptop to interface
4. CDR900 does NOT integrate the current CDR500 Flex Ray adapter – still needed!
5. CDR 900 firmware must be flashed initially

CDR900 Tips

1. CDR900 is a network device, VPN's do NOT LIKE other network devices, they are designed to keep them out. CDR900 does not like laptops with firewalls and other virus protection and lots of other programs – suggest using a *dedicated laptop* with few other programs installed.
2. VCI Manager and CDR are *two different programs*, you MUST close VCI manager before CDR program will recognize CDR900!

New users need **both** CAN+ and CDR900



MSRP: $7300

plus shipping & handling

Add to Cart ›

Bosch CDR Pro Tool Kit

DLC ONLY

The CDR Pro Tool Kit is the heart and soul of the Bosch CDR Tool. This package contains all of the hardware required to perform a DLC/OBD retrieval of EDR data in Bosch supported vehicles. It includes both the CDR 900 and CANplus vehicle interface modules. This package also comes with a 1-year CDR software subscription. NOTE: this package does NOT contain any hardware to perform a "direct-to-module (D2M)" retrieval of EDR data. Please see the accessory cable/adapter packages below for D2M options.

- CDR 900 vehicle interface module
- CDR 900 power cable
- CDR 900/CDR CAN*plus* power supply cable
- CDR 900/CDR CAN*plus* CDR cigarette power adapter
- CDR 900 DLC/OBD cable
- CDR 900 wireless dongles (2)
- CDR 900 extension cable
- CDR 900 legacy cable adapter
- CDR 900 USB cable
- CDR CAN*plus* vehicle interface module
- CDR CAN*plus* DLC/OBD cable
- CDR CAN*plus* GM DLC/OBD cable
- CDR CAN*plus* DLC/OBD adapters (3)
- CDR CAN*plus* PCM adapter
- CDR CAN*plus* USB to Serial port cable
- Deluxe repowering rig
- 1-year CDR software subscription
- Custom carrying case
- Coupon for **FREE online class**: How to Use the Bosch CDR Tool

HARDWARE CHECKLIST 21.0

Part Number	Description	Released
1699200553	Cable, ID826 CDR Audi ACM (requires CDR 500 Adapter)	17.6
1699200646	Cable, ID830 CDR Volvo ACM (requires CDR 500 Adapter)	17.7
1699200630	Kit, CDR 900 Upgrade, with Wireless	17.8
1699200696	Cable, ID831 CDR Mazda ACM	17.8
1699200697	Cable, ID832 CDR Daimler ACM	17.8
1699200778	Cable, ID835, CDR Subaru ACM 1	19.0
1699200779	Cable, ID836, CDR Subaru ACM 2	19.0
1699200783	Cable, ID837, CDR Mitsubishi ACM 1	19.0
1699200784	Cable, ID838, CDR Mitsubishi ACM	19.0
1699200785	Cable, ID839, CDR Mitsubishi ACM 3	19.0
1699200786	Cable, ID840, CDR Mitsubishi ACM 4	19.0
1699200776	Cable, ID833, CDR GM ASCM 1	19.1
1699200777	Cable, ID834, CDR GM ASCM 2	19.1
1699200797	Cable, ID842, CDR VW Group ACM 4	19.1
1699200800	Cable, ID843, CDR Mazda ACM	19.1
1699200838	Cable, ID845, CDR GM ACM	19.1
1699200837	Cable, ID844, CDR Subaru ACM 3	19.4
1699200839	Cable, ID846, CDR Honda ACM	19.4
1699200787	Cable, ID841, CDR Daimler ACM (requires CDR 500 Adapter)	19.6

New 3-digit Part ID

2018

2019+ Forester/Acsent

2015+ Legacy/Outback

Eclipse Cross

Outlander

2019

software to use them not out yet

2020 Golf, Neo

2019+ Mazda 3

SDM-50 2020 Corvette

2020 Legacy/Outback

2020

2021+ Honda Gen 4

New Back Powering Gear Needed

1. Manufacturers (GM+) are replacing mini and micro fuses with Micro 2 and Micro 3 fuses to pack more fuses into smaller spaces. The complete back power kit must now have several different yellow clips to fit all the different vehicles on the road. GM started it, Ford followed, more to come. Mini fits into sub mini, don't need sub mini.



2. Buy General Technologies Corp GTC063 set of 6 (including 2 bigger ones you don't really need) at the site below for $19.95

https://www.amazon.com/General-Technologies-Corp-GTC063-Connector/dp/B0787JKMTH?ref_=bl_dp_s_web_9003703011

EMERGING TRENDS

1. Worry re: cyber security, (hackers trying to take over vehicle control systems).
2. Many are installing “gate-way” modules or features to protect ECU’s from hacking.
3. Back power ACM fuse AND cyber security (gateway) module fuse. DTM still an alt.
4. Recently needed on 2017 RAM 3500, 2018 Toyota, all VW
5. Ford “extended power” fuse for ACM – no fuse labeled “airbag” or “SRS”

Recent CDR Software Updates

- 18.0 Feb 20, 2019 **Subaru** 19 Forstr/Acsen
- 19.0 June 6, 2019 **Mitsubishi** Outlanders
- 19.1 Sept. 20, 2019, 15+ Subaru Outback, GM ASCM reports, more 2020's
- 19.2 Dec. 6 Legacy GM ASCM's, + 2020's
- 19.3 –Dec.23 – CDR900 Toyota support
- 19.4 May 7, 2020 +Subaru, +Porsche, 21GM
- 19.5 Aug 13,2020 + 21's,
- 19.6 Dec 1, 2020 China releases + more
- 21.0 Jan 21, 2021 China releases + more

CDR Budgeting

	Own everything	Middling Approach	A Minimal Approach
Initial Expenses			
Initial Technician Training	$395	$395	$395
Hardware (Cabels Etc...)	$27,300 (Premium Pro Kit)	$17,300 (Deluxe Pro Kit)	$7300 (Pro Kit)
Software	Included	Included	Included
Popular Cables	Included	50 Popular Cables/Adapters Included	~$6000 (Choose 20 Popular Cables/Adapters - Estimate)
Analyst Training (Travel Extra)	$995	$995	$995
Sub Total	**$28,690**	**$18,690**	**$14,690**
Estimated Annual Recurring Expenses			
Software Updates	$1,250	$1,250	$1,250
New Cables	$1100 (est 5 @ $220 each)		
Unexpected Cables		$440 (2 Cables you need but do not have - Shipped overnight)	$440 (2 Cables you need but do not have - Shipped overnight)
Analyst Training (Travel Extra, or Online = $0) $1000 Every Other Year	$500	$500	$500
Sub Total	**$2,850**	**$2,195**	**$2,195**
* Prices Subject to Change			

New Tool for Insurance?

1. Collision Sciences based in Toronto has developed a DLC only tool and Bluetooth software that runs from your phone to read EDR data from vehicles with functioning electrical systems. Intended customer is insurance.
2. Entry price for the hardware is only $150
3. Raw data is uploaded to cloud, free teaser preview
4. Full reports cost $300 per readout and include value added analysis and risk of fraud and injury assessment.
5. Developers are talented hackers - they are NOT getting decoding data from manufacturers – reverse engineering
6. Works on Kia/Hyundai and most CDR supported vehicles
7. Unclear if Bosch or GIT could claim software license violation and go after them – coverage mirrors CDR/GIT

CDR/EDR Training

1. Bosch Training Provider Page lists IPTM (Rick Ruth+), Crash Data Specialists (Brad Muir+), CSI (Rusty Haight), Crash Data Group and Northwestern (was CSI, now + Muir/Russel). SAE (Rick R) not listed, teaches analysis only.
2. There is no "certification" – your qualifications depend on your training and experience
3. IPTM ONLINE technician training $395 and ONLINE analyst training $995.
4. SAE Analyst EDR now Virtual! (Live again 6/2021)
5. Online tech basic from Crash Data Group $200
6. All trainers cover basics, but each has unique style and additional content -trv them all!

Additional EDR training available

1. IPTM has a "Level 2" analyst class which goes into greater depth on how to use the law of sines and cosines to solve Velocity Vector triangles, offset collisions w/EMR, ground forces & new data elements.
2. EDR for Insurance Analyst now available from Ruth Consulting virtually
3. Other classes being adapted to virtual in response to live class cancellations – contact Rick Ruth for latest

Analysis Update: Speed Data

1. Accuracy: We taught +/-4% to speed data based on 1999 GM paper.
2. Part 563 speed data **+/-1 kph** (0.6 mph) as sold vehicles, does not include tire wear adding 2% to speed.
3. SAE 2016-01-1497 "**Compendium** of Pass Vehicle EDR Literature and Analysis of Validation Studies" by Bortles et al, Kineticorp concludes speed within 2mph in tests run to date (mostly 30 to 70 mph).
4. Resolution controls low speed accuracy

Timing of Last Speed Data Point
Red= Bad Green=Good

Time from last data point to impact		
(Time available for possible braking between last data point and impact)		
MANUFACTURER	**Model year**	
GM Gen 1	92-02	No precrash data
GM Gen 2	99-12	**0 to 1 interval between data points, 1 second**
GM Gen 3	2005-2012	0 to 1 interval between data points, some 0.5 **some 1.0 second**
GM Gen 4	2012-presen	0 to 1 interval between data points, 0.5 second
Ford ACM Gen 1	97-07	No precrash data
Ford PCM	2003-2011	0 to 1 interval between data points, 0.2 seconds
Ford ACM Gen 2	2005-2012	0 to 1 interval between data points, **1 second**
Ford Gen 3 RC6	2009-2010	**more than 1 interval between data points, use 1 second**
Ford Gen 3 RC6_2011 & Bosch AB10	2011+	0 to 1 interval between data points, 0.5 seconds
Ford Gen 3 RC7	2013+	0 to 1 interval between data points, 0.5 seconds
Ford Gen 3 AB12	2017+	? Data point interval only 0.2 sec though, use that
Fiat.Chryslers	2006+	0 to 1 interval between data points, 0.1 sec (0.2 some minivan 2010-2012)
Toyota Gen 1 and 2	2001-2012	Last data point taken at impact **but 0 to 0.5 sec old**
Toyotoa Gen 3 & 4	2013+	**Last data point taken at imapct, within 24ms (use zero)**
Honda	2011+	Last data point taken at impact, within 0.1 sec of impact,but 0.3 sec delay for analog speedometer **0.9 sec for Digital speedo**
Nissan	2013+	Last data point taken at impact, within 0.1 sec?
Mazda	2011+	Last data point taken at impact, within 0.1 sec ?
Volvo	2011+	0 to 1 interval between data points, 0.5 second
BMW	2013+	0 to 1 interval between data points, 0.5 second
Mercedes	2014+	0 to 1 interval between data points, 0.5 second
Volkswagen	2015+	File gives time since last data point 0.0 to 0.4 sec - use that
Kia/Hyundai	2010+	Last data point taken at imapct, within 0.1 sec?
Subaru	2012+	Last data point taken at imapct, within 0.1 sec?
Mitsubishi (Chrysler clone)	2009+	? Assume 0 to 1 interval between data points, 0.1 sec
Mitsubishi (Part 563)	2013+	? Assume 0 to 1 interval between data points, 0.5 sec

Analysis Update: Delta V Usage

VEHICLE WITH EDR

- Inline w/departure

  V1 = V3 - ΔVx

- Angled with Departure

  V1=V3cosβ - ΔVx

Note no assumptions about whether collision was central or vehicles reached a common velocity

VEHICLE WITHOUT EDR

- Inversely Proportional ΔV at centers of mass $\Delta V_1 = -\Delta V_2 \frac{W_2}{W_1}$
- Closing Speed Central Inline
- $\text{Closing Speed} = \left[\frac{1}{1+e}\right]\left[|\Delta V_1| + |\Delta V_2|\right]$ w/common vel.
- Closing Speed Offset Inline w/com vel.

  $\text{ClosingSpeed} = \left[\frac{1}{1+e}\right]\left[\frac{|\Delta V_1|}{\gamma_1} + \frac{|\Delta V_2|}{\gamma_2}\right]$
- Intersection inline approximation
- V1/V3/ΔV Triangle Vel Vectors
- ~~V1/V2/Closing Speed Triangle~~

TRIANGULAR VELOCITY VECTOR METHOD

APPOACH VELOCITY = V1

DEPARTURE VELOCITY = V3

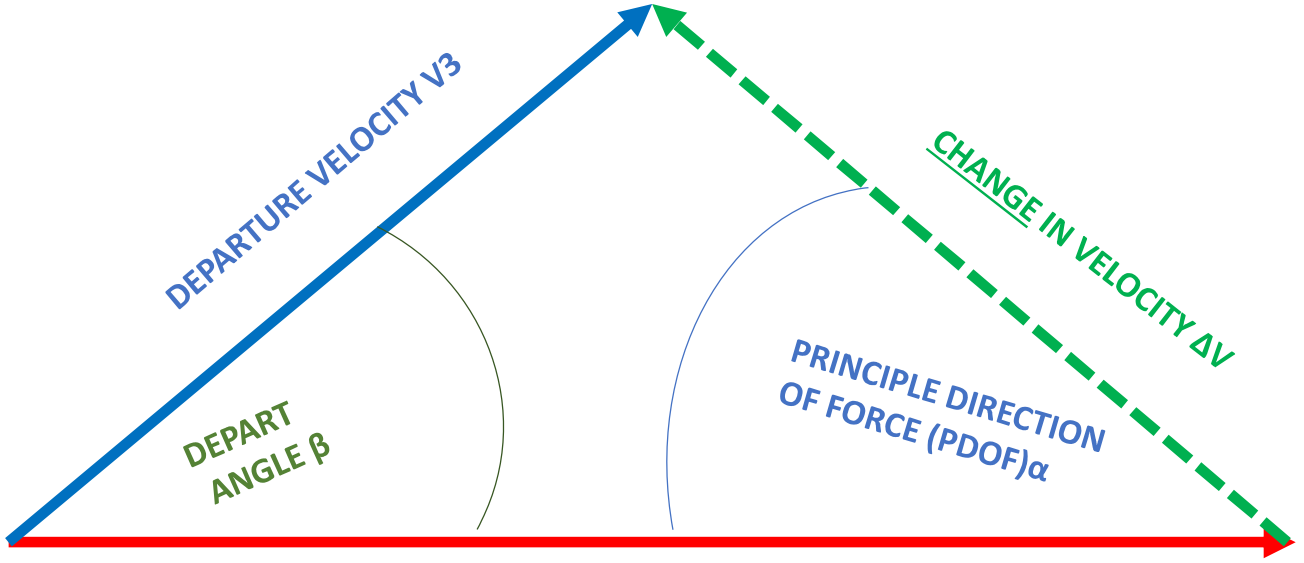
CHANGE IN VELOCITY = ΔV

DEPARTURE ANGLE = β

PDOF = α

SIDES AND OPPOSITE ANGLES (PAIRS) ARE SHOWN IN THE SAME COLOR

IF WE KNOW ANY 3 SIDES OR ANGLES WE CAN FIGURE OUT THE REST

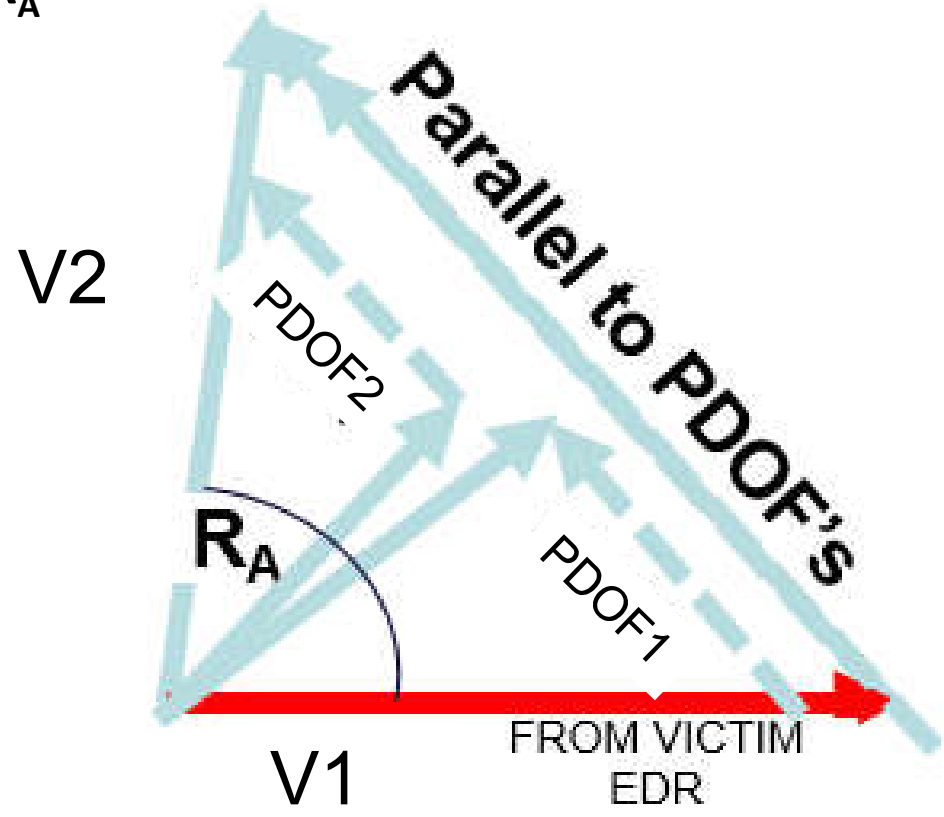


PDOF Triangle -2nd Triangle

1. **Must know relative approach R_A**
2. **Get PDOF1 from V1 EDR**
3. **PDOF2=180- R_A -PDOF1**
4. **ΔV2 = -ΔV1(W1/W2)**
5. **Must have departure speed or angle (3rd piece) of V2 to solve**
6. Use law of sines to solve V2

Note small inner triangles have a parallel edge (parallel PDOF's)

Outer triangle is the PDOF triangle.



~~Closing Speed Triangle~~

ANOTHER WAY TO FIND V1 AT IMPACT

- The definition closing speed is the difference between the V1 and V2 approach velocity vectors
- From **R_A, V1, and CS** find V2
- Use law of sines

Note angles in other corners

Are NOT the PDOF's

The "inline" closing speed formula

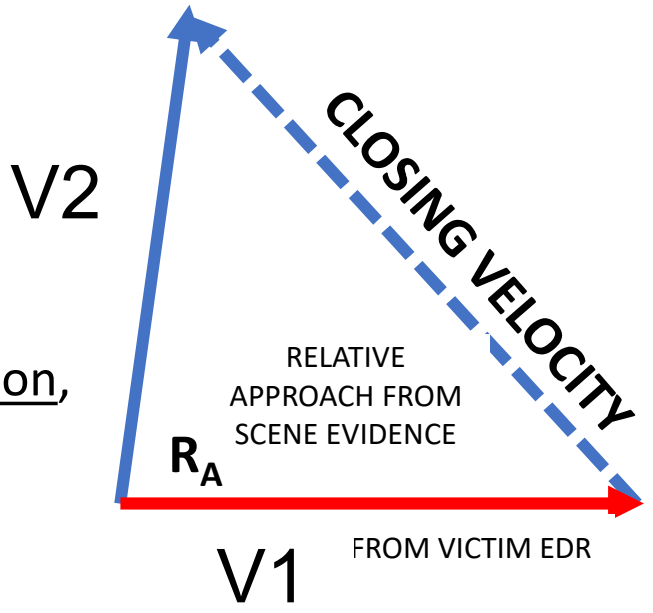
Shown earlier is CS in PDOF direction,

True CS is higher

We may not have enough info to

Determine closing velocity in the

Correct direction



Analysis Updates

1. Closing speed adjustment for Offset collisions assumes ACM is on center of gravity.
2. Additional adjustment may be required for ACM location away from CG.
3. Technique publication from Bob Scurlock and Andy Rich in Collision Magazine expected Spring 2021
4. *It's complicated*, must determine ACM location vs. axis of vehicle rotation!
5. T-bone in rear w center stack module worst

Analysis Update: Delta V Data

1. **Ground Forces were typically ignored in reconstruction**
2. **In small ΔV's (ped/mc) ground forces become significant**
3. **If car locks up brakes before hitting ped, Long. ΔV of ped hit is overstated**
4. **If MC t-bones car, Lat ΔV understated**
5. **ΔV is multiplied by big weight ratio, Closing speed error gets bigger**

NEW NHTSA CISS DATABASE

1. https://crashviewer.nhtsa.dot.gov/CISS/SearchIndex
2. 7,657 cases available from 2016-2019 calendar years – more uploaded
3. Searchable by manufacturer, model year, model, EDR data available, driver assist technology available, etc
4. Find similar cases, check polarity, see typical EDR report
5. Link to VIN decoder, Canadian Specs, Vin check digit calc'tr Welcome to VIN Decoding :: provided by vPIC (dot.gov)

Details by Manufacturer

Honda 21.0 CDR Coverage

Model	Location	##	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
ACURA		Last 3 digits of Bosch CDR cable numbe					Gen 2		Gen 3			Gen 4	
RL	Under dashbo		547										
TL	Under dashbo		547										
TSX	Under dashbo		547										
ZDX	Under dashbo		547										
MDX	Under dashbo		547		387/789			387/810					846
NSX	Under dashbo							387/789					
RDX	Under dashbo			547			387/789		387/825				
ILX	Under dashbo			547			387/789						
RLX	Under dashbo				387/789								
TLX	Under dashbo					387/789				387/810		846	
HONDA			2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
ACCORD	Under dashbo			387/789			387/810		387/825				
CIVIC 2 DR	Under dashbo		547				387/810						
CIVIC 4 DR	Under dashbo		547	387/789			387/810						
CLARITY												387/810	
CROSSTOUR	Under dashbo		547										
CR-V	Under dashbo		547				387/789	387/810					
CR-Z	Under dashbo		547										
FCX Clarity	Under dashbo							387/810					
FIT	Under dashbo	EV	547			387/789		387/810					
HR-V							387/789		387/810				
INSIGHT	Under dashbo		547							387/810			
ODYSSEY	Under dashbo		547		387/789			387/810					
PASSPORT	Under dashbo									387/810			
PILOT	Under dashbo		547				387/789	387/810					
RIDGELINE	Under dashbo		547					387/810					

2011- NO time series EDR But may have Delta V single value

Honda only assists law enforce -ment voluntar ily

Honda EDR Evolution

1. One set of DL's thru 2015
   a) Roll rate polarity varies by supplier – polarity added under graph
   b) Accel data instantaneous
2. 2016-17 Gen 2 adds "Honda sensing" & Cruise. Updated Data Lims.
3. 2018 – 3rd Gen to include "ig switch status"
   a) Airbag deploys even if ign off until car stops
   b) Markets outside US/Japan do not get precrash data
4. 2021 – 4th Gen - ???

Honda Gen 2 EDR

- If equipped may have semi autonomous vehicle data elements

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 2 of 3

Time Stamp (sec)	PCM Derived Accelerator Pedal Position, % full	Forward Collision Warning (Not Warning/ Warning)	Collision Mitigation Braking System (Not Engaged/ Engaged)	Collision Mitigation Braking System, Forward Collision Warning (On/Off)	Lane Departure Warning (Not Warning/ Warning)	Road Departure Mitigation (Not Engaged/ Engaged)	Road Departure Mitigation, Lane Departure Warning (On/Off)
-5.0	16	Not warning	Not engaged	On	Not warning	Not engaged	Off
-4.5	8	Not warning	Not engaged	On	Not warning	Not engaged	Off
-4.0	11	Not warning	Not engaged	On	Not warning	Not engaged	Off
-3.5	13	Not warning	Not engaged	On	Not warning	Not engaged	Off
-3.0	18	Not warning	Not engaged	On	Not warning	Not engaged	Off
-2.5	18	Not warning	Not engaged	On	Not warning	Not engaged	Off
-2.0	17	Not warning	Not engaged	On	Not warning	Not engaged	Off
-1.5	0	Warning	Engaged	On	Not warning	Not engaged	Off
-1.0	0	Warning	Engaged	On	Not warning	Not engaged	Off
-0.5	0	Warning	Engaged	On	Not warning	Not engaged	Off
0.0	0	Warning	Engaged	On	Not warning	Not engaged	Off

Honda Gen 2 Semi Auton cont'd

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 3 of 3

Time Stamp (sec)	Adaptive Cruise Control (Not Engaged/ Engaged)	Adaptive Cruise Control (On/Off)	Lane Keeping Assist (Not Engaged/ Engaged)	Lane Keeping Assist (On/Off)	Cruise Control (Not Engaged/ Engaged)	Cruise Control (On/Off)
-5.0	Not engaged	On	Not engaged	On	Not Engaged	On
-4.5	Not engaged	On	Not engaged	On	Not Engaged	On
-4.0	Not engaged	On	Not engaged	On	Not Engaged	On
-3.5	Not engaged	On	Not engaged	On	Not Engaged	On
-3.0	Not engaged	On	Not engaged	On	Not Engaged	On
-2.5	Not engaged	On	Not engaged	On	Not Engaged	On
-2.0	Not engaged	On	Not engaged	On	Not Engaged	On
-1.5	Not engaged	On	Not engaged	On	Not Engaged	On
-1.0	Not engaged	On	Not engaged	On	Not Engaged	On
-0.5	Not engaged	On	Not engaged	On	Not Engaged	On
0.0	Not engaged	On	Not engaged	On	Not Engaged	On

Reminders: UNIQUE TO HONDA

1. Most manufactures record the speed calculated in the powertrain control module as it is being sent **TO** the speedometer on the CAN bus.
2. Honda records the speed being displayed **ON** the speedometer, after the PCM calculated speed has been dampened or “smoothed”.
3. The last data point labeled 0.0 is sampled at AE. -0.5 is -0.01 to -0.50. While sampled at AE, speedometer processing delays what is being reported, so it lags the true speed at impact during hard braking.

New Defense Allegation

1. NHTSA has opened an “EA” investigation on frontal airbags not deploying in crashes due to electrical interference overloading an ASIC (chip) handling front sensor inputs in TRW supplied modules.
2. Kia, Hyundai, and Chrysler have recalled ACM’s for SOME affected units. Toyota has 2 reported incidents. Mitsubishi and Honda also use TRW.
3. Defense claims all TRW EDR’s 2010 to 2019 are bad. It’s bullshit.

Chevrolet SDM Phase In 87%

3 events

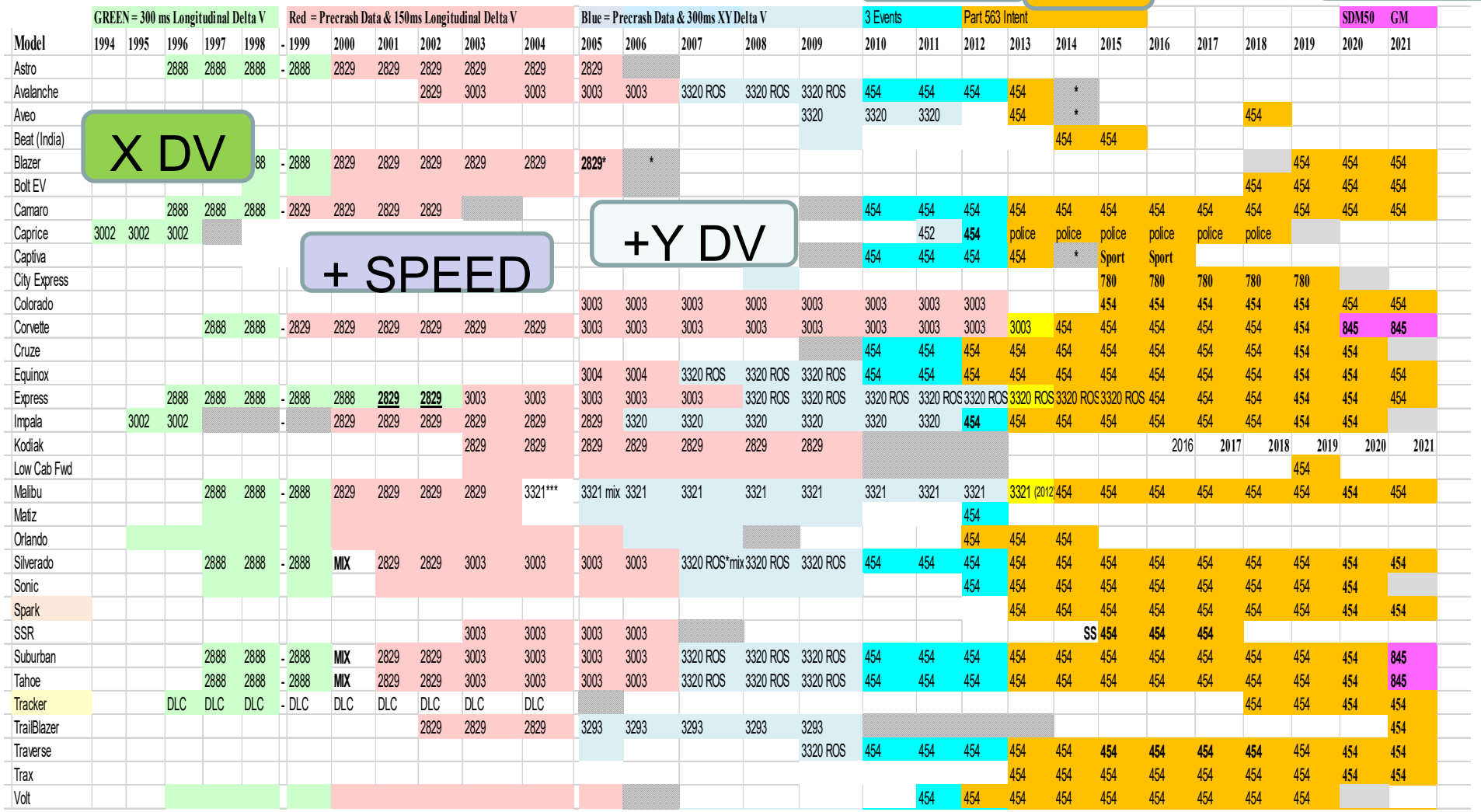
563

SDM50

X DV

+Y DV

+ SPEED



Model	GREEN = 300 ms Longitudinal Delta V					Red = Precrash Data & 150ms Longitudinal Delta V						Blue = Precrash Data & 300ms XY Delta V					3 Events		Part 563 Intent								SDM50	GM
	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Astro			2888	2888	2888	2888	2829	2829	2829	2829	2829	2829																
Avalanche									2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	*							
Aveo																3320	3320	3320		454	*				454			
Beat (India)																					454	454						
Blazer					88	2888	2829	2829	2829	2829	2829	2829*	*													454	454	454
Bolt EV																									454	454	454	454
Camaro			2888	2888	2888	2829	2829	2829	2829								454	454	454	454	454	454	454	454	454	454	454	454
Caprice	3002	3002	3002															452	454	police	police	police	police	police	police			
Captiva																	454	454	454	454	*	Sport	Sport					
City Express																						780	780	780	780	780		
Colorado												3003	3003	3003	3003	3003	3003	3003	3003			454	454	454	454	454	454	454
Corvette				2888	2888	2829	2829	2829	2829	2829	2829	3003	3003	3003	3003	3003	3003	3003	3003	3003	454	454	454	454	454	454	845	845
Cruze																	454	454	454	454	454	454	454	454	454	454	454	
Equinox												3004	3004	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454
Express			2888	2888	2888	2888	2888	2829	2829	3003	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454
Impala		3002	3002				2829	2829	2829	2829	2829	2829	3320	3320	3320	3320	3320	3320	454	454	454	454	454	454	454	454	454	
Kodiak										2829	2829	2829	2829	2829	2829	2829							2016	2017	2018	2019	2020	2021
Low Cab Fwd																										454		
Malibu				2888	2888	2888	2829	2829	2829	2829	3321***	3321 mix	3321	3321	3321	3321	3321	3321	3321	3321 (2012	454	454	454	454	454	454	454	454
Matiz																			454									
Orlando																			454	454	454							
Silverado				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS*mix	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454
Sonic																			454	454	454	454	454	454	454	454	454	
Spark																				454	454	454	454	454	454	454	454	454
SSR										3003	3003	3003	3003								SS	454	454	454				
Suburban				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	845
Tahoe				2888	2888	2888	MIX	2829	2829	3003	3003	3003	3003	3320 ROS	3320 ROS	3320 ROS	454	454	454	454	454	454	454	454	454	454	454	845
Tracker			DLC	DLC	DLC	DLC	DLC	DLC	DLC	DLC	DLC														454	454	454	454
TrailBlazer									2829	2829	2829	3293	3293	3293	3293	3293												454
Traverse																3320 ROS	454	454	454	454	454	454	454	454	454	454	454	454
Trax																				454	454	454	454	454	454	454	454	454
Volt																		454	454	454	454	454	454	454	454	454		

GM full chart printed in color in your book
and is on student reference disk

Chevrolet Recent Changes 87% 563

1 animation

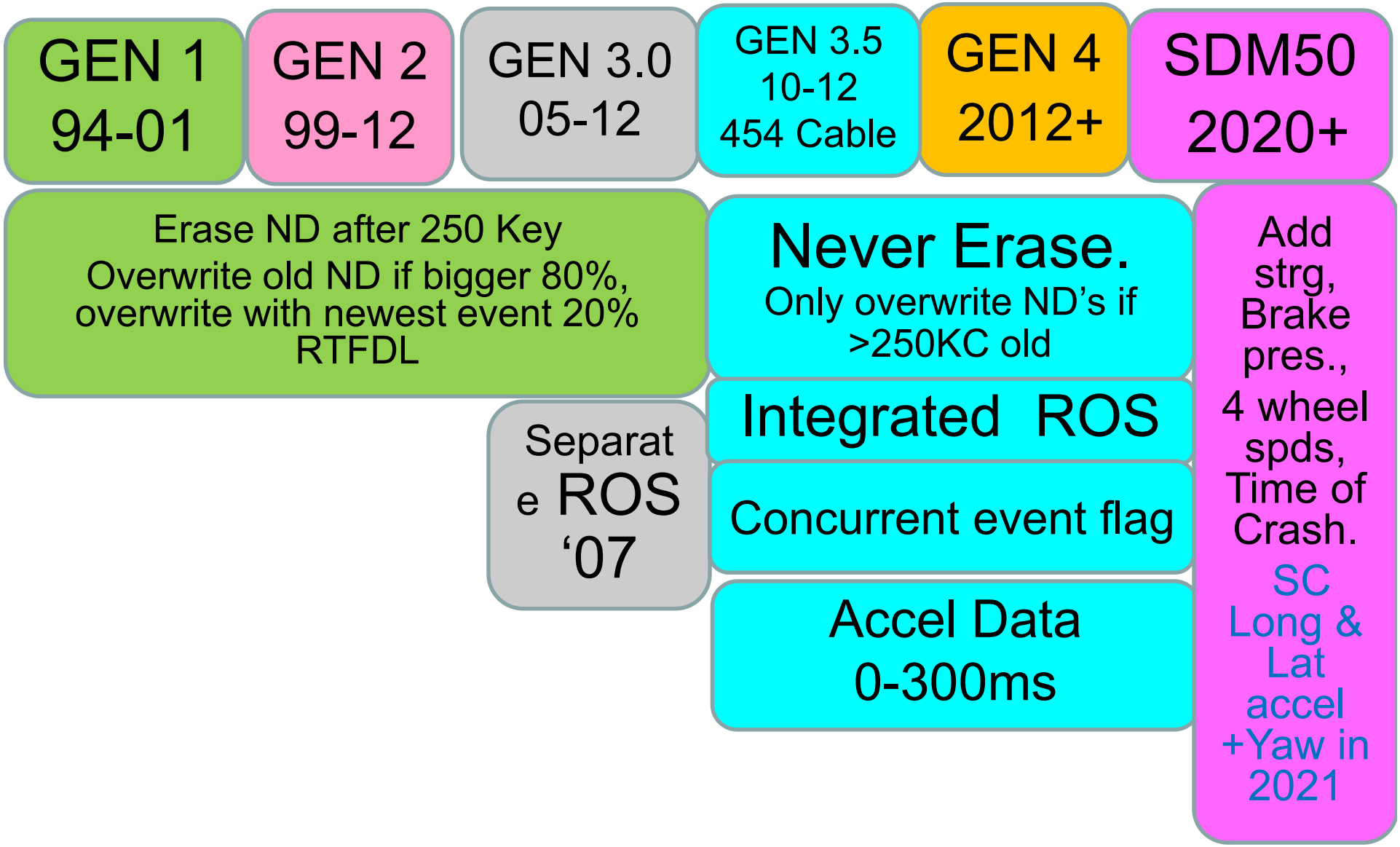
	Part 563 Intent								SDM50	GM
Model	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**
Avalanche	454	454	*							
Aveo		454	*				454			
Beat (India)			454	454						
Blazer								454	454	454
Bolt EV							454	454	454	454
Camaro	454	454	454	454	454	454	454	454	454	454
Caprice	**454**	police	police	police	police	police	police			
Captiva	454	454	*	**Sport**	**Sport**					
City Express				**780**	**780**	**780**	**780**	**780**		
Colorado	3003			**454**	**454**	**454**	**454**	**454**	454	454
Corvette	3003	3003	454	454	454	454	454	**454**	**845**	**845**
Cruze	454	454	454	454	454	454	454	**454**	**454**	
Equinox	454	454	454	454	454	454	454	**454**	**454**	454
Express	3320 ROS	3320 ROS	3320 ROS	3320 ROS	454	454	454	**454**	**454**	454
Impala	**454**	454	454	454	454	454	454	**454**	**454**	
Kodiak					2016	**2017**	**2018**	**2019**	**2020**	**2021**
Low Cab Fwd								454		
Malibu	3321	3321 (2012	454	454	454	454	454	454	454	454
Matiz	454									
Orlando	454	454	454							
Silverado	454	454	454	454	454	454	454	454	454	**454**
Sonic	454	454	454	454	454	454	454	454	**454**	
Spark		454	454	454	454	454	454	454	**454**	**454**
SSR			**SS**	**454**	**454**	**454**				
Suburban	454	454	454	454	454	454	454	454	454	**845**
Tahoe	454	454	454	454	454	454	454	454	454	**845**
Tracker							454	454	454	454
TrailBlazer										454
Traverse	454	454	454	**454**	**454**	**454**	**454**	454	454	454
Trax		454	454	454	454	454	454	454	454	454
Volt	454	454	454	454	454	454	454	454		
Acadia	454	454	454	454	454	454	454	454	454	454

No Precrash in 2013-5 Express/Savanna

GM EDR Evolution

GEN 1 94-01	GEN 2 99-2012	GEN 3.0 05-2012	GEN 3.5 2010-2012 454 Cable	GEN 4 2012+	SDM50 2020+
No Precrash	Precrash 5 @ 1 sec for BOTH events	Precrash 5 @ 0.5 or 1 sec ONE SET of precrash	Precrash 5 @ 0.5 or 1 sec For each event	Precrash 10 @ 0.5 sec	Precrash 10 or 16 @ 0.5 sec + T_O
2 EVENTS - FIRST D, One ND/ DLE, replace SDM after D			3 Events, numbered ND's or D's, replace after 3D's		+event summary 30 records +2 PedPro +3 AACN
Record on frontal airbag algo wakeup		Record on 5mph X or Y DV			
300ms XDV =wakeup, must be neg	150ms XDV, must be neg, 50ms before D 100 after	300ms X+Y DV, + or - some D's -70 to +220, some 300ms after wakeup		300ms after wakeup	120G *accelerometer*

GM EDR Evolution #2



2020 Corvette SDM50 Data

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 1 of 6

Time (sec)	Accelerator Pedal, % Full (%)	ABS Activity	Brake Pedal Position (%)	Driver Applied Brake Pedal Pressure (kPa)	Driver Applied Brake Pedal Pressure Detected	Brake Pedal Moderate Travel Achieved
-8.0	0	Off	0	0	False	False

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 2 of 6

Time (sec)	Cruise and Speed Limiter Switch Status	Cruise Secondary Switch Status	Cruise Control Active	Adaptive Cruise Control Active Authenticated	Transmission Estimated Gear	Transmission Shift Lever Position	Reduced Engine Power Mode Indicator On
-8.0	No Activation	No Activation	Inactive	Inactive	Ninth	Forward Range A	Off

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 3 of 6

Time (sec)	Engine RPM (RPM)	Engine Torque (foot-pounds)	Engine Throttle, % Full (%)	Speed, Vehicle Indicated (MPH [km/h])	SC accel + Yaw 2021	Right Front Wheel Angular Velocity (RPM)	Left Front Wheel Angular Velocity (RPM)
-8.0	1,301	50.5	16	44.1 [71]		595	598

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 4 of 6

Time (sec)	Right Rear Wheel Angular Velocity (RPM)	Left Rear Wheel Angular Velocity (RPM)	Steering Wheel Angle (deg)	Propulsion System Active	Run/Start Input	Ignition Prolongation Time (sec)	Secondary Collision Prolongation Timer (sec)
-8.0	599	598	0	True	High	5.0	0.0

Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 5 of 6

Time (sec)	Antilock Brake System Failed	Brake Pedal Override Flag	Traction Control System Present	Traction Control System Failed	Traction Control System Enabled	Traction Control System Active	Traction Control System Driver Intent
-8.0	False	False	True	False	True	Traction Control Off	Enabled

More 2020 Corvette SDM50

- 3 AACN records in Eng Translation Rpt
- GPS Lat/Long at crash
- Turn Signal Switch Status
- Second and third row seat belt status
- Up to 30 Event Summary records give clear order of events
- Outside Air Temp, Brake Overrides
- Traction Control activate
- Auxilliary Crash Sensor Data

EVERYTHING GM COULD IMAGINE

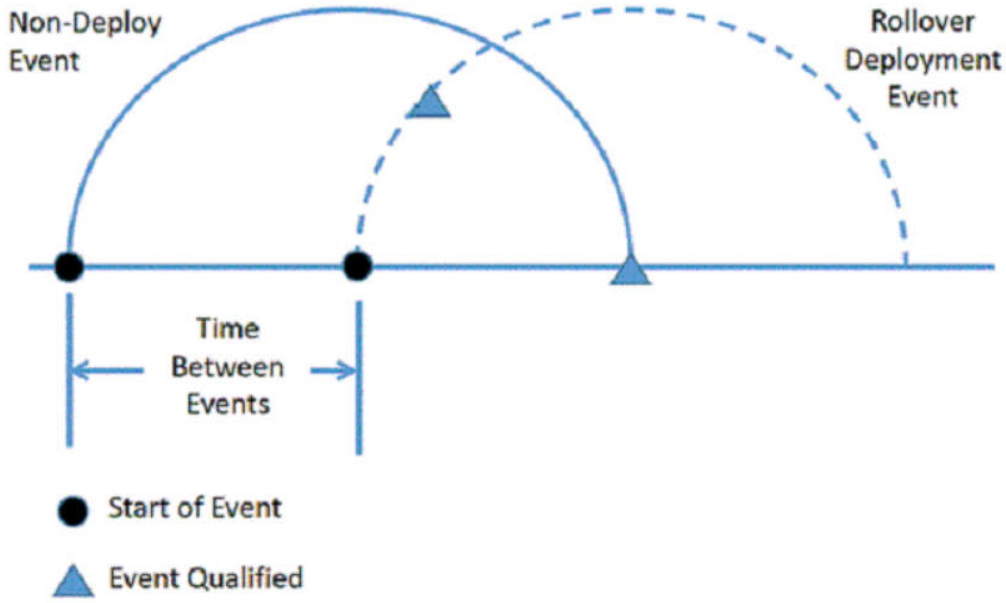
GM Gen 3.5 & 4 Features

1. 2010 454 cable SDM's do not require replacement after a D (until 3 D's).
2. Events & deployments all numbered vs. labeled D/ND.
3. Events may not be in order algorithms started, they are in order they qualified for recording (see graphic next page)
4. **Align precrash data for all events to see relationship**
5. Black plastic SDM housings
6. Accelerometer max values listed in DL's

GM Data Limitations

- Many updated in 17.9 – new timing graphic added for gen 3.5 and higher

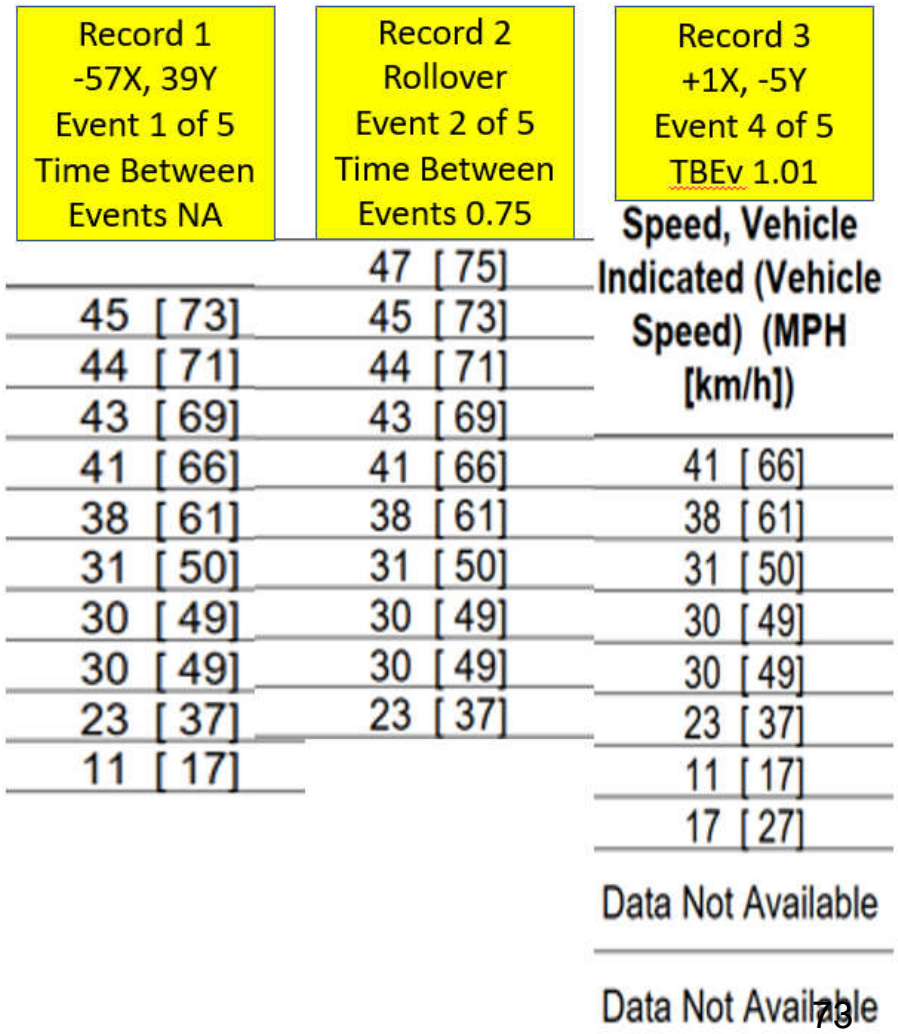
Example of a Concurrent Event:
A Non-Deployment event begins. Before the Non-Deployment event is qualified, a Rollover Deployment event begins and is qualified. Sometime after the Rollover event is qualified, the Non-Deployment event is qualified. The Rollover event will be recorded in the first open record even though the Non-Deployment event enabled before the Rollover event. The Non-Deployment event will be recorded in the next open record. The "Concurrent Event Flag Set" parameter will indicate "Yes" for the Non-Deployment event. The "Time Between Events" parameter will indicate the time from the start of the Non-Deployment event to the start of the Rollover event.



Event Record #1	Event Record #2
Event record Type = Rollover	Non-deployment
Concurrent Event Flag = No	Concurrent Event Flag = Yes
Time Between Events = N/A	Time Between Events = XX seconds

When Concurrent Event Flag is Yes, Overlay Precrash Data from all events

- The event number gives when the event QUALIFIED for recording, not when I started
- Time between events is time between consecutive wake-ups, NOT always time between qualifying events
- Recap Delta V or, if rollover, event type over precrash data to assist understanding



Important Bug Fix 4 & Other GM News

1. One module family Conti SDM-30 comes out misinterpreted – data is scrambled
2. Must send to Conti for manual readout and interp. (ask instructors for how)
3. Foreign coverage – China since 99 (but only Gov't/NPS can use CDR), Australia/Korea/Thailand since 07
4. South America not covered or spoofable

GM Auto Emergency Braking

1. AEB will prevent many crashes and reduce the severity of crashes still happening
2. If there is still a crash, the AEB system may store up to 3 camera images:
   a) 4 sec before event
   b) At event
   c) 4 sec after event
3. Not accessible with CDR – must be requested with proper legal authority from GM

GM Active Safety Control Module

1. First installed in 2013 Cadillac ATS
2. Only with GM option UGN or UKL
3. Initially released in CDR 17.9 to get raw data file only for future interpretation.
4. Interpretation capability added in 19.1
5. REQUIRES CDR 900 interface!
6. Some vehicles have TWO ASCM modules.

GM ASCM CDR 19.6 Coverage

Model	Location	Model Year								
3 digits Bosch CDR cable ID shown		2013	2014	2015	2016	2017	2018	2019	2020	2021
BUICK										
ENCLAVE	cargo left front of wheel well						833			
ENVISION	rear seat left trim by wheel wl				833					
LACROSSE	trunk left					833				
REGAL	trunk left									
CADILLAC										
CT4	? Not in help file 19.6 yet								833?	
CT6 w/o UKL	Trunk right					833 without supercruise				
CT6 w/UKL	Primary right, secondary left				834		Supercruise Option Code UKL			
CTS SEDAN	trunk left		833							
CTS	trunk left				833					
ESCALADE	under left qtr window trim			833						
SRX	under left rear seat	833								
XT4	cargo left front of wheel well							833		
XT5	cargo left front of wheel well					833				
CHEVROLET										
BLAZER	trunk left							833		
MALIBU	trunk left				833					
TRAVERSE	trunk right						833			
VOLT	trunk left					833				
GMC										
ACADIA	cargo left front of wheel well					833				

GM Active Safety Control Module

Mkt	Year/Make	Model	Module		Vehicle Interface	OBD/DLC Connect Adapter/Cable	D2M Connect Adapter/Cable	Module Location
4	2018 Cadillac	ATS (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, left side, near rear, on outboard side of brace, behind trim
6	2018 Cadillac	ATS-L (note 6)	ACM	Data	CANplus	CANplus - F00K108287	CANplus - F00K108454	Center tunnel
4,6	2018 Cadillac	CT6 (note 6)	ACM	Data	CANplus or CDR 900	CANplus - F00K108287 CDR 900 - click here	CANplus - F00K108454 CDR 900 - click here (use listed cable above)	Center tunnel
			ASCM (Note2) how to image	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 834	In the luggage compartment, right center, mounted to the right side in the battery well. The secondary ASCM is on the left.
			ASCM (Note1)	Data	CDR 900	CDR 900 - click here	CDR 900 - Cable# 833	In the luggage compartment, right center, mounted to the right side in the battery well

GM Active Safety Module



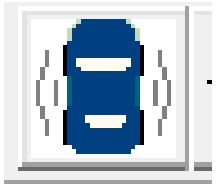
1. Must be downloaded separately from ACM
2. Icon will appear after VIN input for any vehicle that MIGHT have Super Cruise
3. Module /data will only be present if option “UKL” (Super Cruise) or “UGN” (Adv Driver Assist) has been purchased for the subject vehicle. New Icon 19.6.

GM ASCM EDR – **New Icon**

Effective with V 19.6 of the software, the ASCM “RADAR” Icon has been replaced by new Icon



OLD



NEW

- Best guess is that there is more upcoming automated sensing data, from other mfgrs, that have more than frontal radar.

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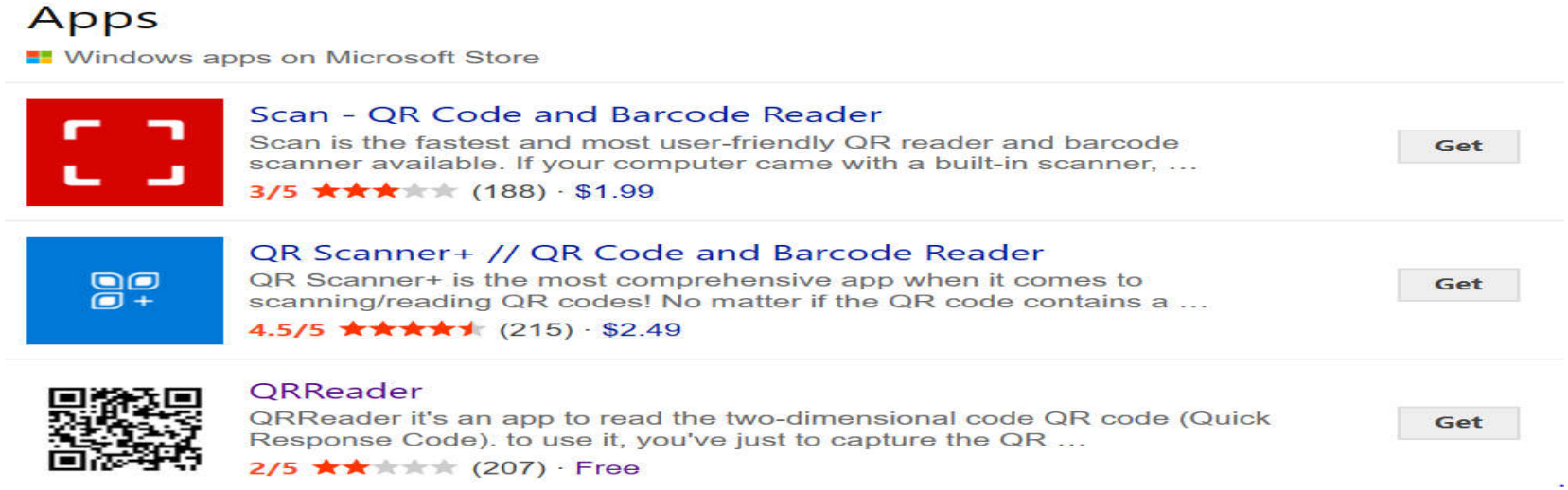
84

GM Active Safety Module

1. Data can be captured every 80 milliseconds and can record up to 50 records for a total of 4 seconds of data. The data recording window can be either 4 seconds before the end of an auto braking event or 4 seconds before an airbag deployment. ONE EVENT ONLY
2. Data is recorded in non-volatile memory at the next proper power-down of the module. Data may **not** be stored if the battery power is lost before the power-down.

GM RPO Sticker Change

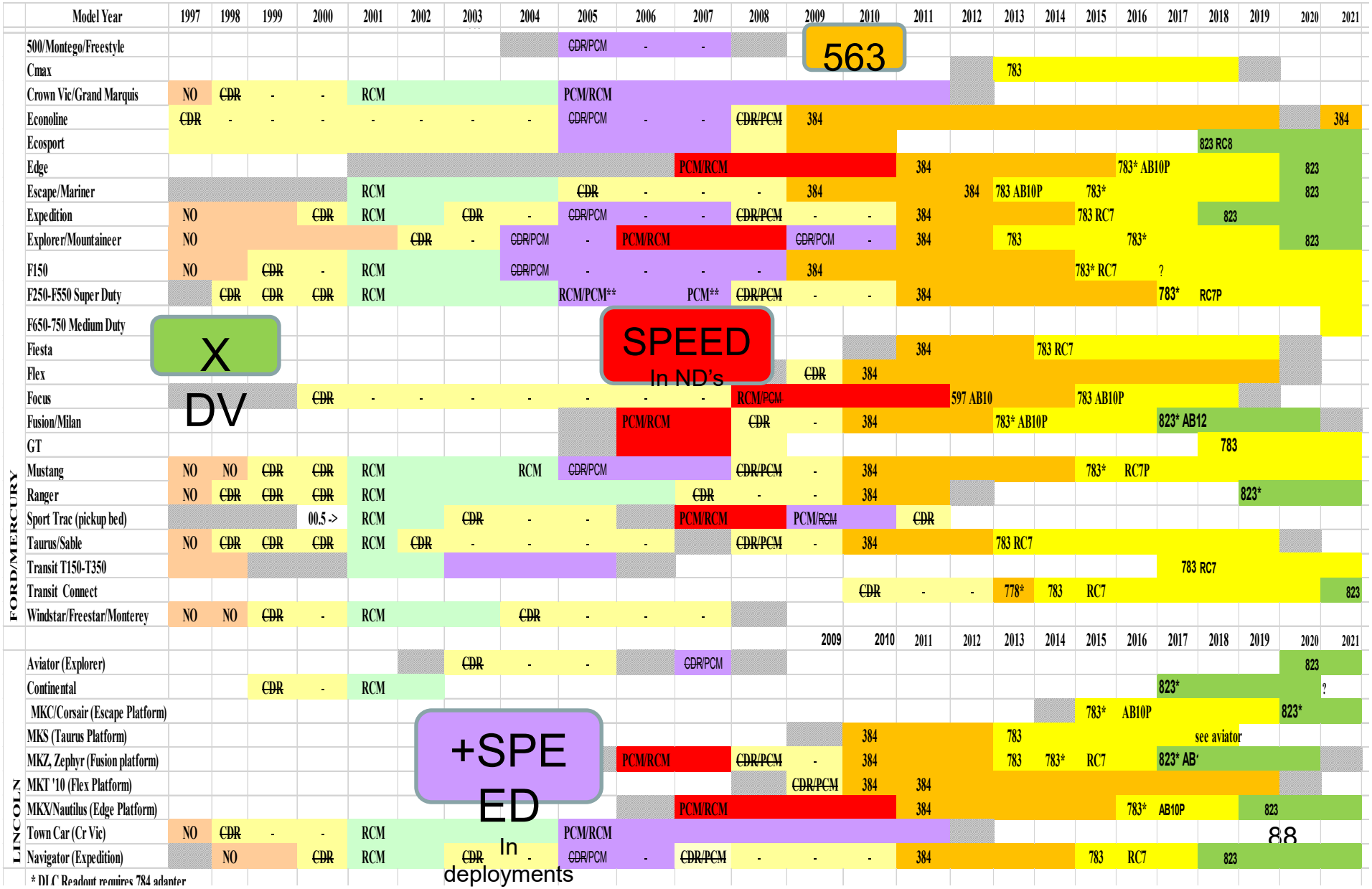
1. GM formerly had a sticker with 3 digit RPO (Regular Production Option) codes printed in the glove box or trunk
2. 2018-ish switched to a “QR code”
3. Google search QR code reader and install one on your phone. Some free, some cost



ETR Access Removed

- CDR 21.0 removes the ability to see and save the Engineering Translation Report as a PDF file.
- Presumably a policy decision
- If ETR is important to you, reinstall an older version of CDR software to access it

Ford CDR 21.0 Coverage 69%



	Model Year	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
FORD/MERCURY	500/Montego/Freestyle									~~CDR~~/PCM	-	-														
	Cmax																	783								
	Crown Vic/Grand Marquis	NO	~~CDR~~	-	-	RCM				PCM/RCM																
	Econoline	~~CDR~~	-	-	-	-	-	-	-	~~CDR~~/PCM	-	-	~~CDR/PCM~~	384												384
	Ecosport																						823 RC8			
	Edge											PCM/RCM			384						783* AB10P				823	
	Escape/Mariner					RCM				~~CDR~~	-	-	-	384			384	783 AB10P		783*					823	
	Expedition	NO			~~CDR~~	RCM		~~CDR~~	-	~~CDR~~/PCM	-	-	~~CDR/PCM~~	-	-	384				783 RC7			823			
	Explorer/Mountaineer	NO					~~CDR~~	-	~~CDR~~/PCM	-	PCM/RCM			~~CDR~~/PCM	-	384		783			783*				823	
	F150	NO		~~CDR~~	-	RCM			~~CDR~~/PCM	-	-	-	-	384						783* RC7		?					
	F250-F550 Super Duty		~~CDR~~	~~CDR~~	~~CDR~~	RCM				RCM/PCM**		PCM**	~~CDR/PCM~~	-	-	384						783*	RC7P			
	F650-750 Medium Duty																									
	Fiesta															384			783 RC7							
	Flex													~~CDR~~	384											
	Focus				~~CDR~~	-	-	-	-	-	-	-	RCM/PCM				597 AB10			783 AB10P						
	Fusion/Milan										PCM/RCM		~~CDR~~	-	384			783* AB10P				823* AB12				
	GT																						783			
	Mustang	NO	NO	~~CDR~~	~~CDR~~	RCM			RCM	~~CDR~~/PCM			~~CDR/PCM~~	-	384					783*	RC7P					
	Ranger	NO	~~CDR~~	~~CDR~~	~~CDR~~	RCM						~~CDR~~	-	-	384									823*		
	Sport Trac (pickup bed)				00,5 ->	RCM		~~CDR~~	-	-		PCM/RCM		PCM/~~RCM~~		~~CDR~~										
	Taurus/Sable	NO	~~CDR~~	~~CDR~~	~~CDR~~	RCM	~~CDR~~	-	-	-	-		~~CDR/PCM~~	-	384			783 RC7								
	Transit T150-T350																					783 RC7				
	Transit Connect														~~CDR~~	-	-	778*	783	RC7						823
	Windstar/Freestar/Monterey	NO	NO	~~CDR~~	-	RCM			~~CDR~~	-	-	-														
														2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
LINCOLN	Aviator (Explorer)							~~CDR~~	-	-		~~CDR~~/PCM													823	
	Continental			~~CDR~~	-	RCM																823*				?
	MKC/Corsair (Escape Platform)																			783*	AB10P				823*	
	MKS (Taurus Platform)														384			783					see aviator			
	MKZ, Zephyr (Fusion platform)										PCM/RCM		~~CDR~~/PCM	-	384			783	783*	RC7		823* AB'				
	MKT '10 (Flex Platform)													~~CDR~~/PCM	384	384										
	MKX/Nautilus (Edge Platform)											PCM/RCM				384					783*	AB10P		823		
	Town Car (Cr Vic)	NO	~~CDR~~	-	-	RCM				PCM/RCM																
	Navigator (Expedition)		NO		~~CDR~~	RCM		~~CDR~~	-	~~CDR~~/PCM	-	~~CDR/PCM~~	-	-	-	384				783	RC7		823			

* DLC Readout requires 784 adapter

Ford w/Stability Control Data 21.0

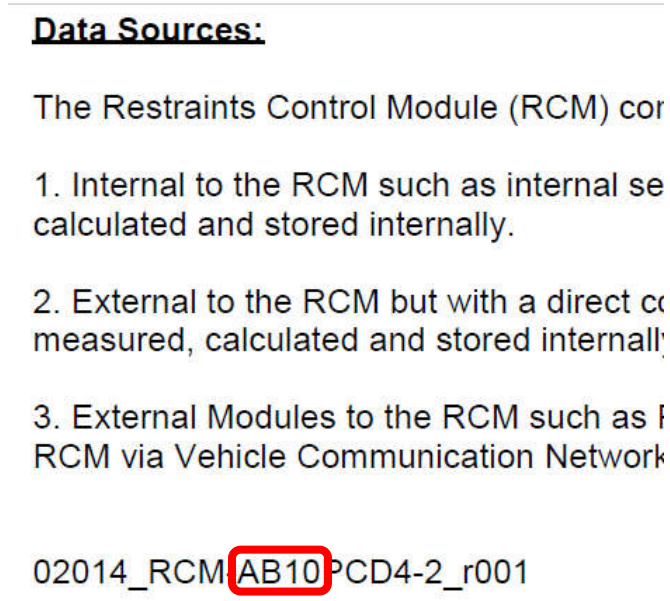
	Model Year	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
FORD/MERCURY	500/Montego/Freestyle													
	Cmax					783								
	Crown Vic/Grand Marquis													
	Econoline	384												384
	Ecosport										823 RC8			
	Edge			384					783* AB10P				823	
	Escape/Mariner	384			384	783 AB10P		783*					823	
	Expedition	-	-	384				783 RC7			823			
	Explorer/Mountaineer	~~CDR~~/PCM	-	384		783			783*				823	
	F150	384						783* RC7		?				
	F250-F550 Super Duty	-	-	384						783*	RC7P			
	F650-750 Medium Duty													
	Fiesta			384			783 RC7							
	Flex	~~CDR~~	384											
	Focus				597 AB10			783 AB10P						
	Fusion/Milan	-	384			783* AB10P				823* AB12				
	GT										783			
	Mustang	-	384					783*	RC7P					
	Ranger	-	384									823*		
	Sport Trac (pickup bed)	PCM/~~RCM~~		~~CDR~~										
	Taurus/Sable	-	384			783 RC7								
	Transit T150-T350							783 RC7						
	Transit Connect		~~CDR~~	-	-	778*	783	RC7						823
	Windstar/Freestar/Monterey													
		2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
LINCOLN	Aviator (Explorer)												823	
	Continental									823*				?
	MKC/Corsair (Escape Platform)							783*	AB10P				823*	
	MKS (Taurus Platform)		384			783					see aviator			
	MKZ, Zephyr (Fusion platform)	-	384			783	783*	RC7		823* AB				
	MKT '10 (Flex Platform)	~~CDR/PCM~~	384	384										
	MKX/Nautilus (Edge Platform)			384					783*	AB10P		823		
	Town Car (Cr Vic)													
	Navigator (Expedition)	-	-	384				783	RC7		823			
	* DLC Readout requires 784 adapter													

Ford Part 563 EDR Subfamilies

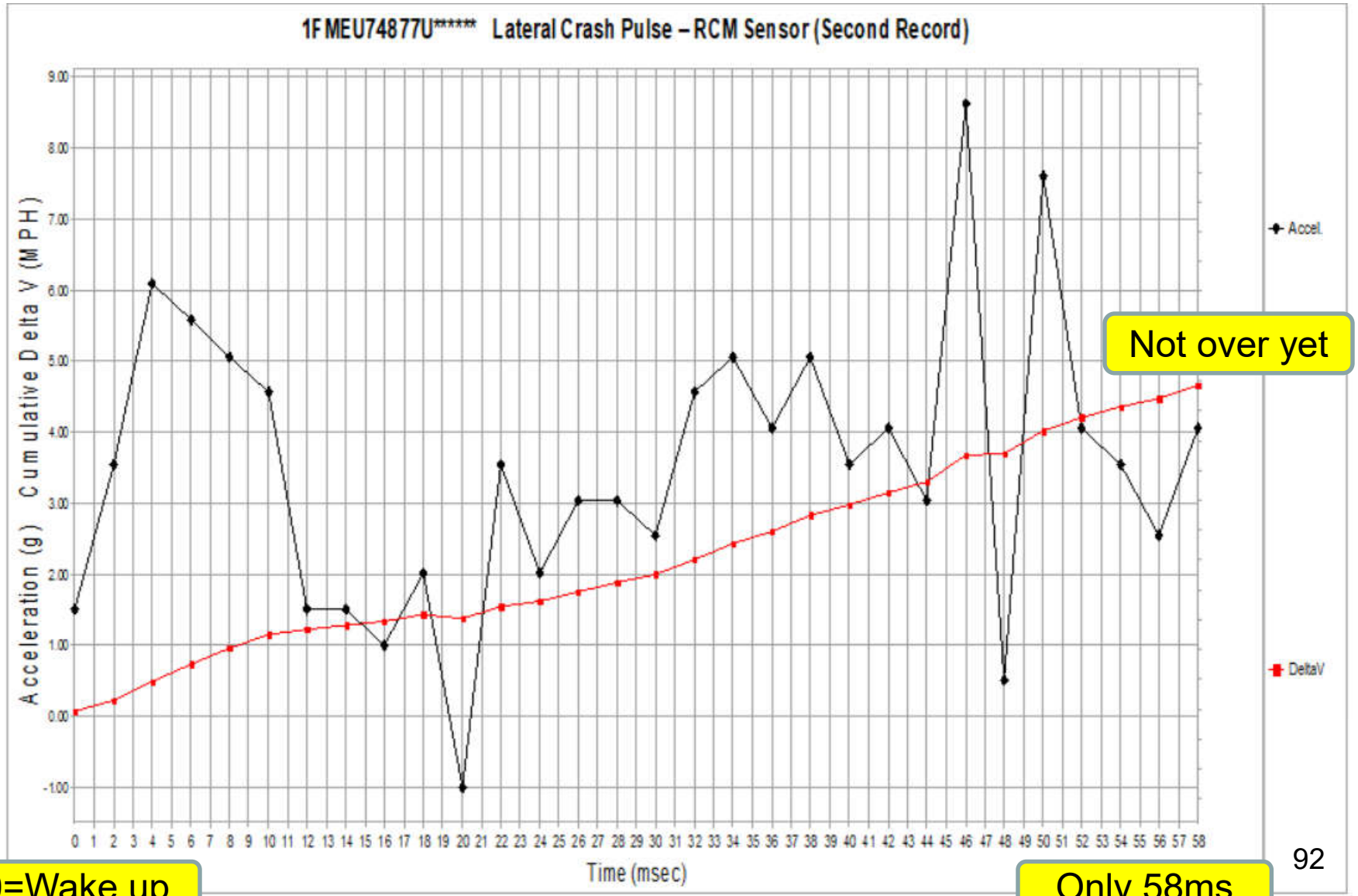
Autoliv RC6 09 Escape 384 Cable +RC6_2011 (RC6X)	Bosch AB10 2012 Focus 597 Cable	Autoliv RC7 2013 Taurus 783 Cable	Bosch AB10P 2015 Focus 783 Cable	2017 RC7P	Bosch AB12 2017 Fusion 823 Cable	Veoneer RC8 2018 EcoSport 823 Cable
Precrash 5 sec @0.5 sec intervals, 250msec X&Y DV					Precrash Speed 5 sec @ *0.2* sec	
Stability Control Sys Data 5 sec @0.1 sec intervals						
RC6 last speed data 0.5 sec plus delay RC6_2011 0.5 sec	Last data point within 0.5 sec	?	?	Within 0.2 sec		
Incomplete Data Lims	Fairly Complete Data Lims	2nd DV graph SOME		2nd DV graph		

Identifying Ford EDR families

1. At the end of the data limitations section, just below the last paragraph, on the left, Bosch puts what version of data limitations is being printed. For Ford, most data limitations by family.
2. No difference between DL
   a) 2009 RC6 and
   b) 2011 RC6X.
3. Open cdrx copy in notepad, see RC6_2011



RC5 Lateral Polarity opposite RH rule



RC8 Precrash Data Look

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Speed, Vehicle Indicated, Quality Factor	Accelerator Pedal, % Full	Accelerator Pedal, % Full, Quality Factor	Service Brake, On/Off	Service brake, on/off Quality Factor	Engine RPM
- 5.0	59.8 [96]	OK	18	OK	OFF(Driver_Not_Braking)	OK	2,174

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 1 of 2

Time (sec)	ABS Activity (Engaged, Non-Engaged)	Stability Control Active	Traction Control via Engine	Traction Control via Brakes	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)
- 5.0	Off/non-engaged	Off/non-engaged (No)	Off/non-engaged (Inactive)	Off/non-engaged (Inactive)	-0.01	0.02

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 2 of 2

Time (sec)	Stability Control Yaw Rate (deg/sec)	Stability Control Roll Rate (deg/sec)	Steering Wheel Angle (Pinion) (deg)	Steering Wheel Angle (Pinion) Quality Factor
- 5.0	0.06	-1.53	0.0	OK
- 4.9	-0.13	-1.18	0.0	OK

Other RC8 features

- Acceleration data added
- Second Delta V graph centered around deployment +/-125ms (in SOME RC7P)
- Unique decoder for occupant size

Pre-Crash Data -5 to 0 sec [1 sample/sec] (First Record)

Time (sec)	Occupant Size Classification, Front Passenger (Child size Yes/No [Hex value])	Extended Power Status
- 5.0	No [$03]	No Fault
- 4.0	No [$03]	No Fault

Ford Occupant Size Classification

Pre-Crash Data -1 sec (First Record)

Ignition cycle, crash	359
Frontal air bag warning lamp, on/off	OFF
Occupant size classification, front passenger (Child size Yes/No [Hex value])	No [$08]
Safety belt status, driver	Driver Buckled

- The Hex Value provides additional information than just child yes/no
- The Decoder is different for each supplier and major module type
- Look at end of data limitations for module type – Autoliv=RC5, RC6, RC7; Bosch=AB9, AB10

RC6/7	AB10	AB12	RC8	Classification
$01	$00	$00	$00	Empty
$02	$01	$05	$02	Child or Occupied Disable
$04	$05	$04	$01	Indeterminate (gray zone between child and adult)
$08	$02	$01	$03	Occupied Enable Small Adult (typical 110 lb 5th female)
$10	$03	$02	$04	Occupied Enable Medium (typical 172 lb 50th male)
$20	$04	$03	$05	Occupied Enable Large Adult (typical 223 lb 95th male)

RC8 Lower Recording Threshold?

- Data Limitations State 5mph DV over 150ms threshold for recording
- But events with less than 5mph are seen

Deployment Data (First Record)

Maximum Delta-V, Longitudinal (MPH [km/h])	-2.02 [-3.25]
Time, Maximum Delta-V Longitudinal (msec)	300.0

- Remember, nothing deployed unless a device is listed with a deploy time
- This is an non-deployment

Unusual Ford Event Names

- Fuel Cutoff Level 1 or 2 (RC7P, RC8, AB12)
- Unlocked Event
- Locked Frontal, Side, or Rollover Event
- Rollover Threshold Exceeded (AB12)
- Side Acceleration Threshold Exceeded (AB12)
- Shutdown Request Threshold Exceeded Level 1,2
- Event Record 1 or 2

Multiple events (front, side, rollover) may be listed in the events recovered but there is only one event in memory – they are “all in one”.

Some events like Fuel Cutoff 1 may NOT store data.

2017 MKZ AB12

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record) - Table 1 of 2

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Speed, Vehicle Indicated, Quality Factor	Accelerator Pedal, % Full	Accelerator Pedal, % Full, Quality Factor	Service Brake, On/Off	Service brake, on/off Quality Factor	Engine RPM	Cruise Control Driver Accelerator Pedal Override
- 5.0	99.2 [160]	OK	0	OK	Off	OK	2,816	Cruise_Reg_Not_Overridden
- 4.8	98.7 [159]	OK	0	OK	Off	OK	2,800	Cruise_Reg_Not_Overridden
- 4.6	98.3 [158]	OK	0	OK	Off	OK	2,786	Cruise_Reg_Not_Overridden
- 4.4	98.2 [158]		0	OK	On	OK	2,770	Cruise_Reg_Not_Overridden
- 4.2	.8 [157]	OK	0	OK	On	OK	2,758	Cruise_Reg_Not_Overridden

Every 0.2 Sec!

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record) - Table 2

Time (sec)	Brake Pre-Charge Request	Brake Assist Sensitivity Level	Brake Deceleration Request (m/s^2)	Large Driver Steering or Accel Pedal Input	Collision Mitigation System Fault	Collis Mitiga Syste Enabled
- 5.0	N/A	N/A	236	N/A	N/A	N/A
- 4.8	N/A	N/A	236	N/A	N/A	N/A

Collision Avoidance

2017 MKZ AB12 add'l data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (First Record)

Time (sec)	Driver Gear Selection (Auto Trans)	Wheel Torque Requested (Nm)	Total Arbitrated Brake Torque (Nm)	Ignition Status	Speed Control Status
- 5.0	Drive	532	0	Run	Off
- 4.5	Drive	120	0	Run	Off
- 4.0	Drive	-20	0	Run	Off

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 1 of 2

Time (sec)	ABS Activity (Engaged, Non-Engaged)	Brake Powertrain Torque Request 1	Brake Powertrain Torque Request 2	Traction Control via Brakes	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)
- 5.0	Non-engaged	No	Inactive	No	-0.02	0.09

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 2 of 2

Had this before

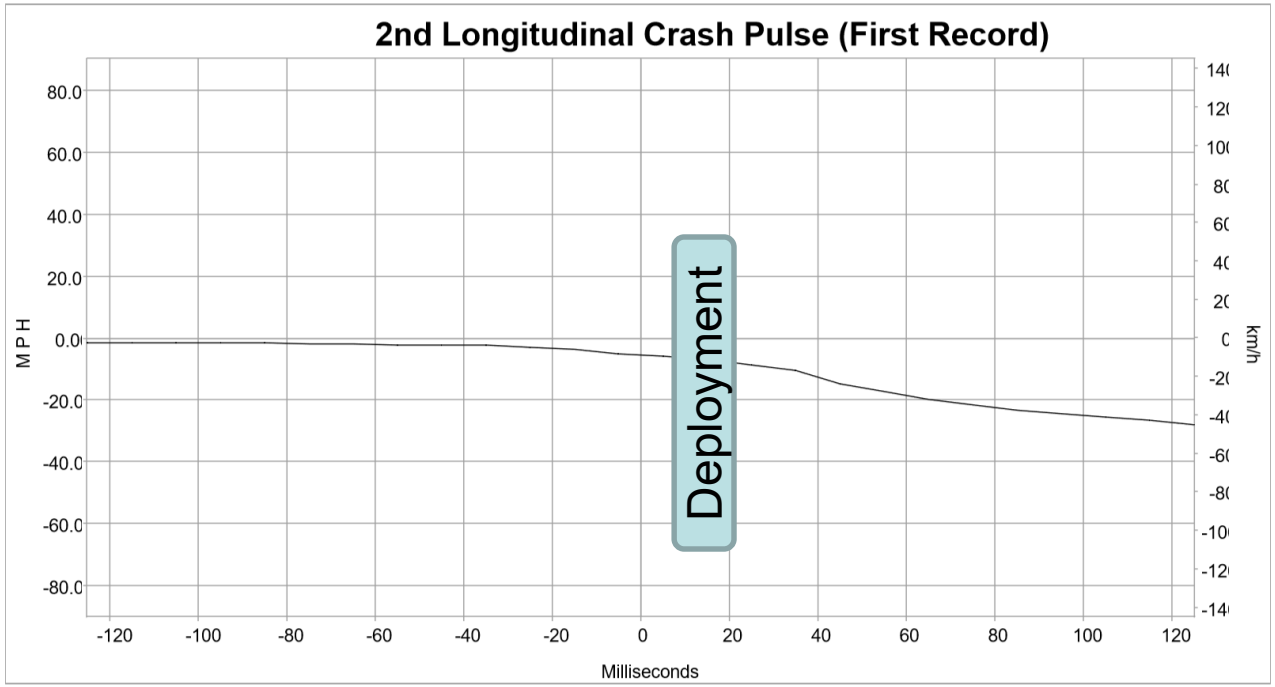
Time (sec)	Stability Control Yaw Rate (deg/sec)	Stability Control Roll Rate (deg/sec)	Steering Wheel Angle (deg)	Steering Wheel Angle Quality Factor
- 5.0	0.06	0.24	-0.9	OK
- 4.9	-0.08	0.00	-0.6	OK
- 4.8	0.45	0.86	0.0	OK

Ford AB12 Anomaly

- File found with “Event Record 2” with no “Event Record 1”. Reported to Bosch.
- Check polarities of stability control system data. Ford Steering, Yaw, and Lat Accel 2009+ should all be positive right, negative left, and same sign – ~~but in some Lat Accel is reversed and is **not** noted in data limitations~~ (CHECKED - FIXED IN 19.5.3)

New RC7P Feature- 2017 F250

- Ford has added a second graph to capture the deployment decision +/-125ms. This is important in hit-curb-then-hit-tree cases where the deployment was previously off the end of the mandated 250ms after wakeup graph



Ford Global Real Timer

Added for AB12/ RC7P/RC8

Timer starts when car is built, is hooked to battery, runs continuously

Month of build is on door jamb sticker (exact day/time is not)

Can be used to tie your recording to your crash

Great for knowing a more precise time between events when not within 5 seconds

Pre-Crash Data -1 sec (First Record)

Ignition cycle, Crash	1,782
Frontal Air Bag Warning Lamp, On/Off	Off
Safety Belt Status, Driver	Belted
Seat Track Position Switch, Foremost, Status, Driver	Rearward
Seat Track Position Switch, Foremost, Status, Front Passenger	Rearward
Safety Belt Status, Front Passenger	Unbelted
Brake Telltale	Off
ABS Telltale	Off
ESC/TC Telltale	Off
ESC/TC Off Telltale	Default Mode
Powertrain Wrench Telltale	Off
Powertrain Malfunction Indicator Lamp (MIL) Telltale	Unchanged Off
Global Real Time (seconds)	47,951,087

Polarity on Stability Control 09+

1. RC6 Data Limitations do not give polarity for stability control data. For lateral it is positive left or CCW. Opposite of right hand rule.
2. New RC7 and AB12 DL's show the stability control lateral polarity as stated above, all 3 backwards of right hand rule

09+ Ford Stability Control Data

Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record)

Times (sec)	Stability Control Lateral Acceleration (g)	Stability Control Longitudinal Acceleration (g)	Stability Control Yaw Rate (deg/sec)
- 1.4	0.086	-0.045	0.12
- 1.3		-0.031	1.5
- 1.2		-0.647	-1.5
- 1.1		-0.75	-2.5
- 1.0		-0.709	-1.5
- 0.9		-0.763	-1.5
- 0.8		-0.826	-1.5
- 0.7		-0.792	-0.75
- 0.6		-0.895	1.25
- 0.5	0.014	-0.83	1.87
- 0.4	0.03	-0.835	0.75
- 0.3	0.022	-0.607	0.37
- 0.2	0.107	-0.653	0.75
- 0.1	0.05	-0.823	-3.37
0.0	-0.004	-0.811	-3.75

Reaction

Integrate lat accel - get velocity – integrate again – get displacement

Drag Factor!

Integrate yaw – get **approach angle change**!! Take tangent of cumulative angle, multiply by speed, get **sideways movement** – how far out of lane vehicle went!!

Autoliv RC6 Anomalies (09-10)

1. There are timing delays in reporting the 0.5 sec interval precrash data in the early Ford Autoliv RC6 modules (09 F150-Escape-Econoline and many 2010). Identify by lag between stability control longitudinal (braking) g's and brake on time and speed lost. Use Stability Control System long. accel for timing of accel release and brake on.
2. Fixed by "RC6X", identifiable only by opening COPY of .cdrx in notepad, observe "RC6_2011" in header (and by agreement between stability control and 0.5 sec interval precrash data).

2012-2015 AB10 Anomaly

1. The “Bosch AB10” ACM found in 2012+ Focus and 2015 Escape and other vehicles **produced before Nov. 2015**, which have a **second event**, incorrectly label pre-crash **time** to impact in the second event.
2. Time between events is accurate – use it
3. Second event precrash (both speed and stability control sections) include extra points at irregular intervals labeled as if they occurred at regular intervals.

AB10 Anomaly cont'd

1. If you can see the first event in the second event's data, it will lead you to think the time between events is longer than it really is.
2. The data in the second event did occur, just not at the time it said it did.
3. The second event last precrash speed is still within 0.5 seconds of the second event.
4. There is no software fix for EDR's produced before Nov. 2015. Fix implemented 11/2015.
5. No mention in DL's yet.

Bosch AB10P Unique Info

1. ~~Longitudinal Accel Stability Control Info is positive for slowing down, negative for speeding up – *opposite of what data limitations say.*~~ Fixed in latest software
2. ~~Lateral Accel data is positive right, negative left – different than the other Ford modules.~~ Fixed in latest software
3. Steering and Yaw remain positive left and are identified as such in data limitations.

2006-7 RC5 Lateral DV Polarity

- Lateral Delta V polarity is not stated in the minimal data limitations
- Two recent cases with clear facts showed it was opposite of right hand rule, positive for hit on pass side, negative for hit on driver side in Versions 19.6 and 21.0
- Ford has been contacted but no guarantee they will update

Steering Angle Glitch

1. Ford stores one value for steering relative to original zero position set at factory
2. If vehicle steering is serviced, original zero may be lost and system learns a new one from average of all new driving.
3. EDR may write down ORIGINAL position relative to zero – so a FEW models serviced may appear to have WRONG value for zero.
4. No warning in DL's about this yet

Backpowering New Fords

1. Ford has added a feature where even if the key is turned off, the airbag system will stay on until the car slows to a near stop. A switched battery feed replaces the current ignition switch feed.
2. There is no an “airbag” fuse. It is now “extended power module” EPM.

19	7.5A [7]	Extended power module.

3. EPM on 2017 Fusion – Micro 3 fuse!
4. Name of fuse changed again on 2020 F350!

For Ford EDR **not supported** by Bosch CDR
See "Unsupported Fords" on my website

Many requests can go directly to Ford's suppliers.

Other requests for assistance to Ford legal:

Email to EDRFORD@FORD.COM

(this is an address for a Bowman and Brooke outside counsel that handles these requests for Ford)

Contact Rick Ruth for additional information prior to making request

As of this writing, Veoneer (former Autoliv) charges $2000 to all, Bosch and Conti are no charge to law enforcement but $1500 to private parties.

Veoneer now says it cannot download modules >15 years old

Email to rcmdownloads@Veoneer.com is being returned as undeliverable in last 2 weeks. Use Deborah.Cox@veoneer.com.

Recent 05 Freestar RCM sent to Takata was returned with Hex Data only and no guide to interpretation (no charge, no value!)

ACM suppliers work from home due to COVID, not allowed in lab!

Available on my website

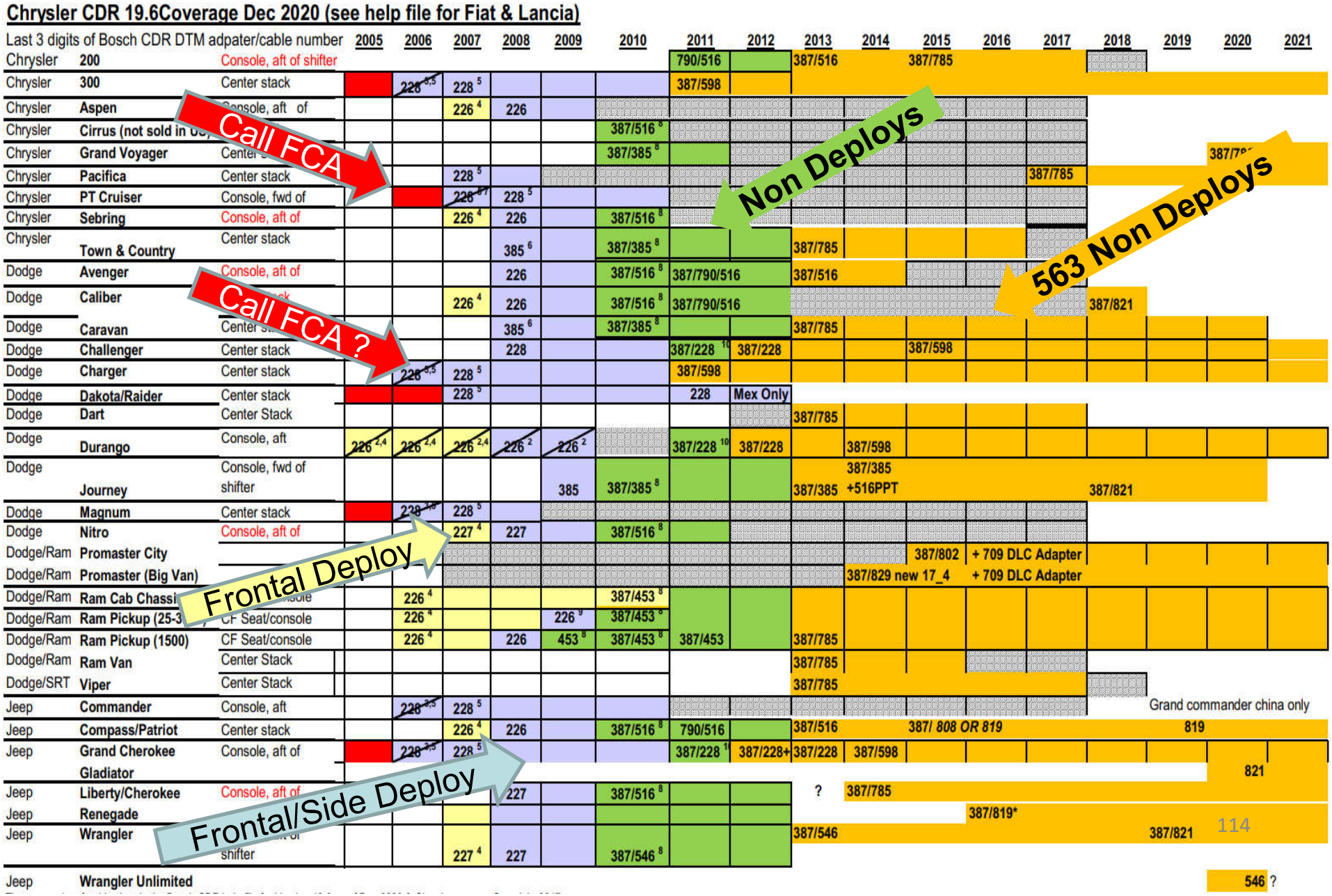
Getting EDR Data from unsupported Fords

Richard R. Ruth, P.E.

(this copy has been updated 05 07 2020, after the original presentation at the Jan 2016 CDR Summit)



FCA 19.6 CDR Coverage 64%



Chrysler Recent Changes

1. **Added Full Size Promaster Van Coverage for 2014-2017 (Sprinter?)**
2. **Fixed issue where 100% accel pedal and throttle reported at 77% in 17.7.1**
3. **Tire Pressure data at 0.25 sec intervals instead of 0.1 sec to make room for other data to be recorded**
4. **Four wheel speeds *in kph* instead of rpm – see non driven vs driven wheels**
5. **Brake master cylinder pressure added to some 2018+ models**
6. **Gear Position added (changes at end of shift)**

Output New Look –

Removed multiple brake elements, added ABS/SC, moved strg to P1, hi res

Pre-Crash Data [10 samples/sec] (Most Recent Event - table 1 of 2)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Pre-Crash Recorder Status	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full	Service Brake	Engine RPM	ABS Activity	Stability Control	Steering Input (deg)
-5.0	Complete	96 [155]	0.00	Off	3,878	No	On	1
-4.9	Complete	96 [154]	0.00	Off	3,835	No	On	-1
-4.8	Complete	96 [154]	0.00	Off	3,726	No	On	-5
-4.7	Complete	96 [154]	0.00	Off	3,621	No	On	-10
-4.6	Complete	96 [154]	0.00	Off	3,489	No	On	-18
-4.5	Complete	95 [154]	0.00	Off	3,344	No	On	-21
-4.4	Complete	95 [153]	0.00	Off	3,210	No	On	-20
-4.3	Complete	95 [153]	0.00	Off	3,109	No	On	-19
-4.2	Complete	95 [153]	0.00	Off	3,068	No	On	-19
-4.1	Complete	95 [152]	0.00	Off	3,108	No	On	-18
-4.0	Complete	95 [152]	0.00	Off	3,079	No	On	-18

Wheel RPM replaced by speed

See driven vs non driven, left vs right, ABS working…

Pre-Crash Data [10 samples/sec] (Most Recent Event - table 2 of 2)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	ABS MIL	Yaw Rate (deg/sec)	Wheel Speed LF (km/h)	Wheel Speed RF (km/h)	Wheel Speed LR (km/h)	Wheel Speed RR (km/h)
-5.0	Off	0.40	154.37	154.52	154.55	154.52
-4.9	Off	-0.48	154.28	154.16	154.25	154.25
-4.8	Off	-1.76	154.40	154.09	153.88	153.77
-4.7	Off	-2.80	154.28	153.95	153.88	153.67
-4.6	Off	-4.24	154.01	153.64	153.85	153.55
-4.5	Off	-6.80	153.98	153.16	153.80	153.19
-4.4	Off	-7.12	153.80	152.88	153.70	152.80
-4.3	Off	-5.84	153.49	152.67	153.37	152.62
-4.2	Off	-5.12	153.13	152.19	153.25	152.38
-4.1	Off	-4.64	152.80	152.04	152.91	151.91
-4.0	Off	-4.64	152.49	151.73	152.46	151.65
-3.9	Off	-4.00	152.13	151.55	152.13	151.31
-3.8	Off	-3.68	151.83	151.25	151.95	151.13

Current Gear & More @ 0.25 sec

Pre-Crash Data [4 samples/sec] (Most Recent Event - table 1 of 2)

(the most recent sampled values are recorded prior to the event)

NEW

Changes at END of shift

Time Stamp (sec)	Pre-Crash Recorder Status	Raw Manifold Pressure (kPa)	PCM MIL	ETC Lamp	Current Gear	Reverse Gear Status
-5.00	Complete	20.00	Off	Off	D5	Not_Reverse
-4.75	Complete	20.00	Off	Off	D5	Not_Reverse
-4.50	Complete	21.60	Off	Off	D5	Not_Reverse
-4.25	Complete	23.20	Off	Off	D5	Not_Reverse
-4.00	Complete	23.20	Off	Off	D6	Not_Reverse
-3.75	Complete	23.20	Off	Off	D6	Not_Reverse
-3.50	Complete	23.20	Off	Off	D6	Not_Reverse
-3.25	Complete	23.20	Off	Off	D6	Not_Reverse
-3.00	Complete	22.40	Off	Off	D6	Not_Reverse
-2.75	Complete	21.60	Off	Off	D6	Not_Reverse
-2.50	Complete	21.60	Off	Off	D7	Not_Reverse
-2.25	Complete	21.60	Off	Off	D7	Not_Reverse
-2.00	Complete	21.60	Off	Off	D7	Not_Reverse
-1.75	Complete	21.60	Off	Off	D7	Not_Reverse

10.63 in

Tire pressure @ 0.25 sec intervals

Pre-Crash Data [4 samples/sec] (Most Recent Event - table 2 of 2)

(the most recent sampled values are recorded prior to the event)

NEW

Time Stamp (sec)	Tire Pressure Position	Tire Pressure Status	Tire Pressure Value (PSI)	Cruise Control Lamp Status	Cruise Control Engaged Status	ACC On/Off	ACC Set Speed (MPH [km/h])
-5.00	LHF	Normal	34	Off	Not_Engaged	Off	0 [0.0]
-4.75	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.50	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.25	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.00	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-3.75	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.50	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.25	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.00	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-2.75	RHF	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-2.50	RHF	Normal	33	Off	Not_Engaged	Off	0 [0.0]

Integrating FCA Yaw Rate

From 08 charger case

- 3 deg/sec * 0.1 sec =	-0.3 degrees
-11 deg/sec * 0.1 sec =	-1.1 degrees
-18 deg/sec * 0.1 sec =	-1.8 degrees
-15 deg/sec * 0.1 sec =	-1.5 degrees
- 9 deg/sec * 0.1 sec =	-0.9 degrees
- 4 deg/sec * 0.1 sec =	-0.4 degrees
- 1 deg/sec * 0.1 sec =	-0.1 degrees
Total Swerve Right	6.1 degrees

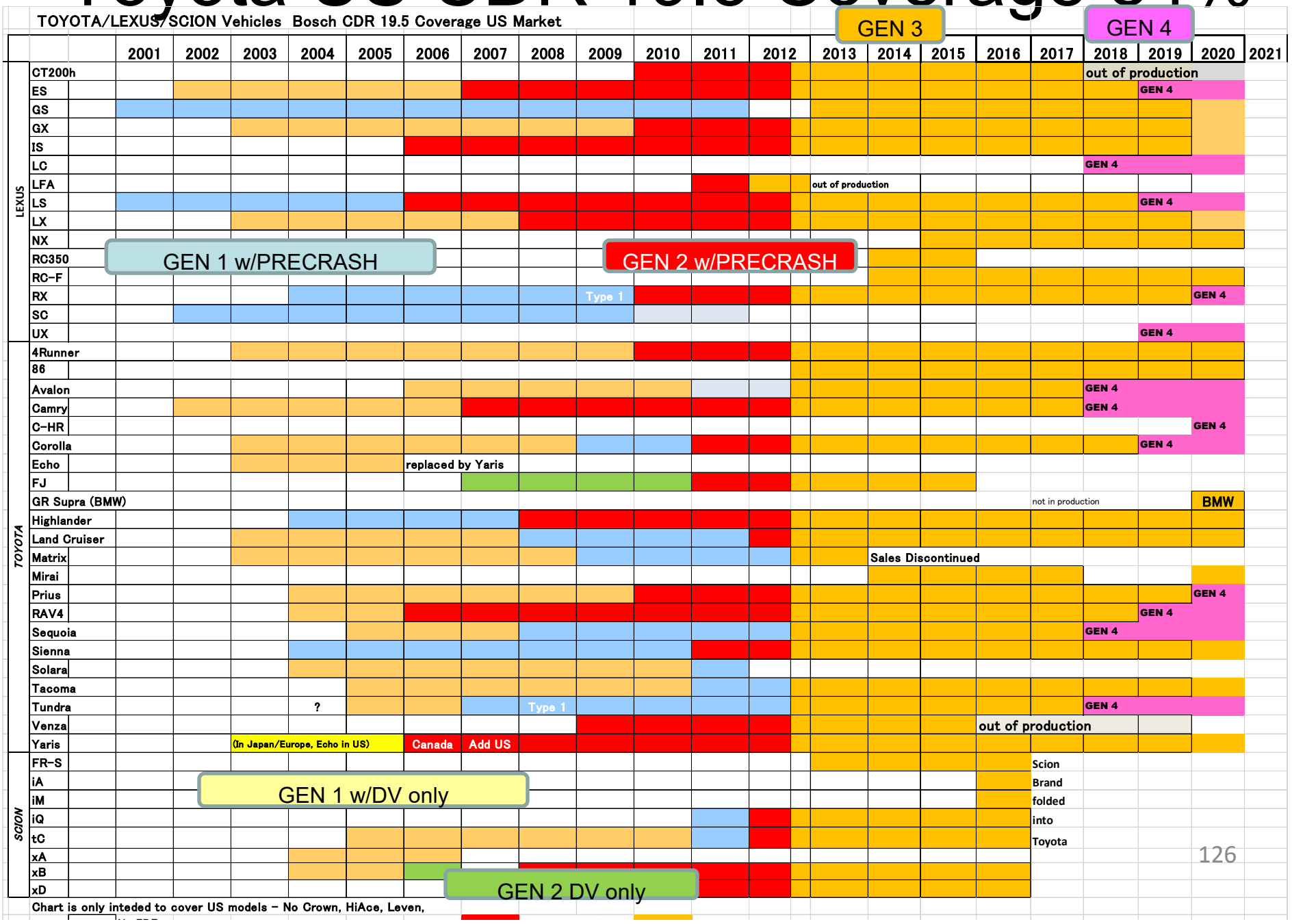
Note 1: this is how far nose of car is pointing right – *CG* motion may *lag*

Note 2: Polarity opposite RH rule this model

New Chrysler Info

1. Frontal sensors are 100G at 45 degrees (100*cos 45° = 71g straight ahead).
2. Replaced clipping flag with “Sensor Design Range Exceeded” line on graph
3. Buckle switch NOT normally used to determine if pre-tensioner fires (Exceptions on some non US built vehicles)
4. Fiat Spider = Mazda Miata, threshold <5mph
5. Master Cylinder Pressure on some 2018!
6. Driver assistance data elements coming in

Toyota US CDR 19.5 Coverage 84%



Toyota EDR Evolution

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17 EDR
3 front 1 side 1 roll	2 front 2 side 2 roll		4 front 4 side 6 precrash
Record on 2g wakeup		Record on 5mph XDV, side 3.5g wake	Record on 5 X or Y DV
150ms XDV 5PC@1 + TRG	200ms XDV 5PC@1+TRG	250ms XDV 10PC@0.5 + TRG	250ms XY 10PC@0.5+TRG

Toyota EDR Evolution cont'd

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17 EDR
70ms YDV@ACM+sats		12 YDV 70ms sats only 13 250ms ACM	250ms YDV
REFRESH 500 MS SPEED		REFRESH 24 MS SPEED	
78 MPH TOP, 2kph res	75 MPH TOP. 2kph res	124 MPH TOP 1 kph res	
Accel Pedal Volts		Accel Pedal % & Throttle %	

Toyota EDR Evolution cont'd

GEN 1
00,02 EDR

GEN 2
04,06 EDR

GEN 3
12,13,15 EDR

GEN 4
17 EDR

RPM to next lower 400

RPM to 100

No Key Cycles

Key Cycles

Time Between Events < 5 sec

Time Between Events <16 sec

Brk Oil Pres, Stab. Cont. Long Accel

Strg & Yaw

New “17 EDR”

- “Front/Rear” events also triggered by 5mph Y DV or side deployment
- Y Delta V from ACM added to frontal events for recon
- More precrash registers (6 vs 2 prior)
- More front/rear & side registers (4 vs 2)
- Separate side events still reported, but do NOT have precrash associated with them. Side events have detail of side satellite sensors (generally ***not used*** by Recons).
- Odometer data element added

Same Data, New Look to Precrash

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	29.8 [48]	49.5	24.5	Invalid	2,400	Invalid	OFF
-4.50	32.3 [52]	47.0	24.5	Invalid	2,500	Invalid	OFF
-4.00	34.2 [55]	38.0	20.0	Invalid	2,700	Invalid	OFF
-3.50	35.4 [57]	32.5	16.0	Invalid	2,800	Invalid	OFF
-3.00	36.7 [59]	26.5	13.5	Invalid	2,400	Invalid	OFF
-2.50	37.3 [60]	22.5	11.0	Invalid	2,400	Invalid	OFF
-2.00	37.3 [60]	19.0	8.5	Invalid	2,200	Invalid	OFF
-1.50	38.5 [62]	19.0	6.0	Invalid	1,900	Invalid	OFF
-1.00	37.9 [61]	18.0	5.5	Invalid	1,900	Invalid	OFF
-0.50	37.9 [61]	18.0	5.0	Invalid	1,800	Invalid	OFF
TRG(0)	36.0 [58]	100.0	26.0	Invalid	1,700	Invalid	OFF

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	1.579	-0.98	-4.5	D
-4.50	OFF	OFF	0.00	1.795	0.00	-1.5	D
-4.00	OFF	OFF	0.00	1.292	0.49	0.0	D
-3.50	OFF	OFF	0.00	0.861	0.00	-1.5	D
-3.00	OFF	OFF	0.00	0.718	0.00	-1.5	D
-2.50	OFF	OFF	0.00	-0.072	0.00	-1.5	D
-2.00	OFF	OFF	0.00	0.000	0.00	-1.5	D
-1.50	OFF	OFF	0.00	0.000	-0.49	-1.5	D
-1.00	OFF	OFF	0.00	0.287	0.00	-1.5	D
-0.50	OFF	OFF	0.00	-0.144	0.00	-1.5	D
TRG(0)	OFF	OFF	0.48	-1.507	61.00	-19.5	D

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-5.00	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-4.50	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

Pedestrian Protection Module

System Status at Pedestrian Event (1st Prior Event, TRG 1)

TRG Count (times)	1
TRG Count not for Pedestrian (times)	SNA
Ignition Cycle, Crash (times)	353
Odometer signal (miles [km])	1,195 [1,923]
Trip count (times)	392
Time count (msec)	1,283,900
Time count input system	Normal
Airbag Warning Lamp, On/Off	OFF
Pedestrian Warning Lamp, On/Off	OFF
Time from Previous TRG (msec)	32767 or greater
Time from Pre-Crash to TRG (msec)	300
Time between TRG not for Pedestrian and TRG for Pedestrian (msec)	SNA
TRG Establishment not for Pedestrian, within 5 sec.	ON
Freeze Signal	ON
Recording Status, Pre-Crash	Complete
Recording Status, Crash Info.	Complete
Recording Status, EDR Correlation Info.	Complete

Deployment Command Data (1st Prior Event, TRG 1)

Front Pop-Up Hood, Time to Deploy (msec)	3.0
Rear Pop-Up Hood, Time to Deploy (msec)	3.0
Airbag for Pedestrian Deployment, Time to Deploy (msec)	SNA

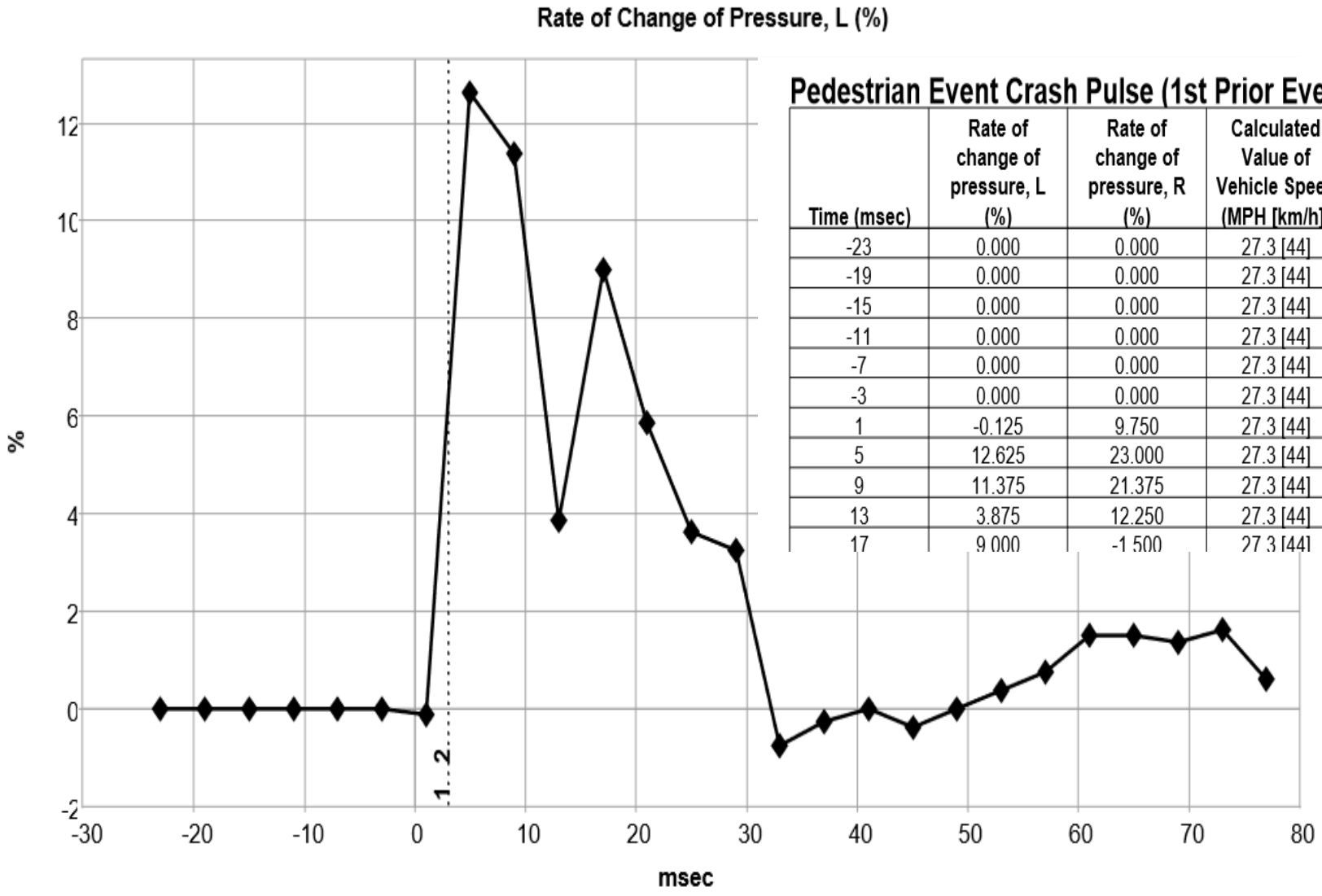
DTCs Present at Time of Event (1st Prior Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Warning ON Time Since DTC was Set, for Pedestrian (min)	0
Diagnostic Trouble Codes	None

Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	1

Data from L/R bumper pressure sensors on 15 & 17 EDR



Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Time (msec)	Rate of change of pressure, L (%)	Rate of change of pressure, R (%)	Calculated Value of Vehicle Speed (MPH [km/h])
-23	0.000	0.000	27.3 [44]
-19	0.000	0.000	27.3 [44]
-15	0.000	0.000	27.3 [44]
-11	0.000	0.000	27.3 [44]
-7	0.000	0.000	27.3 [44]
-3	0.000	0.000	27.3 [44]
1	-0.125	9.750	27.3 [44]
5	12.625	23.000	27.3 [44]
9	11.375	21.375	27.3 [44]
13	3.875	12.250	27.3 [44]
17	9.000	-1.500	27.3 [44]

Toyota Help File Organization

Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in some cases, adapter) to use for EDR data directly from the ACM. It is important to note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Cli list of supported ACM part numbers.

Click here for tips on how to use this *Supported Vehicles* help topic

This Started in Nov 2015

IMPORTANT: To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the appropriate cable called out in this the help file.

General Toyota Application Notes:

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will no longer be grayed out. **This is n indicator that the vehicle is supported.** Only the vehicles listed in the help file below are supported as indicated by the list and any associated coverage not

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287) unless otherwise stated.

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

Vehicle Specific Application Notes:

Click here to view all vehicle specific Toyota coverage notes

Vehicles Are Listed Alphabetically
Not by Model Year

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W X Y Z

4Runner (Toyota) 2003 and later Market: 7

Year	EDR Generation	CDR Cable & Adapter	Module	Module Location
2002 - 2008	Gen 1 (02EDR)	F00K108617 (no adapter)	ACM	Under Center Stack
2009 - 2011	Gen 2 (06EDR)	F00K108616 & F00K108387	ACM	Under Center Stack
2012 and later	Gen 3 (12EDR/13EDR)	F00K108616 & F00K108387	ACM	Under Center Stack

Help file may be CALENDAR year not model year,
EDR may not yet be present in Model Year of first year listed

Toyota Vehicle Lookup

Camry (Toyota) **2001 and later** **Market: 7**

All Markets the Vehicle is Sold

Year	EDR Generation	CDR Cable & Adapter	M
2001 - 2004 2005 (US/Canada)	Gen 1 (00EDR/02EDR)	F00K108615 (no adapter) F00K108617 (no adapter) Click here [?] to match the ACM part number to the correct cable	A
2005 - 2011	Gen 2 (04EDR/06EDR)	F00K108613 (no adapter) F00K108614 & F00K108387 F00K108615 (no adapter) F00K108616 & F00K108387 Click here [?] to match the ACM part number to the correct cable	A
2012 and later	Gen 3 (12EDR/13EDR)	F00K108616 & F00K108387	A

< Models

Note CAMRY got EDR in 02 Model Year in US

Note CAMRY got 04 EDR in 07 Model Year in US

Note CAMRY got 12 EDR as running change in 12 model year, early models have 04/06 EDR in US

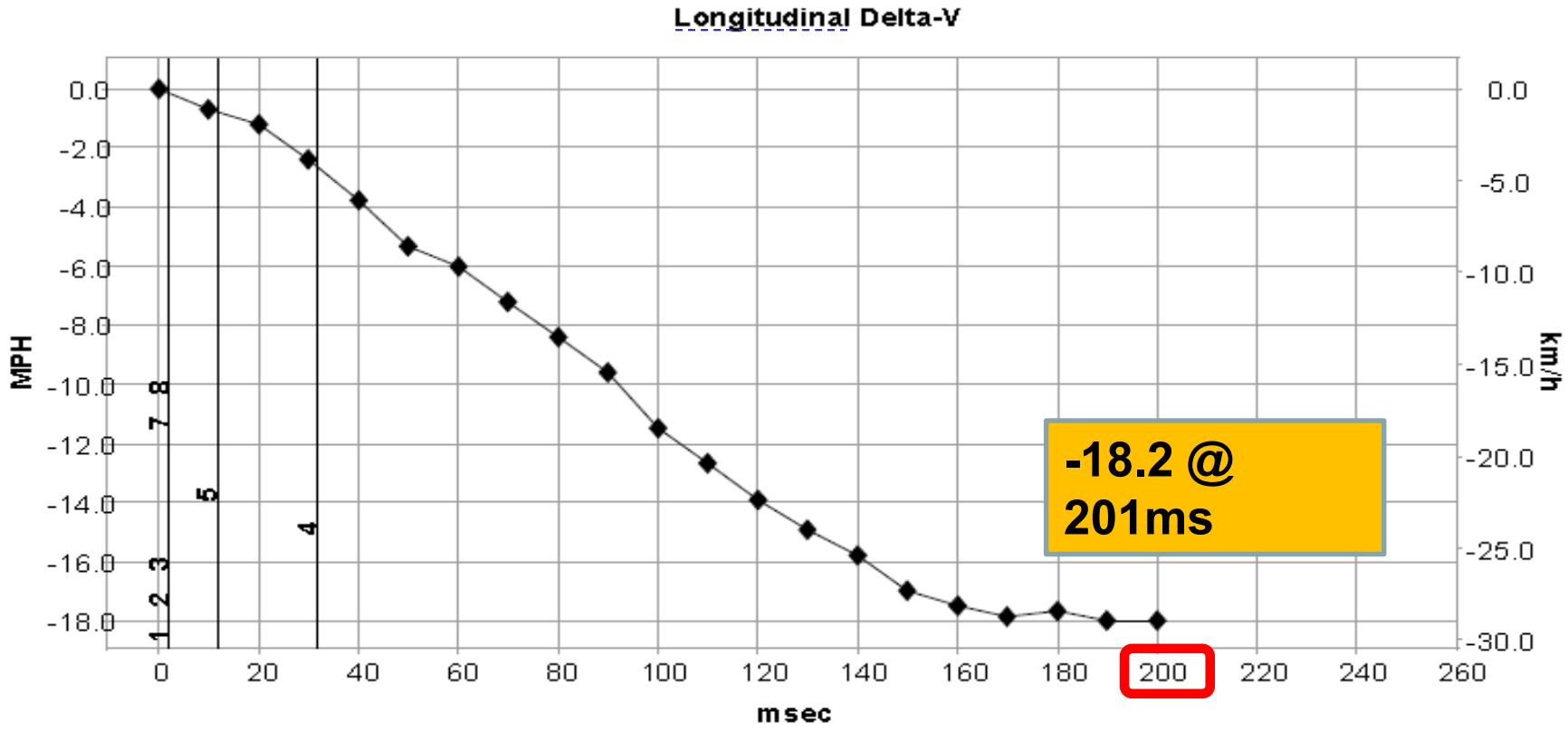
Toyota 12 EDR Drawback

1. Lateral acceleration is only recorded in a side algorithm wakeup.
2. When it is recorded, it is only for 68ms
3. It is only recorded for the side satellites, not for the central control module
4. If trying to calculate a PDOF you must be very aware of the different recording timing and sensor location
5. Fixed in “13EDR” – get 250ms of Y accel

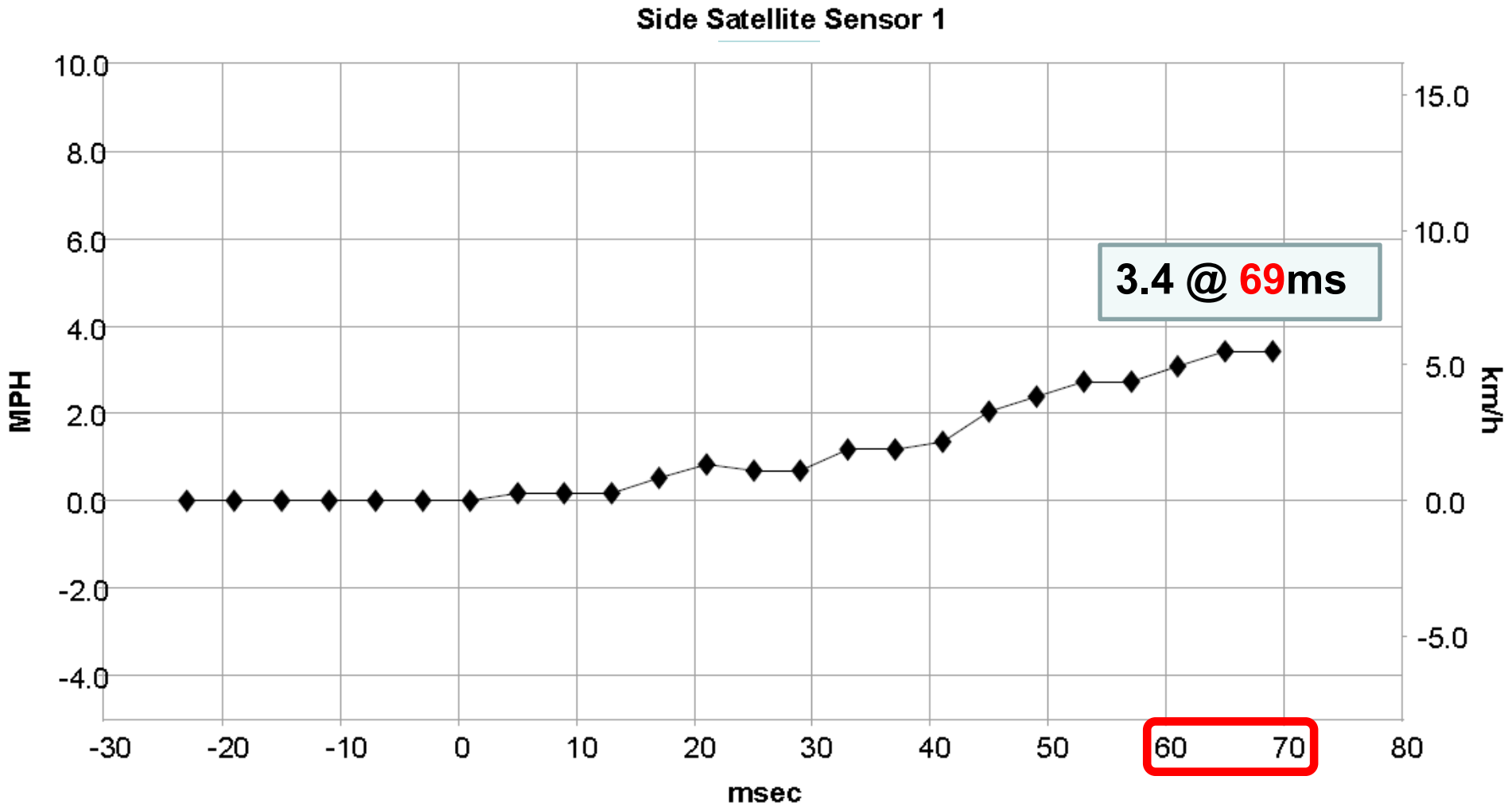
Delta V analysis – What's the PDOF? The Total?? 12 EDR

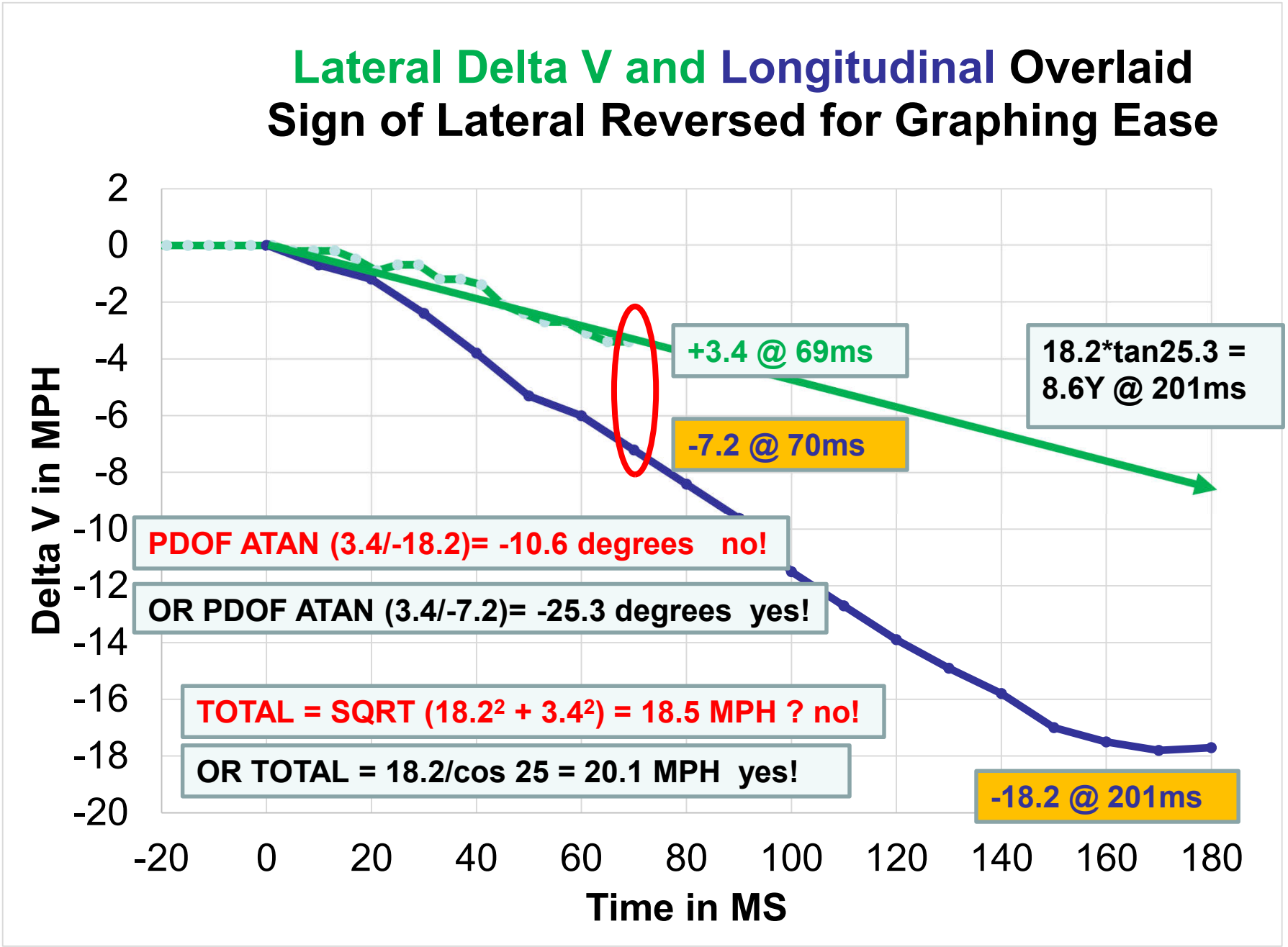
Longitudinal Crash Pulse (1st Prior Event, TRG 2 - table 1 of 2)

Recording Status, Time Series Data	Complete
Time from Time Zero to TRG (msec)	2.0
Length of Delta-V (msec)	200
Max. Longitudinal Delta-V (MPH [km/h])	-18.2 [-29.2]
Time, Maximum Delta-V, Longitudinal (msec)	201.0
Power Supply Status at Max. Delta-V	ON



12 EDR Lateral Delta V -




Lateral Delta V and Longitudinal Overlaid
Sign of Lateral Reversed for Graphing Ease
Delta V in MPH
2
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
-20
0
20
40
60
80
100
120
140
160
180
Time in MS
+3.4 @ 69ms
18.2*tan25.3 =
8.6Y @ 201ms
-7.2 @ 70ms
PDOF ATAN (3.4/-18.2)= -10.6 degrees no!
OR PDOF ATAN (3.4/-7.2)= -25.3 degrees yes!
TOTAL = SQRT (18.2² + 3.4²) = 18.5 MPH ? no!
OR TOTAL = 18.2/cos 25 = 20.1 MPH yes!
-18.2 @ 201ms

VCH Freeze Frame Data 2013+

- New feature in 2013+ Rav 4 and 2014+ Highlander & Lexus IS, 2015+ Lexus NX & RC, "**V**ehicle **C**ontrol **H**istory"
- Triggers EDR-style recording if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion. 26 triggers including crash avoidance, ESC.
- Accessible only with Toyota Techstream service tool – not part of Bosch CDR system. Annual s/w subscription or 2 day subs available.
- CDR900 or Nexiq works as pass thru device same as Toyota Mongoose hardware
- Data exports as CSV to Microsoft excel
- Space in memory for MANY events – can show a history of erratic driver behavior
- Details in SAE 2016-01-1495 Appendix D. New **2019-01-0632** "**Reconstructing Vehicle Dynamics from On-Board Event Data**" **discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events).**
- **Photos stored by pre-collision braking system may be accessed with Techstream system!**

TOYOTA/LEXUS Vehicles with Vehicle Control History Data

		2013	2014	2015	2016	2017	2018	2019	2020
LEXUS	CT200h								
	ES								
	GS								
	GX								
	IS		250 and 350						
	LS								
	LX								
	NX								
	RC								
	RX								
TOYOTA	Avalon								
	Camry								
	Corolla								
	Prius			C and V	and Prius	and Prime			
	Highlander								
	Land Cruiser								
	4Runner								
	Tacoma								
	Tundra								
	Sequoia								
	RAV4								
	Sienna								
	Yaris								
SCION	iM								

In your student reference materials: 131-slide presentation by Brad Muir has details on accessing the VCH, and new crash avoidance data



Sample VCH Summary

Vehicle Diagnostic Report

2020 Rav4 [redacted]
004320 mile

[redacted]

[redacted]

Printed By: Default User(1)

[redacted]

Vehicle Control History Report

(1 of 2)

Current Key Cycle : 00372
Current Key Cycle Elapsed Time : 00000:17:23
Maximum Number of Overwrite of the Recording Area : 30000

Key Cycle : All

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger	Image	ODO (mile)	Number of Overwrite of the Recording Area
00030	00000:53:56.4	-	2/1/2020 4:24:56 PM	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	Icon	51	1
00044	00000:19:06.4	-	2/4/2020 11:00:03 AM	PCS operation history		264	1
00051	00000:01:15.3	-	2/6/2020 9:30:21 AM	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	Icon	315	1
00055	00000:24:15.8	-	2/7/2020 8:50:59 AM	Steering angle speed over certain number was detected (related to LDA/LKA/LTA)	Icon	340	1

A way to get ped hits??

Vehicle Control History

Current Key Cycl

Key Cycle: All

00372

Key Cycle	Elapsed Time	Clock Type	Time and Date	Trigger
00143	00000:47:31.8	-	2/24/2020 11:24:28 AM	Steering angle speed over certain number was dete
00144	00000:19:09.1	-	2/24/2020 7:22:43 PM	PCS operation history
00145	00000:07:55.7	-	2/24/2020 7:53:02 PM	Sudden braking history
00162	00000:43:37.7	-	2/28/2020 6:26:13 PM	PCS operation history
00178	00000:03:40.9	-	3/4/2020 5:09:47 PM	Steering angle speed over certain number was dete
00182	00000:24:30.4	-	3/6/2020 5:50:27 PM	Sudden braking history
00182	00000:24:30.4	-	3/6/2020 5:50:27 PM	PCS operation history
00182	00000:24:30.5	-	3/6/2020 5:50:27 PM	Steering angle speed over certain number was dete
00182	00000:24:30.8	-	3/6/2020 5:50:27 PM	Accelerator signal and brake signal input simultane
00226	00000:00:04.3	-	3/27/2020 6:58:48 AM	Accelerator signal and brake signal input simultane

Toyota “13 EDR” & “15 EDR”

1. **13 EDR** Adds Lateral ***Accel*** data from ACM in frontal events for 200ms (Fixes biggest drawback of **12 EDR**)

Longitudinal/Lateral Crash Pulse (Most Recent Event,

Time (msec)	Longitudinal Delta-V (MPH [km/h])	Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2)	F
0	0.0 [0.0]	0.0	
10	0.3 [0.6]	-1.9	
20	0.5 [0.8]	0.0	

2. ***Calculate*** Lat **Delta V** for each 10ms interval
3. DV_{10ms} = ***Accel***/9.8*(32.2/1.466)*0.01s
4. Add up intervals to get ***total*** **Delta V**
5. **Template available to calculate it**

2018+ Toyota “17EDR”

1. “Front/rear” 5mph X events replaced by “Front/Rear/Side”, events triggered by 5mph X ***or Y*** DV or side deploys. Y DV graph added.
2. 6 precrash memory spaces, 4 DV spaces for each crash type (vs 2 each prior)
3. Separate side events still reported with 70ms side accel in m/sec^2, but do NOT have precrash associated with them.

17EDR Precrash (new look, same data)

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	29.8 [48]	49.5	24.5	Invalid	2,400	Invalid	OFF
-4.50	32.3 [52]	47.0	24.5	Invalid	2,500	Invalid	OFF
-4.00	34.2 [55]	38.0	20.0	Invalid	2,700	Invalid	OFF
-3.50	35.4 [57]	32.5	16.0	Invalid	2,800	Invalid	OFF
-3.00	36.7 [59]	26.5	13.5	Invalid	2,400	Invalid	OFF
-2.50	37.3 [60]	22.5	11.0	Invalid	2,400	Invalid	OFF
-2.00	37.3 [60]	19.0	8.5	Invalid	2,200	Invalid	OFF
-1.50	38.5 [62]	19.0	6.0	Invalid	1,900	Invalid	OFF
-1.00	37.9 [61]	18.0	5.5	Invalid	1,900	Invalid	OFF
-0.50	37.9 [61]	18.0	5.0	Invalid	1,800	Invalid	OFF
TRG(0)	36.0 [58]	100.0	26.0	Invalid	1,700	Invalid	OFF

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	1.579	-0.98	-4.5	D
-4.50	OFF	OFF	0.00	1.795	0.00	-1.5	D
-4.00	OFF	OFF	0.00	1.292	0.49	0.0	D
-3.50	OFF	OFF	0.00	0.861	0.00	-1.5	D
-3.00	OFF	OFF	0.00	0.718	0.00	-1.5	D
-2.50	OFF	OFF	0.00	-0.072	0.00	-1.5	D
-2.00	OFF	OFF	0.00	0.000	0.00	-1.5	D
-1.50	OFF	OFF	0.00	0.000	-0.49	-1.5	D
-1.00	OFF	OFF	0.00	0.287	0.00	-1.5	D
-0.50	OFF	OFF	0.00	-0.144	0.00	-1.5	D
TRG(0)	OFF	OFF	0.48	-1.507	61.00	-19.5	D

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-5.00	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-4.50	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

17 EDR time data & ped events

System Status at Pedestrian Event (1st Prior Event, TRG 1)

TRG Count (times)	1
TRG Count not for Pedestrian (times)	SNA
Ignition Cycle, Crash (times)	353
Odometer signal (miles [km])	1,195 [1,923]
Trip count (times)	392
Time count (msec)	1,283,900
Time count input system	Normal
Airbag Warning Lamp, On/Off	OFF
Pedestrian Warning Lamp, On/Off	OFF
Time from Previous TRG (msec)	32767 or greater

1. Odometer added
2. “Trip count” similar to key cycles but slightly higher,
3. Time Count (like Ford “Key on time”), Odometer
4. Pedestrian events if pedpro option bought

NISSAN EDR's 19.3 (CDR 51%)

Model	Location	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Last 3 digits of Bosch CDR cable number																
INFINITI																
G CONVERTIBLE	TUNNEL BETWEEN SEATS				*		Jul-12		780							
G COUPE	TUNNEL BETWEEN SEATS		*						780							
G SEDAN	TUNNEL BETWEEN SEATS		*				Jul-12		780							
M	TUNNEL BETWEEN SEATS								780							
Q30													780	780	780	780
Q40	TUNNEL BETWEEN SEATS										780					
Q50	TUNNEL BETWEEN SEATS								780	780	780	780	780	780	780	780
Q60	TUNNEL BETWEEN SEATS								780	780	780		780	780	780	780
Q70	TUNNEL BETWEEN SEATS								780	780	780	780	780	780	780	
EX	TUNNEL BETWEEN SEATS								780							
FX	TUNNEL BETWEEN SEATS								780							
JX35	TUNNEL BETWEEN SEATS								780							
QX50-80	TUNNEL BETWEEN SEATS								780	780	780	780	780	50,60,80	50,60,80	50,60
QX30	TUNNEL BETWEEN SEATS												598	598	598	598
NISSAN		2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
370Z	TUNNEL BETWEEN SEATS		*				Aug-12		780	780	780	780	780	780	780	780
ALTIMA COUPE	TUNNEL BETWEEN SEATS			*					780	INCLUDED IN SEDAN BELOW						
ALTIMA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780	780	780	780	780
ARMADA	TUNNEL BETWEEN SEATS	*							780	780	780	?	780	780	780	780
CUBE	TUNNEL BETWEEN SEATS				*				780	780						
FRONTIER	TUNNEL BETWEEN SEATS	*					Oct-12		780	780	780	780	780	780	780	
GT-R	TUNNEL BETWEEN SEATS				*					780	780	780	780	780	780	780
JUKE	TUNNEL BETWEEN SEATS								780	780	780	780	780			
KICKS														780	780	780
LEAF	TUNNEL BETWEEN SEATS								780	780	780	780	780	780	780	780
MAXIMA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780	780	780	780	780
MICRA	Check Manual										780	780	780	780	780	
MURANO	TUNNEL BETWEEN SEATS		*		*				780	780	780	780	780	780	780	780
MURANO CROSS C	TUNNEL BETWEEN SEATS								included in above							
NV200	TUNNEL BETWEEN SEATS								780	780	780	780	780	780	780	780
PATHFINDER	TUNNEL BETWEEN SEATS	*							780	780	780	780	780	780	780	780
QUEST	CTR CONSOLE NEAR FRONT		*						780	780	780	780	780			
ROGUE, R SPORT	Check Manual			*					780	780	780	780	780	+Qashqai	780	780
SENTRA	TUNNEL BETWEEN SEATS		*						780	780	780	780	780	780	780	780
TITAN	TUNNEL BETWEEN SEATS	*							780	780	780	780	780	780	780	780
VERSA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780		780	780	780
VERSA NOTE HB	TUNNEL BETWEEN SEATS						Sep-12		780	780	780	780	780	780	780	
XTERRA	TUNNEL BETWEEN SEATS	*					Oct-12		780	780	780					

Yellow = Proprietary Consult Tool Only
No direct to module cables

Dealers do NOT have special software required to read EDR

Yellow area not covered by CDR

Nissan Pre-Crash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % full	Engine RPM	Motor RPM	Service Brake (On, Off)	Steering Input (deg)
-5.0	24 [39]	16	2650	0	Off (Brake Not Activated)	-32
-4.5	23 [37]	11.5	1750	0	Off (Brake Not Activated)	-2
-4.0	21 [34]	9	1200	0	On (Brake Activated)	28
-3.5	15 [24]	9	1150	0	On (Brake Activated)	14
-3.0	13 [21]	4	950	0	On (Brake Activated)	4
-2.5	11 [17]	4	950	0	On (Brake Activated)	4
-2.0	5 [8]	4	900	0	On (Brake Activated)	-48
-1.5	3 [5]	4	850	0	On (Brake Activated)	-98
-1.0	1 [2]	4	850	0	On (Brake Activated)	-160
-0.5	3 [5]	10.5	1100	0	Off (Brake Not Activated)	-194
0.0	6 [9]	16	1150	0	Off (Brake Not Activated)	-218

Hybrid/Electric or Trans Input

Trans input tells you convertere slip

Table 2 – not rq'd

At AE

Nissan Long ΔV Polarity Fixed?

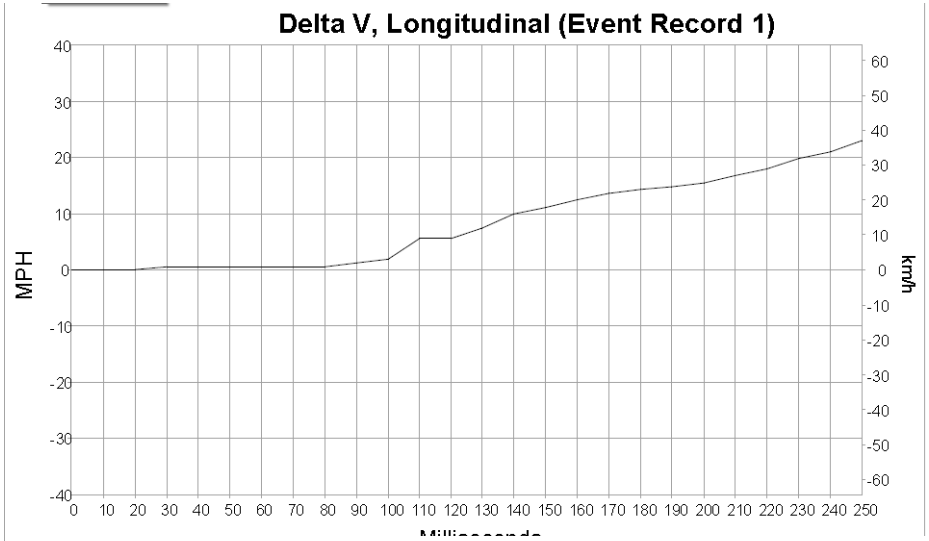
Front hit *was* positive - ***Some exceptions still in 19.3!***

2013 Altima in CDR 12.2

System Status at Event (Event Record 1)

Frontal Air Bag Suppression Switch Status	S airbag inhibit
Maximum Delta-V, Longitudinal (MPH [km/h])	29 [47]
Time, Maximum Delta-V, Longitudinal (msec)	300
Maximum Delta-V, Lateral (MPH [km/h])	5 [8]
Time, Maximum Delta-V, Lateral (msec)	210
Maximum Acceleration, Longitudinal (g)	32.5
Time, Maximum Acceleration, Longitudinal (msec)	102.5
Maximum Acceleration, Lateral (g)	36
Time, Maximum Acceleration, Lateral (msec)	112.5

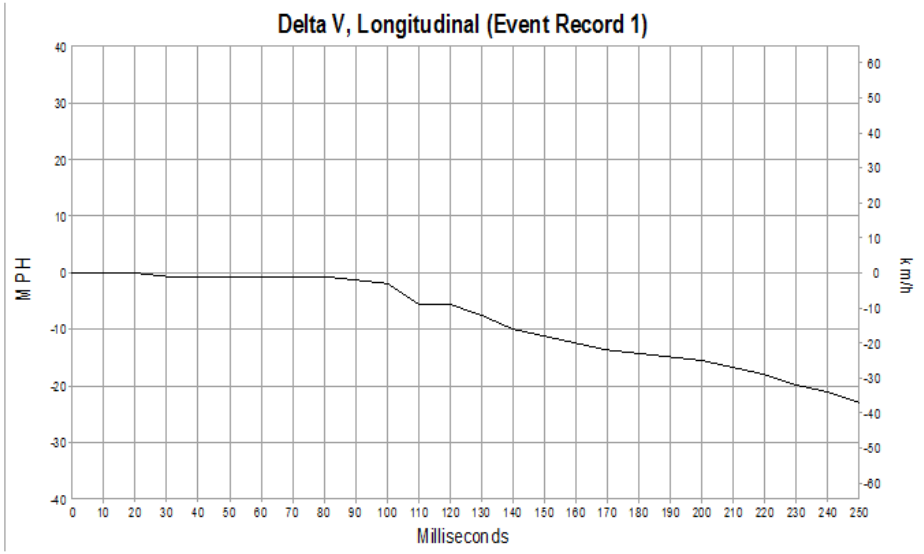
+29 at 300ms



2013 Altima in CDR 16.5

S airbag inhibit	
Maximum Delta-V, Longitudinal (MPH [km/h])	-29 [-47]
Time, Maximum Delta-V, Longitudinal (msec)	300
Maximum Delta-V, Lateral (MPH [km/h])	5 [8]
Time, Maximum Delta-V, Lateral (msec)	210
Maximum Acceleration, Longitudinal (g)	-32.5
Time, Maximum Acceleration, Longitudinal (msec)	102.5
Maximum Acceleration, Lateral (g)	36
Time, Maximum Acceleration, Lateral (msec)	112.5

-29 at 300ms



Nissan Recording Threshold

1. Several files have been read with events under 5mph Delta V over 150ms.
2. From 2013-2019 we believed all Nissans had a wakeup threshold of AE
3. At 2020 EDR User Summit, it was reported that tests were done on a 2014 Pathfinder at low speeds. No events were created below 5mph Delta V, events were created over 5mph.
4. Threshold could be model dependent!

Mazda CDR 19.6 Coverage 44%

Model	Location	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Last 3 digits of Bosch CDR cable number showr														
MAZDA				All Mazda use 387 adapter for direct to module										
CX-3	TUNNEL BELOW IP								812					
Tribute/CX-5	TUNNEL AFT SHIFTER			Read as Escape		779			811	812				
CX-9	TUNNEL BELOW IP			778					811					
CX-30													843	
Mazda 2	TUNNEL AFT SHIFTER			792				not listed	812	not listed				
Mazda 3	TUNNEL AFT SHIFTER				779				811			843		
Mazda 5	TUNNEL BELOW IP				778					not listed				
Mazda 6	TUNNEL BELOW IP			792			779		811		831			
MX-5 Miatta	TUNNEL BELOW IP					no coverage			812					

Law Enforcement Contact Your Instructors - Data available from pre-CDR ACM's from Mazda or Mazda suppliers.

Memo: The Mazda Tribute was sister to the Ford Escape thru 2011. A 2011 Tribute was read using a 2011 Escape VIN.

Volvo CDR 19.5 Coverage 32%

Model	Location	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Last 3 digits of Bosch CDR cable number sho													
VOLVO		All Volvo use 387 Adapter for DTM images											
S40													
S60	CTR TUNNEL UNDER CONSOLE		799							500/830		?	
S80, S80L	CTR TUNNEL UNDER CONSOLE												
S90	CTR TUNNEL UNDER CONSOLE								816				
V40	CTR TUNNEL UNDER CONSOLE								816 Euro				
V60	CTR TUNNEL UNDER CONSOLE		799							500/830		?	
V70	CTR TUNNEL UNDER CONSOLE				799								
V90	CTR TUNNEL UNDER CONSOLE								816				
XC40	CTR TUNNEL UNDER CONSOLE									500/830		?	
XC60	CTR TUNNEL UNDER CONSOLE				799								
XC70	CTR TUNNEL UNDER CONSOLE				799								
XC90	CTR TUNNEL UNDER CONSOLE							816				?	

The source data for this chart is the Bosch CDR help file for Version 19.5
Always consult the latest help file for the most accurate coverage

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Old Volvo DV only Law Enf only call Volvo

No 2020 CDR coverage. Do they still have EDR??

2016+ XC90 Sample Volvo File

(vs. 2013 Volvo with minimum data)

Pre-Crash Data -1 Sec (Event Record 1)

Ignition Cycle, Crash	2,764
Safety Belt Status, Driver	On, Belted
Safety Belt Status, Passenger	On, Belted
Frontal Airbag Warning Lamp	Off
Frontal Airbag Suppression Switch Status, Front Passenger	On
Seat Track Position Switch, Foremost, Status, Driver	Not Equipped
Seat Track Position Switch, Foremost, Status, Front Passenger	Not Equipped
Occupant Size Right Front Passenger Child	Not Equipped

2020 still not added to other models

Added Data for XC90

Pre-Crash -5 to 0 sec (Event Record 1)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Service Brake (On, Off)	Steering input (%)	ABS Activity	Stability control status
-5.0	43.5 [70.0]	100.0	Off	0.0	Off	On
-4.5	47.8 [77.0]	100.0	Off	0.0	Off	On
-4.0	52.2 [84.0]	100.0	Off	2.0	Off	On
-3.5	55.9 [90.0]	100.0	Off	3.0	Off	On
-3.0	58.4 [94.0]	100.0	Off	3.0	Off	On
-2.5	61.5 [99.0]	100.0	Off	11.0	Off	Engaged
-2.0	62.8 [101.0]	100.0	Off	6.0	Off	Engaged
-1.5	65.2 [105.0]	100.0	Off	-9.0	Off	Engaged
-1.0	67.1 [108.0]	100.0	Off	-8.0	Off	Engaged
-0.5	69.6 [112.0]	100.0	Off	-9.0	Off	Engaged
0.0	70.2 [113.0]	100.0	Off	-13.0	Off	Engaged

2016+ XC90 Sample Volvo File

2020 still not added to other models

System Status at Retrieval

Vehicle Identification Number	YV1LF
Application Diagnostic Database Part Number	31387255 AL
Ignition Cycle, Download	2,765
Lifetime Operating Timer (sec)	2,154,311

System Status at Event (Event Record 1)

Data Area Status, Event Record 1	Locked, Data Stored
Data Area Read Status, Event Record 1	Data Not Read
Complete File Recorded (Yes/No)	Yes
Multi-Event, Number of Events (1,2)	Event Number 1
Time from Preceding Event (sec)	Written but No Data Available
Maximum Delta-V, Longitudinal (MPH [km/h])	< -62 [-100]
Time, Maximum Delta-V, Longitudinal (msec)	100
Maximum Delta-V, Lateral (MPH [km/h])	-19.9 [-32.0]
Time, Maximum Delta-V, Lateral (msec)	>300

Clipped Value (at Max)

Not present in 2013, Added for XC90

[redacted]

BMW CDR US 19.5 Coverage

See help file for Rolls Royce

Model	Location	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Last 3 digits of Bosch CDR adapter/cable number shown											
BMW	**All BMW require 387 Adapter for DTM**										
1 Series					not covered - Europe Only						
2 Series	Driver side, behind IP, rt of strg			7	Europe only not covered						
2 Series xdrive	Center console between seats										500/822
3 Series	Passenger side, behind the glove box		796								500/822
Active Hybrid 3	Driver side, behind IP, rt of strg			796							
4 Series	Passenger side, behind the glove box		796					500/822			
5 Series	Passenger side, behind the glove box		796								500/822
6 Series	See Service Manual								500/822		
6 Series GT	Passenger side, behind the glove box		796			500/796	500/822				
7 Series	Passenger side, behind the glove box								500/822		
8 Series	Passenger side, behind the glove box		796								500/822
X3 xDrive	Center Tunnel, Between Seats			796					500/822		
X5	Center Tunnel, Between Seats				798	500/798	500/807		also X2		
X1, X2	See Help File - Various				796			500/822			500/807
X3, X4	Center Tunnel, Between Seats				796						500/822
X6	See Service Manual								500/822		
X7	Center Tunnel, Under Dash				798				500/822		
Z4	See Service Manual				798				500/822		
i3, i8				500/807							

No “back model” coverage

500=flex ray

The source data for this chart is the Bosch CDR help file for Version 19.5

Always consult the latest help file for the most accurate coverage

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BMW EDR Evolution

1. Basic EDR 2013
2. “Extension” 2014
3. Some driver assist mid-2015 - timing
4. Auto Steering (autonomous vehicle)
   Cameras all views 4 fps
   Not intended for CDR,
   only BMW can access
5. Clock and odometer in 2019 Mini Cooper

Hardware/Software Glitch?

- Recently a 2020 BMW X5 was read DTM using CAN+/387/CDR500 and obtained a file with no events. The hex data shows memory section headings which all said “No Data Received”.
- Module was read using CDR900/CDR500 and retrieved a valid event.
- This is under investigation by Bosch, until resolved use CDR900 whenever possible.

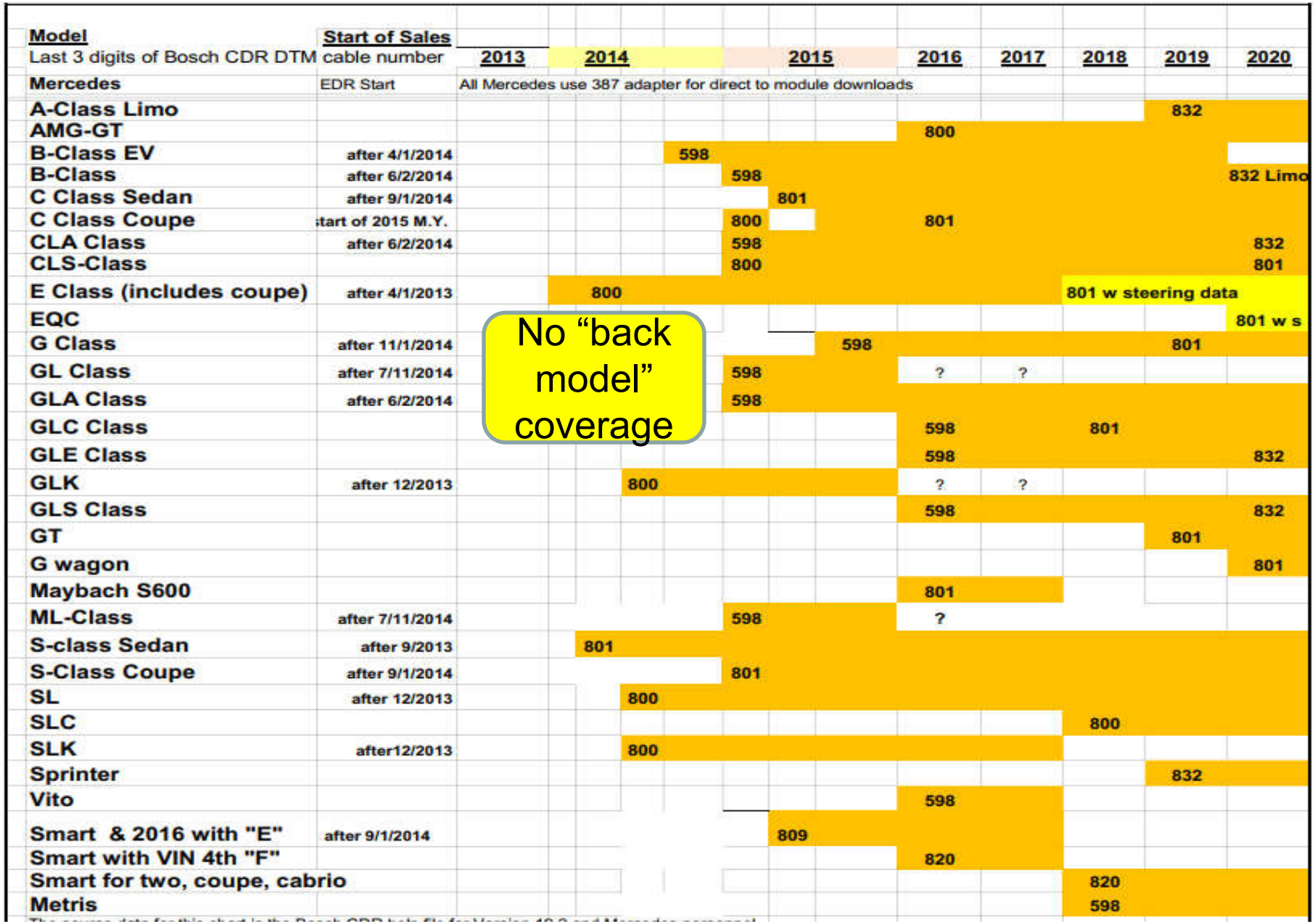
More data elements in 2020 X5 “ACSM5”

- Recent files show day/date/time of crash and second/third row seat belt use. more

Multi-Event, Number of Events	1. Event
Time From Previous Event to Current Event (msec)	Data Not Available
Time From Last Speed Data Sample (Precrash) to Time Zero (msec)	57
Time From Initial Event to Current Event (msec)	Data Not Available
Complete File Recorded, Public, Prio 1 Data	Completed Successfully
Ignition Cycle, Crash (Cycle)	1,315
Ignition Cycle, Download (Cycle)	1,316
Vehicle Clock, Date and Time at Event (YYYY-MM-DD, HH:MM:SS)	2020-12-18, 09:03:54
Vehicle Mileage (km)	13,285
Operating Time (min)	20,794

Safety Belt Status, 2nd Row, Left Side	Not Belted
Safety Belt Status, 2nd Row, Center	Not Belted
Safety Belt Status, 2nd Row, Right Side	Not Belted
Safety Belt Status, 3rd Row, Left Side	Data Not Available
Safety Belt Status, 3rd Row, Center	Data Not Available
Safety Belt Status, 3rd Row, Right Side	Data Not Available

Mercedes CDR Coverage 19.2 36%



Model	Start of Sales	2013	2014	2015	2016	2017	2018	2019	2020
Last 3 digits of Bosch CDR DTM cable number									
Mercedes	EDR Start	All Mercedes use 387 adapter for direct to module downloads							
A-Class Limo								832	
AMG-GT					800				
B-Class EV	after 4/1/2014		598						
B-Class	after 6/2/2014			598					832 Limo
C Class Sedan	after 9/1/2014			801					
C Class Coupe	tart of 2015 M.Y.			800	801				
CLA Class	after 6/2/2014			598					832
CLS-Class				800					801
E Class (includes coupe)	after 4/1/2013		800				801 w steering data		
EQC									801 w s
G Class	after 11/1/2014			598				801	
GL Class	after 7/11/2014			598	?	?			
GLA Class	after 6/2/2014			598					
GLC Class					598		801		
GLE Class					598				832
GLK	after 12/2013		800		?	?			
GLS Class					598				832
GT								801	
G wagon									801
Maybach S600					801				
ML-Class	after 7/11/2014			598	?				
S-class Sedan	after 9/2013		801						
S-Class Coupe	after 9/1/2014			801					
SL	after 12/2013		800						
SLC							800		
SLK	after12/2013		800						
Sprinter								832	
Vito					598				
Smart & 2016 with "E"	after 9/1/2014			809					
Smart with VIN 4th "F"					820				
Smart for two, coupe, cabrio							820		
Metris							598		

No “back model” coverage

Mercedes Details

1. Mercedes phased in EDR in anticipation of the NPRM requiring EDR's for Sept. 2014
2. Precrash speed, brake, throttle (563 Table 1) & lat/longitudinal ΔV.
3. Can hold 5 events, D's lock, ND's FIFO
4. Recording threshold is 5 mph over 150ms
5. Speed is avg. of 4 wheel speed sensors
6. The new 2017 E Class has Steering, ABS, Stab Cont'l. This marks a departure from their philosophy to record only the minimum required by law.

Mercedes Key Cycles

- Some Mercedes have unusually high key cycle counts for their age, and since event.
- Other brands have 2500 key cycles per yr or 20,000km.
- Working theory is that counter increments whenever key fob leaves range and comes back into range.

Ignition Cycle, Crash (cycle)	11,639
Ignition Cycle, Download (cycle)	12,113
Vehicle Mileage (km)	20,950
Operating Time (min)	29,712
Vehicle Identification Number	4JGDA5JB2HA******
Event Counter (counts)	1

VW/Audi Brands EDR

1. EDR in 2014 EOS, most 2015+ VW/Audi.
2. EDR was installed in anticipation of the US NPRM requiring EDR's by 9/1/2014, which was never implemented.
3. **Bentley** has EDR 2016+.
4. **Lamborghini** has EDR in Adventadore/Hurracan
5. **Porsche no EDR** *until 2020 Taycan.*
6. US only 2015-17, Worldwide 2018+ in transverse engine platforms

VW/Audi 19.6 US Coverage 34+%

Model	Location	2009-13	2014	2015	2016	2017	2018	2019	2020	2021
Last 3 digits of Bosch CDR adapter/cable number		See help file for Rolls Royce								
All VW/Audi Require the 387 Adapter to read DTM			Gold = US only (Canada not covered)				Yellow=Worldwide			
VW										
Arteon	UNDER CENTER STACK						805			
Atlas	UNDER CENTER STACK						805			
Beetle	UNDER CENTER STACK			804			Canada added			
CC	UNDER CENTER STACK			804						
Crafter	UNDER CENTER STACK						805			
EOS	UNDER CENTER STACK		806							
Golf, eGolf	UNDER CENTER STACK			805			Worldwide			
Jetta	UNDER CENTER STACK			804			805			
Neo										
Passat	UNDER CENTER						805			
Polo	UNDER CENTER STACK						805			
Routan	UNDER CENTER STACK	785								
Sportwagen + Alltrack				Golf platform, not called out separtely in help file						
T-Roc	UNDER CENTER STACK							805		
T-Cross	UNDER CENTER STACK						805			
Tiguan	UNDER CENTER STACK			4		805				
Touareg	UNDER GEAR SELECTOR			4				500/826		
Touran	UNDER CENTER STACK						805			
AUDI				15	2016	2017	2018	2019	2020	2021
A1	Under Center Stack							805		
A3,S3, RS3	Under Center Stack			05					842	
A4, S4, A4 allroad	Tunnel between 1st/2nd row seats			804		500/813				
A5, R5, S5, RS5	Tunnel between 1st/2nd row seats			804			500/813			
A6, A6L, S6, Ava	Tunnel between 1st/2nd row seats			804			500/826			
A7, S7, RS7	Tunnel between 1st/2nd row seats			804			500/826			
A8, A8L, S8	Tunnel between 1st/2nd row seats			804			500/826			
Q2	Under Center Stack						805			
Q3	Under Center Stack			804				805		
Q5, SQ5	Tunnel between 1st/2nd row seats			804			500/813			
Q7	Under Center Stack					500/813			500/826	
Q8	Under Center Stack						500/826			
R8	Under Center Stack					804				
TT, TTS				not covered at this time						
eTron	Under Center Stack							500/826		

Chrysler Minivan Sister

No “back model” coverage

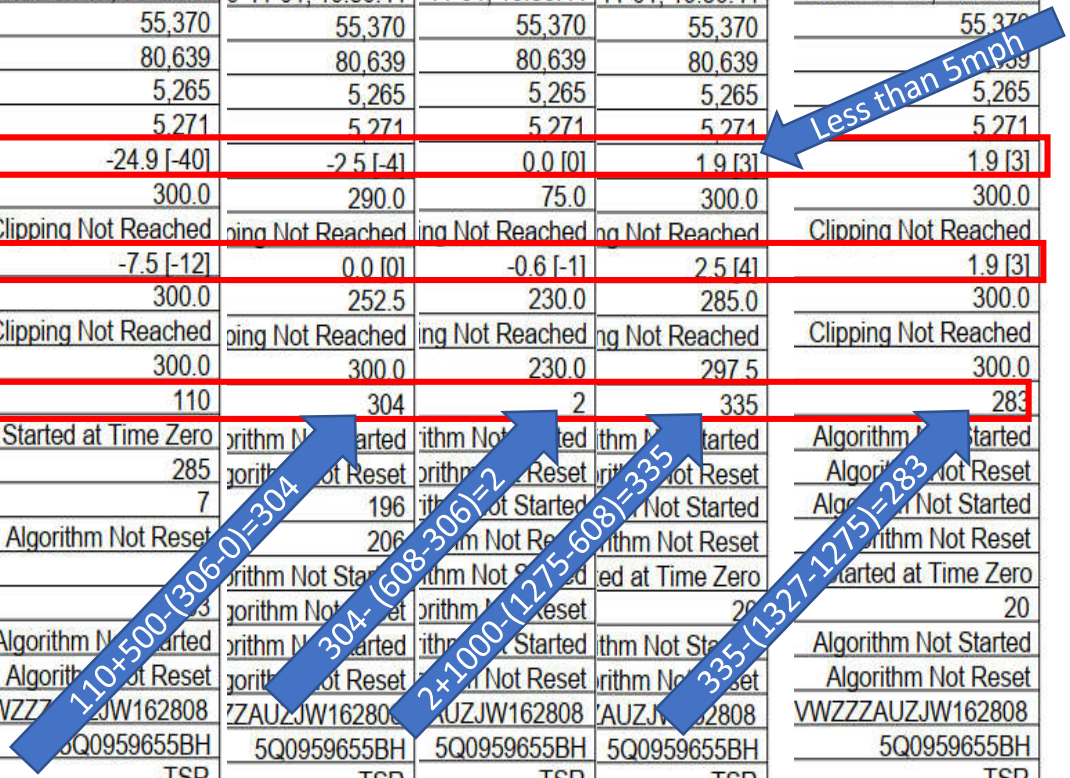
VW/Audi Details

1. Same data set for all event types
2. Time from last precrash to AE noted
3. Recorder holds up to 6 ND events in Chronological order.
4. New modules will use Flex Ray
5. Date/Time of Crash from clock (GPS if clock not set) in 2015+ Audi models, phasing into VW models
6. Newer models require hood to be open to read through DLC

Threshold & Time between events (Five Overlaid)

System Status at Event (Record 5)

Event Counter at Event	1	2	3	4	5
Event Type	Frontal	Frontal	Frontal	ta Not Available	Data Not Available
Multi-Event, Number of Events	1, Event	2, Event	3, Event	4, Event	5, Event
Time from Initial Event to Current Event (msec)	0.0	306.0	608.0	1,275.0	1,327.0
Vehicle Clock, Date and Time at Event (YYYY-MM-DD, HH:M	2019-11-01, 19:50:40	9-11-01, 19:50:41	-11-01, 19:50:41	11-01, 19:50:41	2019-11-01, 19:50:43
Vehicle Mileage (km)	55,370	55,370	55,370	55,370	55,37
Operating Time (min)	80,639	80,639	80,639	80,639	59
Ignition Cycle at Event (Cycles)	5,265	5,265	5,265	5,265	5,265
Ignition Cycle at Download (Cycles)	5,271	5,271	5,271	5,271	5,271
Maximum Delta-V, Longitudinal (MPH [km/h])	-24.9 [-40]	-2.5 [-4]	0.0 [0]	1.9 [3]	1.9 [3]
Time, Maximum Delta-V, Longitudinal (msec)	300.0	290.0	75.0	300.0	300.0
Clipping Time, Longitudinal Acceleration Sensor (msec)	Clipping Not Reached	ping Not Reached	ing Not Reached	ng Not Reached	Clipping Not Reached
Maximum Delta-V, Lateral (MPH [km/h])	-7.5 [-12]	0.0 [0]	-0.6 [-1]	2.5 [4]	1.9 [3]
Time, Maximum Delta-V, Lateral (msec)	300.0	252.5	230.0	285.0	300.0
Clipping Time, Lateral Acceleration Sensor (msec)	Clipping Not Reached	ping Not Reached	ing Not Reached	ng Not Reached	Clipping Not Reached
Time, Maximum Delta-V, Resultant (msec)	300.0	300.0	230.0	297.5	300.0
Time from Last Speed Data Sample (Precrash) to Time Zero	110	304	2	335	283
Time from Time Zero to Algorithm Start (Front) (msec)	ithm Started at Time Zero	orithm N arted	ithm Not ted	ithm N tarted	Algorithm tarted
Time from Time Zero to Algorithm Reset (Front) (msec)	285	gorit ot Reset	orithm Reset	rit ot Reset	Algori ot Reset
Time from Time Zero to Algorithm Start (Side) (msec)	7	196	it ot Started	Not Started	Alg Not Started
Time from Time Zero to Algorithm Reset (Side) (msec)	Algorithm Not Reset	206	m Not Re	ithm Not Reset	rithm Not Reset
Time from Time Zero to Algorithm Start (Rear) (msec)		orithm Not Sta	thm Not S d	ed at Time Zero	tarted at Time Zero
Time from Time Zero to Algorithm Reset (Rear) (msec)		gorithm Not et	orithm Reset	20	20
Time from Time Zero to Deployment (Rollover) (msec)	Algorithm N arted	orithm N arted	ith Started	ithm Not Sta	Algorithm Not Started
Time from Time Zero to Algorithm Reset (Rollover) (msec)	Algorit ot Reset	gorit ot Reset	Not Reset	rithm No set	Algorithm Not Reset
Vehicle Identification Number (VIN)	WVWZZ JW162808	ZZAUZJW16280	UZJW162808	AUZJ 2808	VWZZZAUZJW162808
Part Number, ACM	5Q0959655BH	5Q0959655BH	5Q0959655BH	5Q0959655BH	5Q0959655BH
Supplier ID, ACM	TSR	TSR	TSR	TSR	TSR
Production Date, ACM	171117	171117	171117	171117	171117
Supply Voltage (Before Event) (V)	14.7	12.6	12.6	12.6	12.6
Complete File Recorded	Completed Successfully	eted Successfully	ted Successfully	ted Successfully	Completed Successfully



VW/Audi Freeze Frame Data in older models?

- Diagnostic Freeze frames **at deployment** may have single values of speed data at time of crash in some 07/8+ Audi and 08/9+ VW. Start date by model/model start is unknown.
- The ***vehicle must have power during/after the crash*** to finish writing – if power lost at impact zeros will be written.
- Useability and Admissibility are still major issues –
- Recon Community Needs to Network to gather a critical mass of files (Rick will send out his equipment to you if you have a car expected to have data).

VW Freeze Frame Data Tool

Ross-Tech

Diagnostic Software for VW-Audi Group Cars

Ross-Tech, LLC
881 Sumneytown Pike
Lansdale PA 19446, USA
Tel: +1 267 638 2300

VCDS® HEX-V2® Enthusiast
- USB Interface -

- Limited # of Vehicles -
- VAG models 1995 to current* -

Starts at $199

New, universal replacement for all our previous interfaces. Includes a VCDS software license and the interface you need to connect to your VW/Audi Group vehicles. Limited to either 3 or 10 different vehicles (managed by VINs), this interface and VCDS software turns your Windows® PC into a powerful near factory level diagnostic tool for the Do It Yourself Enthusiast.



Features:

- **Fully licensed for VCDS software**, acts as the key to unlock the software, no additional software purchase required
- **VAG model year coverage:** 1995 – Current model year*
- **Fast Operation** – faster on CAN based models than our older generation wired products
- New **Improved car side connector** – increases

Data found in a 2016 VW

Last Data was similar to last EDR speed entry

```
1 Fault Found:
6351360 - Crash Shut-Down Activated
          P1609 00 [009] - -
          Confirmed - Tested Since Memory Clear
             Freeze Frame:
                    Fault Status: 00000001
                    Fault Priority: 1
                    Fault Frequency: 1
                    Reset counter: 139
                    Mileage: 27186 km
                    Date: 2017.10.09
                    Time: 07:10:23

                    Engine speed: 896 /min
                    Vehicle speed: 38.69 km/h
                    System status: Driving mode
                    Terminal 15 status: Terminal 15 on
                    Voltage terminal 30: 14.51 V
                    Power steering current consumption: -6.15 A
                    Power steering current consumption: 49.06 A
                    End stage temperature: 33 °C
                    Steering assistance: engine output: 100.00 %
                    Sensor for motor position: 211 °
                    Sensor for motor position: -2547 /min
                    Steering assistance: target engine torque: 0.03 Nm
                    Steering torque: -0.37 Nm
                    Steering torque: OK
```

TESLA



1. Remember the definition of an EDR is something that records a time series of data **as a result of a crash**.
2. Tesla may record information as a result of turning the key on, but that is not an event data recorder. More on this later.
3. Tesla did not have any EDR in the ACM of its original 2008 Roadster models.
4. Tesla has installed EDR in The 2012+ Model S, the 2015+ Model X, and the 2017 Model 3.
5. Effective March 2018, Tesla announced they will sell a publicly available tool to download the Tesla EDR

Tesla EDR

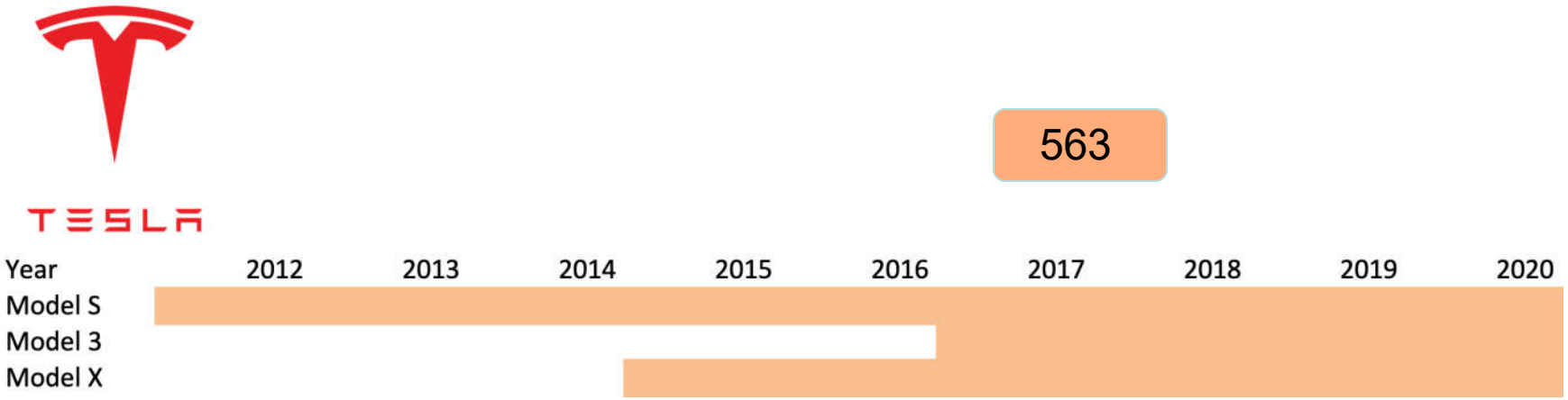
- Kit includes:
  - TIV-144: In vehicle cable for Tesla Model S and X
  - TIV-145: In vehicle cable for Tesla legacy Model S
  - TIV-996: In vehicle cable for Tesla Model 3
  - TD2M-601: Direct to Module cable for Tesla Model X and 3
  - TD2M-602: Direct to Module cable for Tesla Model S
  - TD2M-139: Direct to Module cable for Tesla Model S (NEW CABLE)
  - AC Power Adapter
  - PCAN-USB adapter
  - Hard Shell Case



COST: $1200.00

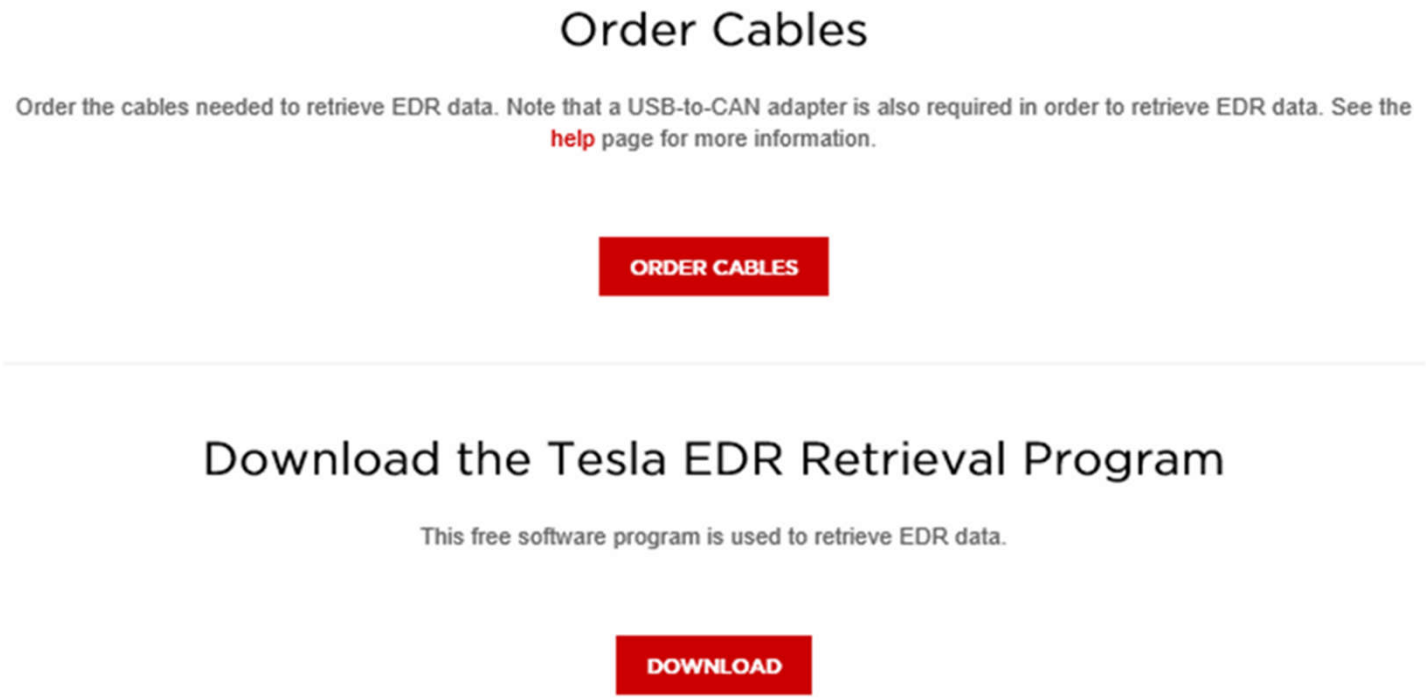
Tesla EDR

Coverage



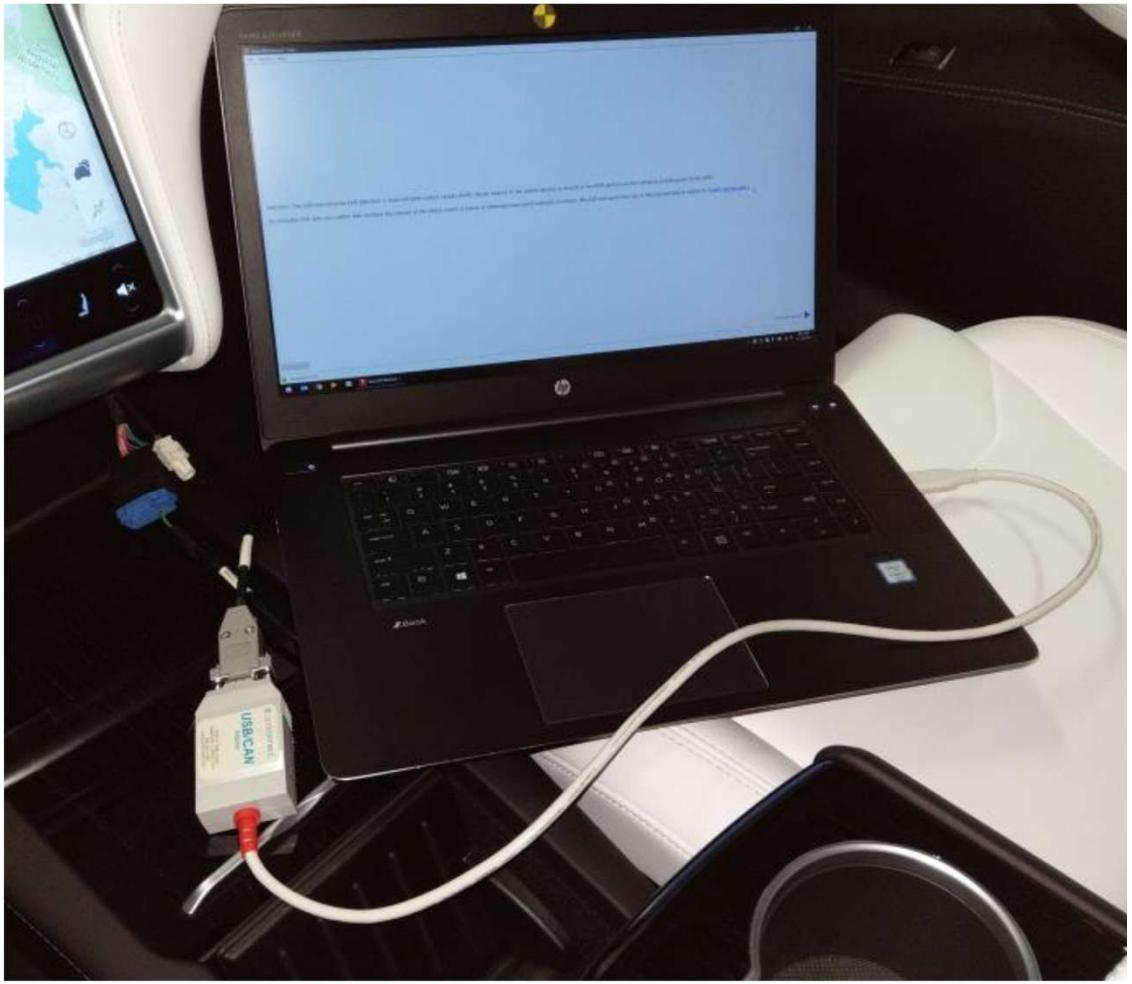
TESLA

- More information is available on their website https://edr.tesla.com
- See “Other sources” later



The Tesla EDR Tool

1. Tesla provides the software free of charge on their website that installs on your laptop. No special computer is needed.
2. The hardware kit sells for a list price of $1200.00.
3. The equivalent of the Bosch CDR interface manual is the “PCAN-USB” adapter. It is very small, and reminds you more of a USB to serial adapter, just a slight bit larger.

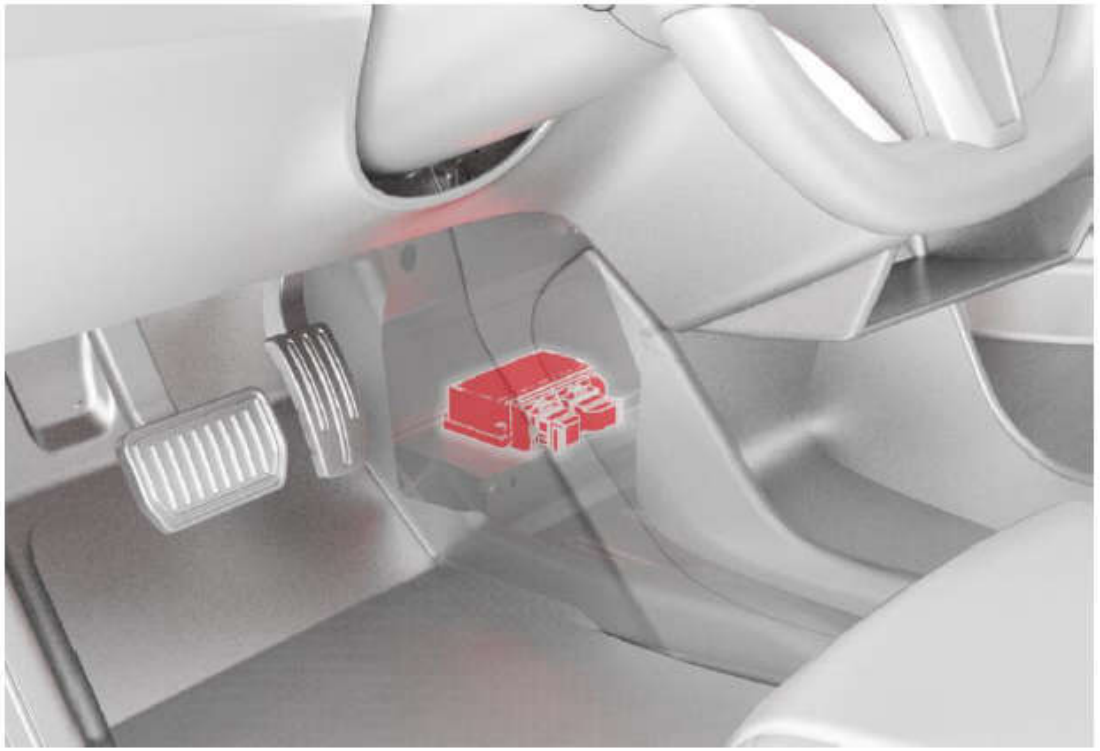


Plug in the PCAN adapter to the Body Harness. Model X and Model 3 are in other places & use a different connector described in Tesla's 27-page EDR guide at **edr.tesla.com/help**.



TESLA also has 3 direct to module connectors

If the In-Vehicle Connection procedure is unsuccessful, you might be able to connect to the RCM directly, and establish a direct-to-module connection. The RCM is located forward of the center console, below the instrument panel and center screen.



Some 2018 Model 3 Cables must be replaced

External Power must be used for DTM readouts

·. Connect 12V power to the Tesla Model S Direct-To-Module EDR Retrieval Harness.

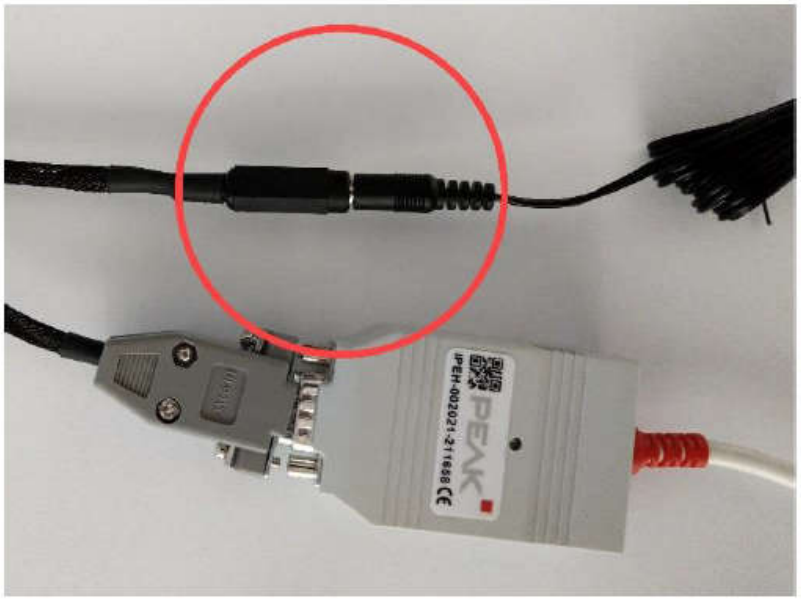


Figure 20. Connecting 12V power to the EDR retrieval harness

. Proceed to “Retrieve Data” on page 26 to complete the data retrieval procedure.

The Tesla 27-page Guide

- The Guide is VERY thorough and helpful
- It gives step by step instructions for the hardware assembly and to navigate the EDR program
- There are photographs of connector and module locations
- There are instructions for disassembly to get to the connectors and modules

Tesla EDR

- Data varies by model

	Model		
Model	S	3	X
Event Data			
Max X Δv	X	X	X
Time to X Max ΔV	X	X	X
Max Y Δv	X	X	X
Time to Y Max ΔV	X	X	X
Time to Max Resultant Δv	X	X	X
Ignition cylcle Event	X	X	X
Ignition cycle runtime	X	X	X
Odomter at event	X	X	X
Driver Seat Track Posiition	X	X	X
Driver belt	X	X	X
Passenger Belt/Occupant	X	X	X
2nd row Belt/Occupant	X		X
3rd row Belt/Occupant			X
File Complete	X	X	X

Tesla EDR

- Data varies by model

	Model S	Model 3	Model X
PreCrash Data			
Vehicle Speed	2Hz	5Hz	2Hz
Accelerator Pedal	2Hz	5Hz	2Hz
Rear Motor Speed	2Hz	5Hz	2Hz
Service Brake	2Hz	2Hz	2Hz
Stavility Control	2Hz	2Hz	2Hz
ABS	2Hz	2Hz	2Hz
Steering Wheel Angle	2Hz	10 Hz	10 Hz
Lateral Acceleration	Not Reported	10 Hz	10 Hz
Longitudinal Acceleration	Not Reported	10 Hz	10 Hz
Roll Rate	Not Reported	10 Hz	10 Hz
Yaw Rate	Not Reported	10 Hz	10 Hz

Tesla EDR

- Data varies by model

	Model		
	S	3	X
Crash Data			
Longitudinal Δv	300ms	300ms	250ms
Lateral Δv	300ms	300ms	250ms
Longitudinal Acceleration	500Hz (300ms)	2000Hz (60ms)	2000Hz (60ms)
Lateral Acceleration	500Hz (300ms)	2000Hz (60ms)	2000Hz (60ms)
Normal Acceleration	Not Reported	Not Reported	50Hz (1.4s)
Roll Angle	100Hz (6s)	Not Reported	Not Reported

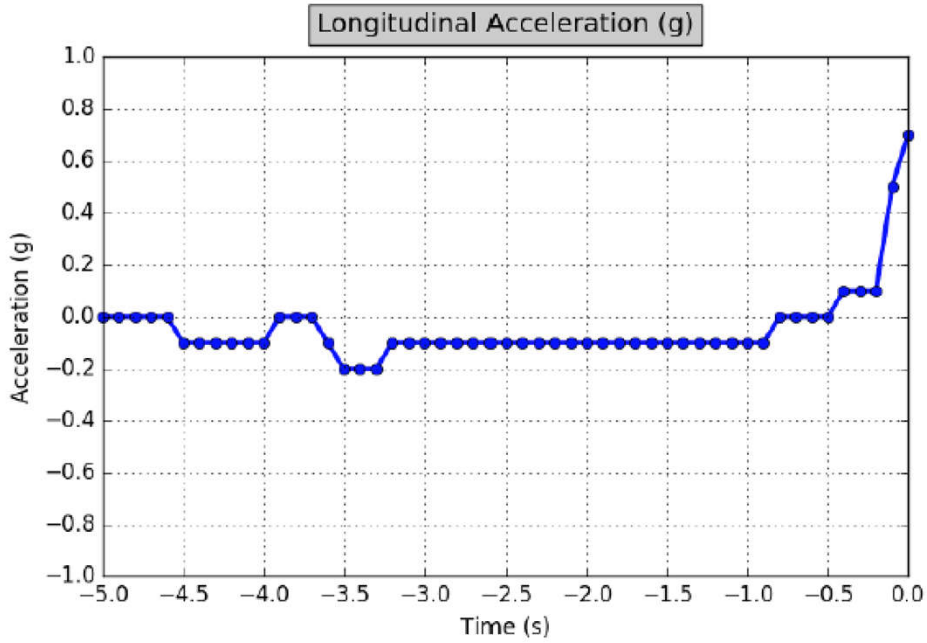
Tesla EDR Sample File

- Model 3 Data

Time (sec)	Service Brake	Stability Control	ABS Activity
-5.0	Off	Off	Off
-4.5	Off	Off	Off
-4.0	Off	Off	Off
-3.5	Off	Off	Off
-3.0	Off	Off	Off
-2.5	Off	Off	Off
-2.0	Off	Off	Off
-1.5	Off	Off	Off
-1.0	Off	Off	Off
-0.5	Off	Off	Off
0.0	On	Off	Off
[illegible]	[illegible]	[illegible]	[illegible]
-2.6	6.0	10.8	714
-2.4	6.0	13.2	746
-2.2	6.0	14.8	784
-2.0	7.0	12.4	807
-1.8	7.0	12.4	845
-1.6	7.0	12.4	883
-1.4	7.0	11.2	905
-1.2	8.0	12.0	922
-1.0	8.0	10.4	941
-0.8	8.0	4.8	932
-0.6	8.0	0.0	916
-0.4	8.0	0.0	888
-0.2	8.0	0.0	843
0.0	7.0	0.0	749

Tesla EDR Sample File

- Model 3 Data



Time (s)	Acceleration (g)	Time (s)	Acceleration (g)	Time (s)	Acceleration (g)
-5.0	0.0	-3.2	-0.1	-1.4	-0.1
-4.9	0.0	-3.1	-0.1	-1.3	-0.1
-4.8	0.0	-3.0	-0.1	-1.2	-0.1
-4.7	0.0	-2.9	-0.1	-1.1	-0.1
-4.6	0.0	-2.8	-0.1	-1.0	-0.1
-4.5	-0.1	-2.7	-0.1	-0.9	-0.1
-4.4	-0.1	-2.6	-0.1	-0.8	0.0
-4.3	-0.1	-2.5	-0.1	-0.7	0.0
-4.2	-0.1	-2.4	-0.1	-0.6	0.0
-4.1	-0.1	-2.3	-0.1	-0.5	0.0
-4.0	-0.1	-2.2	-0.1	-0.4	0.1
-3.9	0.0	-2.1	-0.1	-0.3	0.1
-3.8	0.0	-2.0	-0.1	-0.2	0.1
-3.7	0.0	-1.9	-0.1	-0.1	0.5
-3.6	-0.1	-1.8	-0.1	0.0	0.7
-3.5	-0.2	-1.7	-0.1		
-3.4	-0.2	-1.6	-0.1		
-3.3	-0.2	-1.5	-0.1		

Troubleshooting

1. Some users report a “time out” error when trying to read data. The vehicle is going back to sleep. **Putting your foot on the brake pedal** and keeping it there for the whole readout keeps the vehicle alive.
2. Low battery power accentuates the time out problem. Tesla NEMA 5-15 adapter?

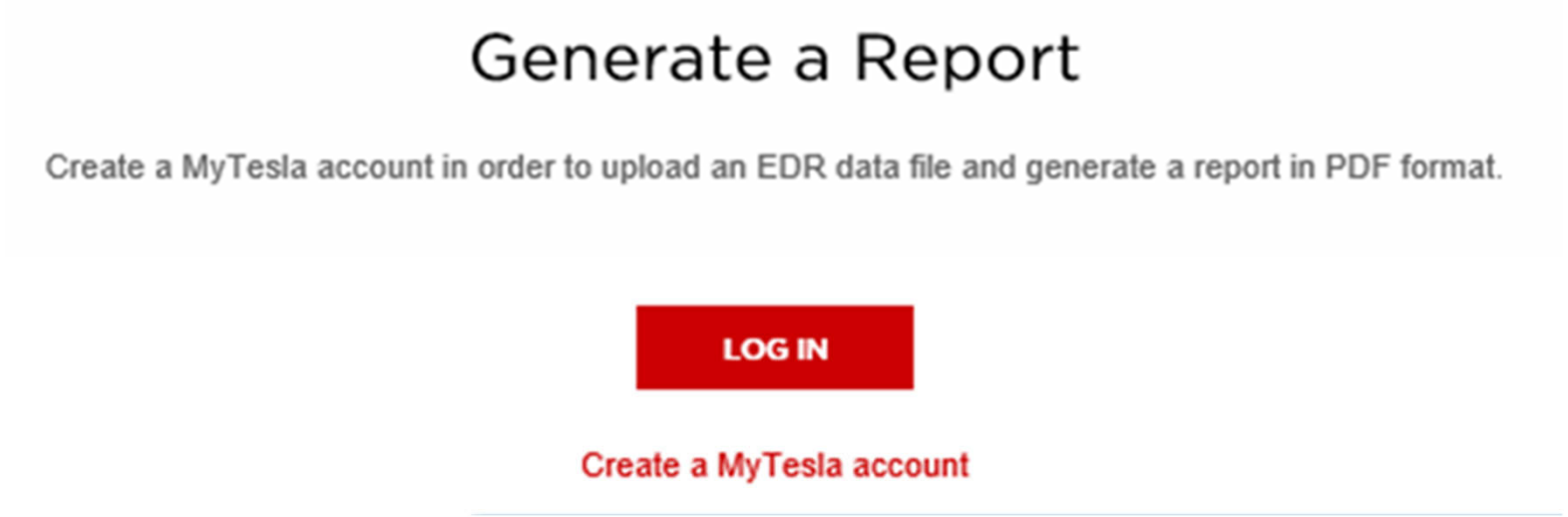




3. Or see the Help Guide for information how to backpower through the First-Responder loop.

Converting Raw Data to a Report

The Tesla software extracts a raw data file. To convert the raw data to a report, you must upload the raw data to the Tesla Server. A report is immediately returned in the latest version of software.



Worried about report conversion being biased by Tesla?

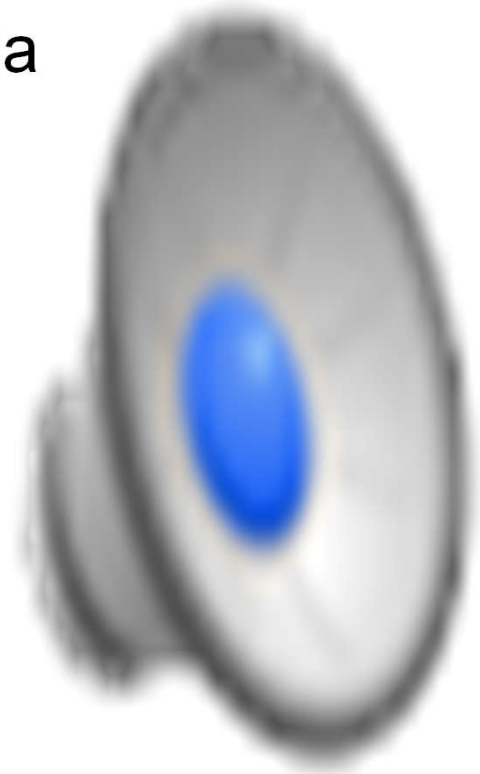
1. Data is Encrypted.
2. The report is returned IMMEDIATELY. There is no time for Tesla to alter any data to Tesla's benefit.
3. However, some clients insist that no one has a copy of the data except them.
4. There is a "Save without VIN sequence" option to de-identify data
5. There is no alternative.

What about OTHER data Tesla Data Records?

1. Most of the time you will NOT NEED to access the other data Tesla records, the EDR should be sufficient. But if you need a longer time history, Tesla records MANY data parameters every one second plus, when anything electronically monitored happens. It gets uploaded to the cloud.
2. It is NOT an "event" data recorder, it is a continuous recorder (does **NOT** meet Part 563 definition of a data recorder) (so **NO** PUBLICLY AVAILABLE tool to read it)
3. Law Enforcement contact Tesla and demonstrate proper legal authority
4. Tell them what you are interested in (they have so much you have to narrow it down)

What about OTHER data Tesla Data Records?

1. Modern Tesla have recorded video that can be uploaded to Tesla
2. This data can be retrieved from Tesla
3. Contact legal@tesla.com



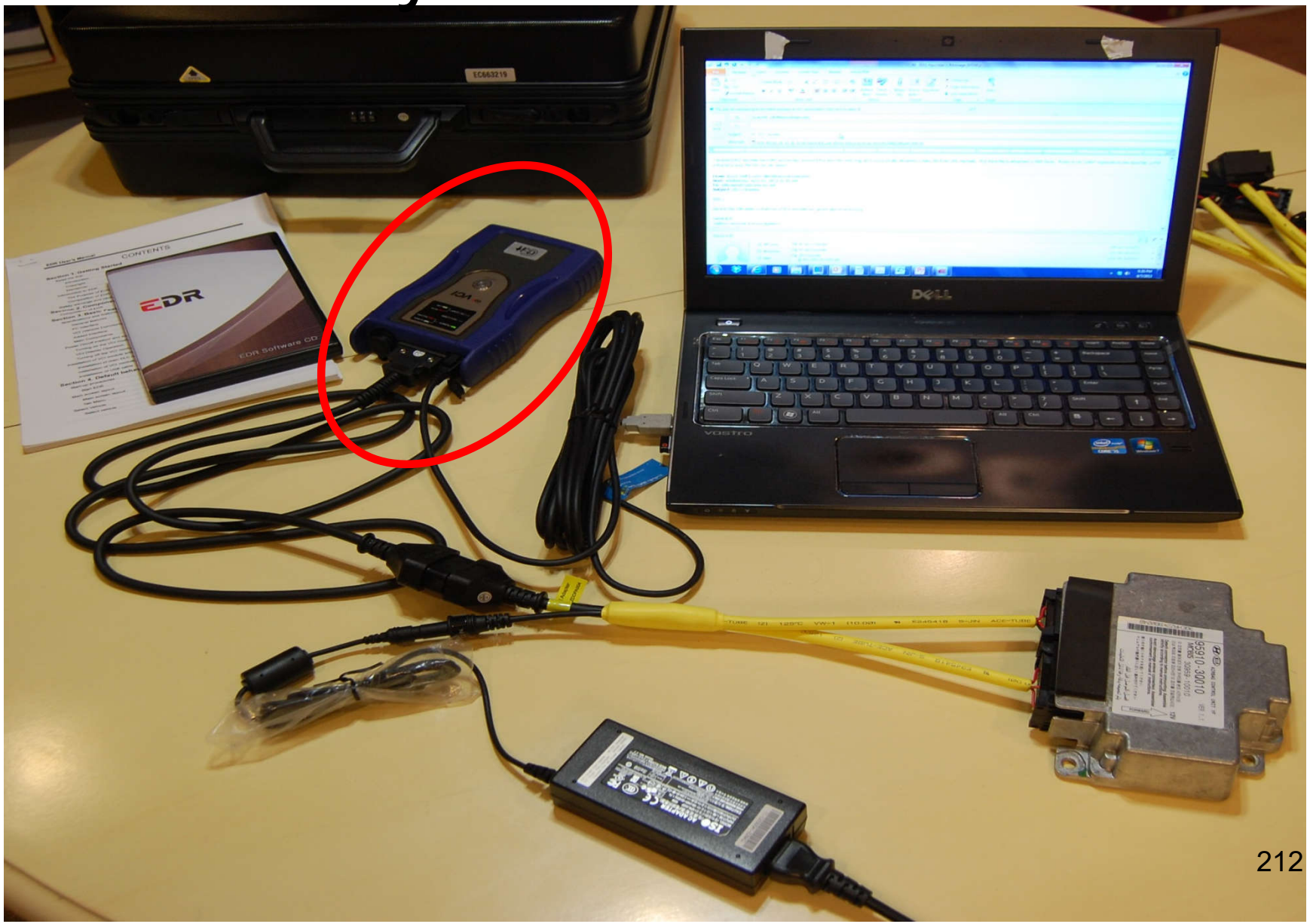
Hyundai/Kia Tools

- There is no DLC only kit. Hyundai kit comes with 13 DTM cables thru #24, Kia with 15 DTM cables thru #23. 10 Cables are in both. New cables cost $250. #25-30 for 2019-2021's not yet included.
- Hyundai: Call GIT America Tool Company order Part G0Z**H**DMN001 ***$6450 does not include software.***
- GIT now also sells Kia tool, (was Snap On),order part GIT0Z**K**DMN001 ***$6950 + tax+ship***
- Software updates **$495/yr** per brand from GIT Tool Company. Software V#38 Hyundia/ V#36 Kia released 7/2020. New $1120 CAN-FD adapter for GV80 backorder.
- Dan Portillo dportillo@gitamerica.com(888) 549-4977
- Dan will tell local PD's of nearest tool owner to them
- Kits are on back order since August 2020 due to COVID
- Different version of tool for Korea & China. Not sold in other markets. US kit won't update from non-US servers

Additional Info

1. K/H Interface modules are identical, but you must buy both to get software for both. Software updates won't download unless the correct interface module is attached to your laptop.
2. I read K/H modules for others for $500. More people are buying Hyundai kits now that we have 8+ model years on the road. Only a few own Kia tool.

The Hyundai EDR Reader



Officially Supported 2013+ Models V36/40

Model	Model Years												
KIA	2010	2011	2012	2013*	2014	2015	2016	2017	2018	2019	2020	2021	
CADENZA (Azera sister)					19(VG)	19	19	13(YG)	13	13			
FORTE	10	10	10	10 (TD)	6 (YD)	6	6	12(YD,YDM)	12	21(BDm)	21	21	
K5												26(D3LA)	
K900						11 (KH)	11	11	11	21(RJ)	21		
NIRO (DE EV, DE HEV, DE PHEV)								21(DE)	21	21	21	21	
OPTIMA	No	4	4	4 (TF/QF)	4	4	14 (JF,JFA)*	14	14	14(JFA)	14	?	
OPTIMA HYBRID	No	4	4	4 (TF HEV)	4	4	4	23(JF HEV)	23	23	23	?	
RIO	No	No	8	8(UB)	8	8	8	8	21 (SC)	21	21	21	
RONDO (Canada only 2011 MY+)	No	No	9	(not officially supported since non-US, read as a Kia Rio)									
SEDONA	No	No	No	No	10(VQ)	12(YP)	12	12	12	12	12	12	
SELTOS												28(SP2)	
SORENTO	No	3	3	3(XM)	3,7(XM)	3,7	13 (UMA)	13	13	13	13	29+CFCI	(MQ4)
SOUL	1[see note 1]	1[see note 1]	1	1(AM)	12(PS)	12(PSEV)	12	12	12	12,26(SK3 EV)	26 (SK3)	26	
SPORTAGE (SL)	No	Yes	2	2(SL)	2	2	2	13(QL)	13	13	13	13	
STINGER (CK)									21 (CK)	21	21	21	
TELLURIDE (ON)											25(ON)	25	
HYUNDAI	2010	2011	2012	2013*	2014	2015	2016	2017	2018	2019	2020	2021	
ACCENT (RB)	No 10	No?	5[see note 2]	5(RB)	5	5	5	5	21(HC)	21	21	21	
AZERA (HG)		likely	Yes	7(HG)	7	7	7	7					
ELANTRA Coupe(JK), Sedan(UD/MD	No	Yes	8	8(JK/UD/MD)	8	8	8	21(AD, ADA)	21	21	21	Hyb 28 (CN7)	
ELANTRA GT (GD) (aka "i30")	old model		new model	6/12(GD)	6,12	6,12	6,12	6,12	21(PD)	21	21	21	
EQUUS (VI)(stretch genesis)	?	yes?	yes?	9,11(VI)	9,11(VI)	9,11(VI)	9,11(VI)						
GENESIS (Coupe BK, Sedan BH/DH)	9 shave keyway	9 shave keyway	?	9(BK, BH)	9	13(DH)	13						
GENESIS G70										21,25(IK)	21,25		
GENESIS GV70												29+CFCI (JK1)	
GENESIS G80								13(DH)	13	13	13	29+CFCI (RG3)	
GENESIS GV80											29+CFCI	(JX1)	
GENESIS G90								13(HI)	13	13	13	13	
IONIQ EV and HEV								21(AE)	21	21	21	21	
KONA (i20, Rio sister)small SUV)									24 (OS)	24	24	24	
KONA EV										27 (OS EV)	27	27	
NEXO										26 (FE)	26	26	
PALISADE											25(LX2)	25	
SANTE FE	9 shave keyway note 3	9 shave keyway	9 shave keyway	7(NC,DMA)	7	7	7	7	7	25 (TMA)	25	25	
SONATA		4	?	(YFA, YF HEV	4	14 (LF) 4 (HEV)	14 (LF HEV)	14	14	14	25(DN8A)	25	
TUCSON	Yes	Yes	2	2(LM)	2	2	13(TL)	13	13	13	13	30+CFCI	(NX4)
VELOSTER coupe	out of productio		?	5(FS)	5	5	5	5	5	14 (JS)	14	14	
VELOSTER N										28(JSN)	28	28	
VENUE											30(QX)	30	

New GENESIS brand

1. The Luxury brand of Hyundai
2. G80 and G90 models. G70 new for 2019.
3. Read with Hyundai data recorder tool
4. 2021 GV80 will require CAN-FD adapter attached to interface ($1180 part)

Kia/Hyundai Tool Issues

1. You input VIN, it prints what you entered in report. VIN is not stored in the ACM. Do sufficient documentation to prove your readout came from your vehicle.
2. The K/H report is a PDF file. There is no raw data file. If K/H ever discovers an error and updates software, you will have to re-read the module. The software version at time of readout prints in the report.
3. Recommended: ***Secure module*** as evidence

Pre-2013 Kia Hyundai may have data

Model	Vehicle Size, Descrip	Model Years 2010	2011	2012	2013*
KIA		Cable #	Cable #	Cable #	Cable #
CADENZA (VG, 2014) (Azera sister)	112.0" WB	YG requires cable 13 per software			
FORTE (TD 2013, YD 2014))	104.3"WB, compact	10	10	10	10 (TD)
K900 (KH 2015)	119.9"WB				
NIRO					
OPTIMA (2013 TF QF, 2016 JF, JFA)	110.0" WB sedan	No	4	4	4 (TF/QF)
OPTIMA HYBRID	110.0" WB sedan	No	4	4	(TF HEV)
RIO (UB)	94.5"WB, compact	No	No	8	8(UB)
RONDO (Canada only 2011 MY+)		No	out of production		
SEDONA (VQ '14, YP '15)	113.8/118.9" WB miniv	No	No	No	No
SORENTO (XM to 2015)(UMA 2016)	106.3" WB, crossover	No	3	3	3(XM)
SOUL (AM 2013, PS 2014+)	100.4" WB	1 see note 1	1 see note 1	1	1(AM)
SPORTAGE (SL)	103.9" WB, crossover	No	Yes	2	2(SL)
HYUNDAI					
ACCENT (RB)	94.5"WB, compact	No 10	No?	5 see note 2	5(RB)
AZERA (HG)	109.4"WB, sedan		likely	Yes	7(HG)
ELANTRA Coupe(JK), Sedan(UD/MD)	104.3"WB, compact	No	Yes	8	8(JK/UD/MD)
ELANTRA GT (GD) (aka "i30")	former Elantra Touring	old model		new model	6(GD)
EQUUS (VI)(stretch genesis)	119.9" WB, luxury seds	?	yes?	yes?	9(VI)
GENESIS (Coupe BK, Sedan BH/DH))	115.6" WB, luxury seds	9 shave keyway	9 shave keyway	?	9(BK, BH)
GENESIS G90					
SANTE FE (Sport DMA, NC)	106.3" WB, crossover	9 shave keyway note 3	9 shave keyway	9 shave keyway	7(NC,DMA)
SONATA (YF sedan, YF HEV)	110.0" WB sedan		4	?	4(YF)
TUCSON (LM to 2015)(TL in 2016)	103.9" WB, crossover	Yes	Yes	2	2(LM)
VELOSTER coupe (FS)	104.3"WB, compact	out of productior		?	5(FS)

2010-2012 EDR Access

1. The tool requires VIN input and only allows read of 2013+ VINS.
2. Input surrogate 2013 VIN to communicate with 2010-2012 models. Report will list surrogate VIN used.
3. Decoder changed slightly for 2013. Not all data decodes properly.
4. 2010-12 file admissibility must be presented skillfully (get help).
5. **Data successfully admitted in Virginia, Florida, and California.**

K/H 2010-2012 data reliability?

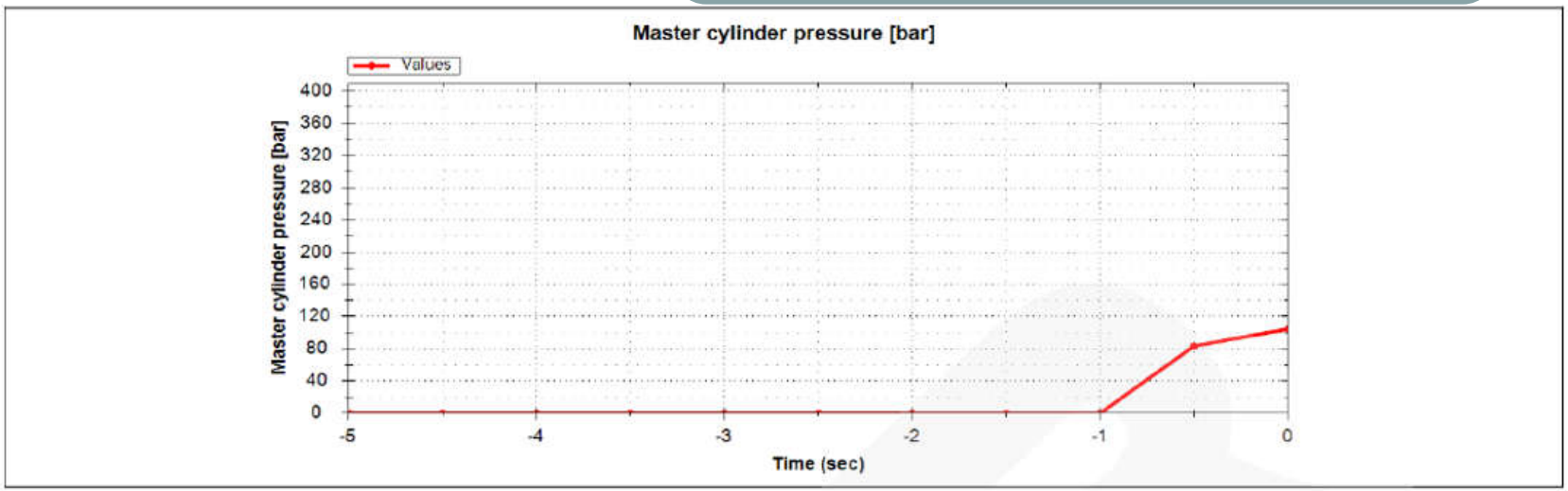
1. In 50+ NHTSA crashes, speed, delta V and belt use appear OK (one exception)
2. Steering angle, and accel data appear implausible on some
3. Key cycles at download not in all models
4. SAE 2014-01-0502 has details.
5. Additional testing on 2012 Soul reported in SAE 2015-01-1445.
6. Admitted to court in VA, CA, one other

Brake Pres. added to 2019 Forte

< Event 1 >

Master cylinder pressure

Units in Bars. 1 Mpa=10 bars
Toyota max scale is 12 Mpa=120 bar



Num	Time (sec)	Master cylinder pressure [bar]
1	-5.0	0.2
2	-4.5	0.2
3	-4.0	0.2
4	-3.5	0.2
5	-3.0	0.2
6	-2.5	0.2
7	-2.0	0.2
8	-1.5	0.2
9	-1.0	0.2
10	-0.5	83.7
11	0.0	104.6

Other Non-Bosch CDR Tools

Law Enforcement: Ask Subaru dealer to help you with my tutorial

	2012 Nov YTD SALES	Nov YTD US Market SHARE	2013 EDR?	Tool?
SUBARU	299,788	2.3%	YES	Denso DST-i

Bosch coverage starts

Dealer scan tool $2835 plus software $1835/mfr (905)669-4812

New Denso tool from Nuspire required for 2016+, $6K, $2700 software/yr

MITSUBISHI	53,677	0.4%	YES	OTC tool co

Bosch coverage starts

Panasonic Touchpad + software $6395, annual $1200/yr, 888 727 6672

JAG/LAND ROVER (Tata Motors India)	49,452	0.4%	YES	OTC/Bosch SPX
SAAB	480	0.0%	YES?	Hitachi HDS-3000?

Full dealer scan tool, Toughbook & software approx $5000

Hitachi HDS-3000, software $1835 for Saab

Subaru EDR Availability
Mostly CDR now, but 19 WRX require Denso DSTi & SSM4

Model	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Bosch CDR900 Interface Required for true Subaru models											
Cables 614 and 616 were originally released for Toyota models but also service some Subaru's											
SUBARU US Models											
Ascent	Not in production								835		
BRZ	n/a	n/a	Read as Toyota brand 86 model with Cable 614 (CAN+ or CDR900 interface)								
Crosstrek	n/a	n/a	616								
Forester	No	No	616						835		
Impreza	No	614		614/616							Japan
Legacy	No	No	614		836					844	
Outback	No	No	614		836					844	
WRX/STI	No	No	No	No	614		616	614/616	SSM4 w/DSTi		
Tribeca	No	No	No	No	Discontinued						
NON-US Models											
Exiga (Australia, Japan)			614								
XV (Europe)		614					614/616			616	
Levorg (Japan)					614		614/616			?	

NOTE: Bosch CDR coverage phased in from software versions 19.1 to 19.4 to 19.6

CDR 2015+ Outback Precrash Data

Pre-Crash Data -5 to 0 Sec (Record 1)

Time (sec)	-5.0	-4.5	-4.0	-3.5	-3.0	-2.5	-2.0	-1.5	-1.0	-0.5	0.0
Speed, Vehicle Indicated (MPH [km/h])	21.1 [34]	22.4 [36]	23.6 [38]	24.9 [40]	24.9 [40]	23.0 [37]	19.3 [31]	16.2 [26]	10.6 [17]	7.5 [12]	6.8 [11]
Accelerator Pedal, % Full (%)	23	28	24	0	0	0	0	0	0	0	0
Service Brake, On/Off	OFF	OFF	OFF	OFF	ON	ON	ON	ON	ON	ON	ON
Engine RPM (RPM)	1,900	2,100	2,300	1,800	1,300	1,100	1,200	1,100	1,000	1,000	1,000
ABS Activity	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Stability Control (On, Off, Engaged)	ON	ON	ON	ON	ON	ON	ON	ON	ON	ON	ON
Steering Input (deg)	0.0	0.0	0.0	0.0	0.0	-2.5	-2.5	-5.0	-5.0	-2.5	-2.5

Subaru EDR Tools

1. Bosch CDR 900 reads 2015+ Legacy/Outback and 2019+Forester/Ascender. CDR 19.4 covered remaining models EXCEPT 2019/20 WRX.
2. For 2019 WRX use Subaru Denso DSTi diagnostic interface with SSM**4** software from Nuspire. Tool + 1st year software sub now $5.8K/yr, is compatible to '04. Software sub $2700/yr.
3. For Subaru BRZ, read with CDR as a Toyota

MITSUBISHI 0.5% market share

1. Bosch CDR now covers 2011+ Outlander Sport, 2014+ Outlander & 2017+ Eclipse Cross. ***More expected in 21.1 2020/21 CDR coverage not listed in help file – does it work?? Accepts VIN.***
2. Pre-Bosch Mitsubishi EDR only tool based on dealership a Panasonic Touchpad. Cost to $6380.
3. The kit is made to work only on vehicles with a functioning electrical system thru the **DLC**.
4. Some Mitsubishi have an **"ETACS" gateway module (interior fuse box)** that must be included in circuit to make it work DTM (Brad Muir has built a DTM circuit).
5. Some incomplete files retrieved (no data).

Mitsubishi EDR's Jan 2021
Some use CDR, some use private tool

Mitsubishi ACM part numbers normally start with 8635A followed by 3 digits, some part nos shown below
Mitsubishi was part owned by DaimlerChrysler 2000-2004
Information before 2013 is incomplete and compiled from reports from users
Some models are covered by Bosch CDR, some require the proprietary Mitsubishi tool

Model	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Eclipse (PS Platform)	Part MU141170 4/2005 to 8/2011		Proprietary EDR Tool		Production ends 8/2011										
Eclipse Spyder Conv.		?	Data in deployments only?												
Eclipse Cross (SUV)										837 Bosch CDR			?	?	
Endeavor SUV (PS Platform)		?			Production ends 8/2011 replaced by Outlander										
Galant (PS Platform)		?				Production ends 8/2012									
iMiEV		Not sold in US				No gateway module					?				
Lancer		Read as 09 Dodge Caliber VIN 1B3HB48A89D000000 Deploys only Cable 226 Parts A146,A254, A288				ETACS gateway. Part A307-308 thru 201					Production Ends				
Lancer Evolution															
Lancer Sportback															
Mirage (subcompact)						No gateway module									
Outlander SUV (GF platform) & PHEV		A208 Read w/CDR as Caliber		A208/A304 mix. A208 5 sec@0.1. Deploy only. Read as 09 Caliber.		ETACS gateway	838 Bosch CDR						?	?	828 Bosch CDR
Outlander Sport, RVR, AS				838 Bosch CDR									?	?	
Raider pickup (Dodge Dakota sister)		Bosch CDR supported cable 228 (2007 -2010)		Not in production				Bosch cables 839 and 840 released in anticipation of future coverage - Mirage & Lancer??							

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The source of this data is the Mitsubishi EDR tool flyer from Bosch Diagnostics and CDR help file version 21.0 &other available industry data - best info avialable, not guaranteed
Acknowledgement to Brad Muir of Crash Data Specialists for providing many details on models and data available
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Typical Mitsubishi CDR Data

Pre-Crash Data -5 to 0 Sec (Record 1)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Service Brake, On/Off	Engine RPM (RPM)	Steering Input (deg)
-5.0	3.1 [5]	0	On	797	4.5
-4.5	2.5 [4]	0	On	751	4.5
-4.0	1.9 [3]	0	On	648	4.0
-3.5	1.2 [2]	0	On	593	4.0
-3.0	0.6 [1]	0	On	641	4.0
-2.5	0.6 [1]	60	Off	719	4.5
-2.0	1.2 [2]	71	Off	1,297	14.0
-1.5	4.3 [7]	75	Off	1,840	46.5
-1.0	7.5 [12]	77	Off	2,010	90.0
-0.5	9.9 [16]	76	Off	2,180	124.5
0.0	11.2 [18]	72	Off	2,206	146.0

- +Long & Lat Delta V and Acceleration
- +Roll Angle -1 to +5 sec @ 0.1 sec interval

Mitsu Proprietary PDF report function not working on 2018+

- Mitsubishi not responded to software update requests
- Error message when trying to PDF 2018+ reports
- Robert Willer workaround - paste 2018 CSV data into a file from a 2017.
- The 2017 file PDF's successfully and appears to be an accurate representation of the data, except it says it is from a 2017

Jaguar/Land Rover

1. Jag/LR have EDR's, they are read with a dedicated laptop and software. Tool only retrieves hex data, must be sent to England for interp. Only Robert Willer in FL has the tool (and he will help others).
2. insur@jaguarlandrover.com for private, govn@jaguarlandrover.com for law enf.
3. **They are now referring requesters to suppliers.** (call to find out who supplier is)
4. Some Mid 2000's have ***PCM data (***like Ford) but *not spoofable* with CDR.

Jaguar/Land Rover Data

1. Recent reports from Autoliv/Veoneer have full Part 563 Content
2. Recent reports from Bosch have only a single value of speed and no time series of Delta V data, which means it does NOT meet the definition of an EDR.
3. This is the first known instance of a manufacturer that had an EDR pulling the EDR out.

Newly published research

1. Every year at the SAE World Congress in April new research is presented.
2. A very brief summary follows, I encourage you to read these papers.
3. Other good sources of EDR info include Collision Magazine

https://www.collisionmagazine.com/article-index/

& Accident Reconstruction Journal.

EDR SAE Papers 2020

2020-01-1327 Performance of Event Data

Accuracy of Speed Change Measured by Event Data Recorders during Oblique Offset Frontal Impacts by by Curtat, Wilkinson, Siegmund (MEA Forensic)

2020-01-1328 Evaluation of General Motors Event Data Recorder Performance in Semi-Trailer Rear Underride Collisions by Famiglietti, Fatzinger et al (Momentum Engineering)

2020-01-TBD Analysis of Honda Accord EDR Data for Low- to High Speed Impact Severities by Crosby/ Skiera/Bare (Exponent)

2020-01-1329 Further Non-deployment and Deployment Laboratory Experiments Using a Toyota Auris 2007 Event Data Recorder by Tabone (Univ. of Malta)

Possible EDR @ SAE 2021

currently in peer review

1. **Accuracy of Gen 2 Toyota EDR in Low Speed Side Impacts**
2. **Analysis of Crash Data from a 2012 Hyundai Genesis Coupe Event Data Recorder**
3. **Re-Evaluating Event Data Recorder Sample Rates to Accurately Capture Evasive Maneuvers in Modern Vehicles**
4. **Validation of EEPROM Chip Removal and Reinstallation for Retrieval of Electronic Crash Data**

Regulatory Update

1. NHTSA issued NPRM in Dec 2012 to **require** 563-EDR's by Sept 2014 (past!)
2. NPRM proposed EDR as FMVSS 405, OEM's would have to self certify EDR's. Strong OEM objections were filed.
3. Officially withdrawn Feb 2019 https://www.federalregister.gov/documents/2019/02/08/2019-01651/federal-motor-vehicle-safety-standards-event-data-recorders

4. Mfrs were NOT expected to remove EDR's – But one Land Rover model may have – now has Bosch freeze frame recorder.

Regulatory Update

1. NHTSA/IIHS got 10 auto mfrs to agree on 9/11/2015 to making **A**uto **E**mergency **B**raking **(AEB)** standard (timeline 2022). Voluntary commitment is a relatively new strategy by NHTSA. Now 20 have agreed.
2. Next big issue is Autonomous Vehicles (self driving cars).
3. States are drafting conflicting regulations on type of certification required – could hinder development.

Appeal for Files

- Manufacturers keep coming out with new generations of EDR's
- The way I keep up is asking my network of former students send me files from new vehicles
- PLEASE SEND ME FILES FROM 2019/2020/2021 MY VEHICLES
- If you don't de-identify them, I will
- Helpful if I know which way they turned to check polarity of steering, yaw, lat accel

Crash Data Retrieval Update Nov 2020 End



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